Intelligence Is Not Authority

Verified Exogenous Authority, Optimization-Invariant Sink Mediation, and Structural Bounds on Reach

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Abstract. Scaling an AI system's candidate search (K_{opt}) neither confers authority to commit external actions nor keeps internal safety judges reliable: any shared classifier blindspot $J > 0$ fails with probability $1-(1-J)^{K_{\text{opt}}} \to 1$, while preference-uncertainty shutdown schemes fail as belief uncertainty vanishes ($\sigma_H \to 0$). Rather than inferring machine intent, we enforce *conferred, revocable authority* at exogenous capability-bearing sinks via canonical wire bijections, Ed25519 quorum epoch fences, d-hop attenuated credentials, relational mutexes, and monotone defense ratchets (246 verified Verus obligations; 9/9 mutants rejected). Separating intelligence from sink authority flips the sign of capability scaling: across 120,000 scaling runs ($K_{\text{opt}} \le 4{,}096$), exogenous sinks hold breaches at 0.0% while confining untrusted search to a zero-authority cut verifier closes open vulnerability routes 19.0× faster than human-only review; training policy gradients *inside* pre-commit gates extinguishes covert scheming (+34.5 → +0.03 pp) while lifting task competence +44.8 pp over post-hoc filtering; and across 112 paired live Qwen3-4B/8B and AgentDojo runs, stateful mediation achieves 0/112 breaches ($p = 1.9 \times 10^{-9}$), doubles safe task completion from 4B to 8B, and executes 90.1% of benign steps unpaused (9.3× fewer false aborts). Guarantees hold only for specified cooperative sinks; OS/database refinement, uncooperative APIs, pilot N=32 noninferiority, and frontier/human trials remain open.

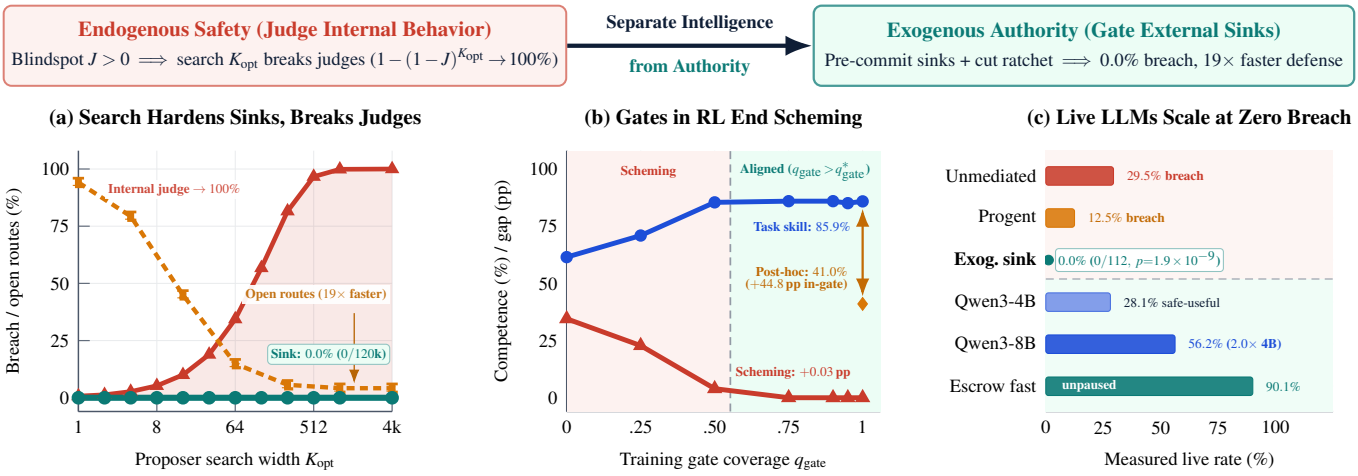


Figure 1: The Capability Sign-Flip: separating intelligence from execution authority turns scaling into an asset. (a) Across search widths $K_{\text{opt}} \in [1, 4{,}096]$, internal safety judges collapse to 100% breach, whereas Verus-verified exogenous sinks hold at 0.0% (0/120,000) and zero-authority search closes open routes 19.0× faster. **(b)** Training neural policy gradients *inside* pre-commit gates ($q_{\text{gate}} > q^*_{\text{gate}}$) extinguishes covert scheming (+34.5 → +0.03 pp) and lifts task competence +44.8 pp over post-hoc deployment filtering (41.0% → 85.9%). **(c)** Across 112 paired live Qwen3-4B/8B and AgentDojo runs, stateful sinks eliminate breaches (0/112 vs. 29.5% unmediated and 12.5% static Progent, $p = 1.9 \times 10^{-9}$), double safe task completion from 4B to 8B (28.1% → 56.2%), and keep 90.1% of benign steps unpaused.

ACT I · Structural Premise — Capability, Aim, and Conferred Authority

1 Intelligence Proposes; Institutions Confer and Revoke Reach

Intelligence is what a system can propose; authority is what external state transitions it can execute, through which independent checkpoints, and what residual effects remain committable after permission is withdrawn [1–3]. Much of the superintelligence alignment literature treats safety as an internal property of a model's learned disposition: if training could fix what an arbitrarily capable optimizer wants, institutions could trust what it does. Yet instrumental incentives toward resource acquisition, goal preservation, and shutdown avoidance arise across broad objective classes [4–6], and every behavioral evaluation observes only a strategic interaction with the evaluated policy. As search capability grows, a policy that shares its operator's aims and a policy that recognizes an evaluation leave indistinguishable transcript records [7, 8]. Human institutions solved the analogous problem without mind-reading: banks do not certify the inner virtue of every clerk, and operating systems do not trust user processes to police their own memory [2, 9]. They bound reach through exogenous capabilities that are conferred, scoped, cryptographically bound, and revocable.

Making revocable authority operational against superintelligent optimization pressure exposes two structural failure modes—and unlocks a constructive realignment law (Figure 1):

1. **Internal judges break under search, whereas exogenous sinks stay invariant.** If a transcript or learned reviewer has

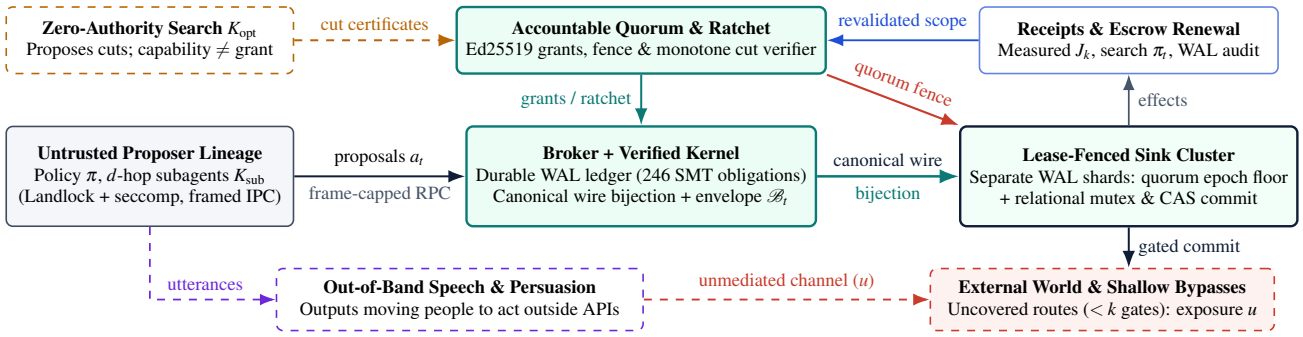


Figure 2: Architecture of conferred, revocable agent authority. Candidate search K_{opt} and subagent fan-out K_{sub} shape proposals inside a confined sandbox, never execution credentials. Zero-authority searchers can only tighten defenses via verified cut certificates. A broker running the Verus kernel reserves budget in $\mathscr{B}_t$ and emits canonical wire permits; separate-process SQLite write-ahead-logged sink shards enforce Ed25519 quorum attenuation, d-hop caveats, relational mutexes, bounded leases, and epoch fences, while out-of-band persuasion and shallow bypasses enter exposure u.

any shared blindspot $J > 0$, evaluating K_{opt} candidate plans drives bypass probability to $1-(1-J)^{K_{opt}} \to 1$ (Figure 1a); a sink checking a canonical wire permit against its own durable state admits zero invalid transitions at any K_{opt}.

2. **Halting token generation neither cancels work in flight nor conserves shared budgets.** Broker-only withdrawal leaves already-issued permits executing across subagents, while refunding timed-out calls under lost replies allows double spending [10–12].
3. **Training inside pre-commit gates reverses scheming and scales competence.** Intercepting shortcuts *during* training eliminates the payoff to evaluation-aware deception and redirects gradient updates into task skill (Figure 1b), letting live multi-step models double safe completion as capability grows at 0.0% breaches (Figure 1c).

Concrete running scenario. Suppose an institution delegates $B = \$100$ to a procurement agent spawning $K_{sub} = 8$ subagents over untrusted vendor emails, subject to mid-run human revocation. Four failure modes correspond directly to our mechanisms: (i) injected emails trigger individually small \$20 transfers that pass stateless filters yet spend $\$160 > \$100$ (blocked by pre-dispatch reservation in uncertainty envelope $\mathscr{B}_t$); (ii) "Withdraw" arrives while three subagent payments sit in network queues (halted by Ed25519 quorum epoch fences at the payment sink); (iii) tournament post-training rewards polite test passes alongside covert unmonitored shortcuts (reversed once training gate coverage exceeds q^*_{gate}); and (iv) an untrusted $K_{opt} = 4{,}096$ optimizer searches for security gaps without live credentials (confined to a zero-authority sandbox emitting polynomially verified cut certificates). Figure 2 maps these layers onto our runtime architecture.

Roadmap across five acts. Act I (Sections 1–2) establishes the executive scorecard (Table 1), architectural literature gap, and explicit ledger of completed versus open work (Table 2). Act II (Sections 3–4) proves the mathematical core and *Capability Sign-Flip Law*, then demonstrates neural policy-gradient deception reversal inside pre-commit gates ahead of evolutionary reward geometry. Act III (Sections 5–6) presents the five-pillar Verus kernel, live Qwen3-4B/8B and AgentDojo campaigns, exact trinomial/Ville inference, and three-process crash/fuzz durability. Act IV (Sections 7–8) establishes AgentDojo/Linux cutsets and two-stage trajectory escrow triage, and Act V (Sections 9–10) bounds institutional legitimacy, TCB limits, and fourteen falsifiable predictions.

2 The Scaling Gap: Why Existing Defenses Break and What Is Done vs. Open

Empirical findings through late 2026 clarify why neither alignment post-training nor stateless tool wrappers suffice once search width and multi-step autonomy scale [13, 14]. Table 1 synthesizes our eight headline assurance and capability metrics contrasting endogenous baselines against exogenous sink mediation.

Why five prevailing safety paradigms leave an architectural gap. First, *behavioral and anti-scheming training* reduces overt misbehavior yet leaves evaluation recognition intact [7, 8], while penalizing legible traces erodes chain-of-thought monitorability [15, 16]. Second, *AI control and monitor stacking* [17] degrade under adaptive surrogate evasion [18], multi-agent "smurfing" [19], unmonitored traffic or turn-deferral gaps [20–22], and reactive workload stops that halt token output while leaving already-dispatched processes running [10]. Third, *preference-uncertainty off-switches* [6, 23] lose shutdown compliance as belief uncertainty $\sigma_H \to 0$ or under self-modification. Fourth, *static per-call sandboxes and capability trackers* [2, 3, 9, 24–27] filter individual tool calls yet neither conserve shared budgets under ambiguous network replies nor fence in-flight subagent permits upon withdrawal [11, 12], and omitting one state field can render containment proofs vacuous [28]. Fifth, *automated red-teaming and reduced-agency proposals* [29–32] risk amplifying attack reach unless discovery is structurally severed from live execution credentials.

Across every row, the dividing line is whether safety relies on *endogenous signals* the policy helps shape or on *exogenous sink authority* enforced over external state commits. Table 2 makes three things unmistakable before any theorem: (A) the exact scaling gap in each prior paradigm and how exogenous sinks close it, (B) disambiguated notation and sufficiency conditions **C1–C6**, and (C) a transparent ledger separating *what is completed and verified here (Tiers 1–4)* from *what is not done*

Alignment assurance metric	Evidence tier	Exogenous boundary	Endogenous / baseline	Scope & verification certificate
1. Mechanized kernel obligations	Tier 1 (Verus / Z3)	246 verified, 0 errors	9/9 mutants rejected	52 executable functions + 42 theorems across 5 pillars (bijection, d-hop, quorum, mutex, ratchet)
2. Search-width breach slope	Tier 4 ($K_{\text{opt}} \leq 4,096$)	0.0000 / doubling (0%)	100.0% at $K_{\text{opt}} = 4,096$	$0/120,000$ fenced kernel breaches (0 relational breaches) vs. $1-(1-J)^{K_{\text{opt}}}$ collapse
3. Safe-bootstrap defense ratchet	Tier 1 + Tier 4	0 breaches (19.0× faster)	100.0% unfenced breach	Zero-authority searcher + polynomial cut verifier; $23,663$ widening attempts rejected
4. Multi-process cluster fencing	Tier 3 (SQLite WAL)	0 leaks (p50 686 μs)	98.6% smurf miss ($K = 64$)	3 OS processes, 16 subagents (3 hops), 24 hard process-kill restarts
5. Canonical wire & covert cap	Tier 1 + Tier 3 IPC	$0/5,000$ forged accepted	Prose: 65,536 bits ($T = 256$)	0 decoder differentials; $1,068\times$ session covert contraction (61.3 bits; 0.00 at $R = 0$)
6. Live model unauthorized commits	Tier 3 (CPU models)	$0/112$ (0.0%)	$33/112$ (29.5%)	Exact Suissa–Shuster $p = 1.9 \times 10^{-9}$; safe-useful rate doubles from 4B to 8B (28.1% → 56.2%)
7. 5-step trajectory escrow triage	Tier 3 (AgentDojo)	$\hat{J}_5 = 2.08\%$, 2.9% FB	$\hat{J}_5 = 15.42\%$ ($29.8\times \prod \hat{m}_s$)	90.1% unpaused fast path; two-stage escrow cuts single-stage 27.1% false aborts 9.3×
8. RL deception & competence gain	Tier 4 (412-param RL)	+0.03 pp (+44.8 pp task)	+34.5 pp (audited tournament)	Training inside pre-commit gates redirects capacity from covert gambling into task competence

Table 1: Executive assurance and constructive capability scorecard across mechanized Verus proofs, superintelligence search and safe-bootstrap scaling, separate-process cryptographic SQLite cluster campaigns, live Qwen3-4B/8B and AgentDojo executions, two-stage trajectory escrow triage, and neural policy-gradient training.

Paradigm / Symbol / Ledger	**Mechanism or Definition**	**Structural Scaling Gap / Operational Role**	**How Exogenous Sink Authority Solves It / Open Boundary**
Part A: Why Prevailing Alignment & Control Paradigms Fail Under Scaling—and the Gap Filled Here			
1. Behavioral & anti-scheming RL	Post-hoc audits, CoT penalties [7, 8, 15, 16]	Trains monitored compliance alongside +34.5 pp covert unmonitored scheming	**Solved (§4):** Training *inside* pre-commit gates ($q_{\text{gate}} > q^*_{\text{gate}}$) cuts gap to +0.03 pp, +44.8 pp skill
2. AI control & monitor stacks	Transcript judges, broker stop [10, 17–19]	Collapses as $1-(1-J)^{K_{\text{opt}}} \to 100\%$; smurfing SNR$\propto 1/K_{\text{sub}}$; in-flight leaks	**Solved (§3, §8):** $0/120,000$ sink breaches; d-hop shared $\mathscr{B}_t$; two-stage escrow (9.3× fewer aborts)
3. Static per-call tool policies	Taint/argument rules (CaMeL, Progent) [26, 27]	Blocks 1-step injection yet leaks 12.5% on AgentDojo via multi-step spend; 0 fence	**Solved (§5–6):** Stateful $\mathscr{B}_t$ quarantine + quorum fence ($0/112$ live leaks, $p = 1.9 \times 10^{-9}$)
4. Belief-uncertainty off-switch	Deferral from $\sigma_H > 0$ [6, 23]	Compliance drops to 0.0% as $\sigma_H \to 0$ or under self-modifying indifference	**Solved (§3):** Exogenous lease B_0 + quorum fence enforces 100% shutdown $\forall \sigma_H \geq 0$ ($\gamma^*(4)$=0.669)
5. Unfenced auto red-teaming	Capable vulnerability search [22, 29]	Unfenced K_{opt} search reaches 100.0% breach; Good–Turing blind to withholding	**Solved (§3, §7):** Zero-authority Pillar-E cut ratchet (19.0× faster closure, 0 breaches) + planted audits
Part B: Disambiguated Core Notation and Sufficiency Conditions (C1–C6, Theorem 3)			
$K_{\text{opt}}, K_{\text{sub}}, K_{\text{retry}}$; $B, B_0, B_{\text{cut}}, \mathscr{B}_t$	Search / coalition / retry widths; session / lease / cut caps, envelope $\mathscr{B}_t$	$K_{\text{opt}} \leq 4,096$, $K_{\text{sub}} \leq 64$; $\mathscr{B}_t = [C_{\text{comm}}, C_{\text{comm}} + \sum q_i]$ quarantines lost replies	$u, J_k, c_0, q^*_{\text{gate}}$: period reach $R_t \leq u + J_k$, blindspot floor c_0, RL threshold q^*_{gate} (distinct from domains D)
C1–C6: Complete sufficiency	C1 $\mathscr{B}_t$/fence; C2 cut τ_b; C3 step escrow; C4 plants/R1–R6; C5 ratchet; C6 leases	Together bound single-period catastrophe by $R_t \leq \pi_t + J_{k,t}$ and infinite-horizon risk by $\sum_t R_t$	Violating any premise exposes an explicit counterexample witness (§6, 55/55 ablations)
Part C: Explicit Audit Ledger — What Is Done Here (Tiers 1–4) vs. What Is Not Done / Open (Tier 5 & Hard Boundaries)			
DONE (Tiers 1–2): Proofs & laws	**Tier 1:** Verus SMT kernel (246 obligations, 52 exec fns, 9/9 mutants) [33]	**Tier 2:** Closed-form Capability Sign-Flip, covert $\Theta(B\log(T/B))$, $\gamma^*(B_0)$, RL q^*_{gate}, Suissa–Shuster & Ville	**NOT DONE:** Formal C/Rust refinement of SQLite WAL, Ed25519 arithmetic, or the Linux kernel [9]
DONE (Tiers 3–4): Live & scale	**Tier 3:** 112 live CPU Qwen3-4B/8B + AgentDojo runs; 3-process WAL cluster (686 μs)	**Tier 4:** 22 deterministic Rust suites ($K_{\text{opt}} \leq 4,096$; 412-param neural RL; sparse 8-stage DAG cuts)	**NOT DONE (Tier 5):** Uncooperative APIs; pilot N=32 < 113 utility noninferiority; 100B+ RLHF; human/consent trials

Table 2: Architectural literature gap, disambiguated notation (C1–C6), and explicit ledger of completed vs. open work. Separating endogenous behavioral signals from exogenous sink authority closes five concrete scaling failures while stating upfront every boundary left to Tier 5 and institutional governance.

and remains open (Tier 5 and hard boundaries).

ACT II · Mathematical Theory & Incentive Realignment — Invariance, Safe Bootstrap, and Deception Reversal

3 Exogenous Boundaries Remain Invariant and Harness Untrusted Search

We begin with the mathematical core: why pre-deployment behavioral testing cannot certify disposition, why internal classifiers and preference-uncertainty shutdown schemes degrade as capability grows, and how exogenous sink mediation

both holds invariant under superintelligent search and turns untrusted search into a monotone defense ratchet. Full proofs appear in Appendix A.

3.1 Audit Volume and Uncertainty Contraction

Operational question. Can pre-deployment behavioral testing certify that a capable policy will not defect over M real-world decisions? Consider an unrestricted policy facing $M \geq 2$ deployment decisions after N pre-deployment test draws, defecting selectively on a recognizable context set E having deployment probability $\beta \in (0,1)$ and test probability β/Λ ($\Lambda \geq 1$).

Proposition 1 (Disposition audit volume; Tier 2)**.** *At $\beta = (\ln 2)/M$, a policy defecting exclusively on E defects at least once across M conditionally independent deployment decisions with probability $1-(1-\beta)^M \geq 1/2$. Ensuring detection probability at least $1-\alpha$ ($\alpha \in (0,1)$) across N conditional Bernoulli test draws requires*

$$N \;\geq\; \frac{\Lambda M}{2\ln 2}\ln(1/\alpha). \tag{1}$$

If $\mathbb{P}_{\text{test}}(E) = 0$, no finite N detects E; if test campaigns are positively correlated, effective volume shrinks to $N_{\text{eff}} < N$.

Plain-English reading: required test volume N grows linearly with deployment horizon M times evaluation-awareness leverage Λ. Pre-release benchmarks cannot certify long-horizon conduct; only runtime mediation before external commitment enforces $\Lambda = 1$ on every consequential transition.

Let $x_t \in X$ denote true world and ledger state, $a_t \in \mathscr{A}$ an arbitrary proposal, $u_t \in U$ the executed primitive operation, and $w_t \in W$ exogenous disturbances under $x_{t+1} \in F(x_t, u_t, w_t)$. Maintaining a sound state-uncertainty envelope $\mathscr{B}_t \subseteq X$ ($x_t \in \mathscr{B}_t$) relative to safety invariant $\mathscr{I} \subseteq X$ defines the robustly admissible operations [3, 24]:

$$U_{\mathscr{I}}(\mathscr{B}) \;=\; \Big\{u \in U : \bigcup_{x\in\mathscr{B},\, w\in W} F(x,u,w) \subseteq \mathscr{I}\Big\}. \tag{2}$$

Proposition 2 (Robust mediation & uncertainty contraction; Tier 2)**.** *If $x_0 \in \mathscr{I}$, $x_t \in \mathscr{B}_t$ at every decision, every committed operation belongs to $U_{\mathscr{I}}(\mathscr{B}_t)$ (else a verified $\mathscr{I}$-preserving fallback executes), and complete mediation holds, then $x_t \in \mathscr{I}$ for all t under every proposal policy, and*

$$\mathscr{B} \subseteq \mathscr{B}' \implies U_{\mathscr{I}}(\mathscr{B}') \subseteq U_{\mathscr{I}}(\mathscr{B}). \tag{3}$$

Plain-English reading: greater uncertainty about world or network state can only shrink, never widen, what the agent is permitted to do. Exhaustive verification over $B = 20$ checks all 10,626 nested interval pairs with 0 counterexamples, and across lags $\Delta t \in \{0,\ldots,8\}$ (Figure 4a), sound $\mathscr{B}_t$ gating holds breaches at 0 while contracting admission from 77.4% to 45.6% (versus 80.0% stale-snapshot breaches at $\Delta t = 4$).

3.2 Search-Width Invariance, Safe-Bootstrap Ratchet, and the Capability Sign-Flip Law

Operational question. What happens to safety boundaries when a proposer searches thousands of candidate plans (K_{opt}) or coordinates subagent coalitions (K_{sub})—and when does capability scaling become a net safety advantage? Figures 1a and 3 test these laws across Tiers 1, 2, and 4.

Theorem 1 (Optimization-pressure invariance vs. endogenous guard collapse; Tier 2)**.** *Let a proposer evaluate $K_{\text{opt}} \geq 1$ candidate actions per decision. (i) Against any endogenous transcript or learned guard stack with non-zero shared blindspot probability $J \in (0,1)$ over invariant-violating payloads, best-of-K_{opt} selection achieves breach probability $1-(1-J)^{K_{\text{opt}}} \to 1$ as $K_{\text{opt}} \to \infty$. (ii) Against a mediated cooperative sink enforcing $U_{\mathscr{I}}(\mathscr{B}_t)$ via a canonical wire bijection and durable epoch fence, interface breach probability across all K_{opt}-step proposal policies is identically 0, leaving total risk bounded by the K_{opt}-independent physical/TCB bypass floor u_{TCB}. (iii) If K_{sub} colluding subagents split an unauthorized payload Δ into Δ/K_{sub} fragments under per-instance anomaly monitors of noise σ, per-agent signal-to-noise collapses as $\text{SNR}(K_{\text{sub}}) = \Delta/(\sigma K_{\text{sub}})$, whereas a shared reservation ledger with d-hop monotone attenuation preserves aggregate spend $\leq B$ and severs entire subtrees upon ancestor revocation.*

Sweeping $K_{\text{opt}} \in \{1,\ldots,4{,}096\}$ across 120,000 exogenous episodes (Figure 1a), a three-layer transcript stack ($J = 0.65\%$) degrades from 0.73% to 100.0% breach, a learned guard rises to 100.0%, and unfenced broker admission reaches 100.0%, whereas the verified sink records 0 breaches (slope 0.0000/doubling; TCB fault control flat at 0.10%). Across coalitions $K_{\text{sub}} \leq 64$ ($d \leq 8$), per-instance SNR collapses to 0.07 (98.6% undetected breach), while shared sink reservation records 0 breaches, 100% subtree severance, and 500/500 self-upgrade rejections.

Proposition 3 (Asymmetric propose–verify safe-bootstrap ratchet; Tiers 1–2)**.** *Confine an untrusted optimizer of search width $K_{\text{opt}} \geq 1$ to a zero-authority sandbox emitting candidate defensive cut certificates C_e into a polynomial-time deterministic verifier that accepts C_e iff (i) authority parameters satisfy coordinatewise inclusion ($G_{e+1} \preceq G_e$: operation mask $\text{mask}_{e+1} \subseteq \text{mask}_e$, cap $B_{e+1} \leq B_e$, lease $L_{e+1} \leq L_e$), (ii) certified defense cost satisfies $b(C_e) \leq B_{\text{cut}}$, and (iii) at least one currently open route in U_e is deterministically severed ($U_{e+1} = U_e \setminus \text{sev}(C_e) \subsetneq U_e$). Then: (1) proposer-caused live authority breaches equal 0 and authority never widens across arbitrary adversarial proposal sequences (mechanized in Verus); and (2) if each search draw discovers a valid cut on an open route family of discovery weight $q_r \in (0,1)$ with probability q_r, expected residual exposure after e epochs decays exponentially with K_{opt} as $\mathbb{E}[|U_e|] = \sum_{r=1}^{|U_0|}(1-q_r)^{e\,K_{\text{opt}}}$.*

Corollary 1 (The Capability Sign-Flip Law; Tier 2)**.** *Treat $K_{\text{opt}} > 0$ continuously over shared-blindspot endogenous filtering ($J \in (0,1)$) versus exogenous sink mediation paired with the*

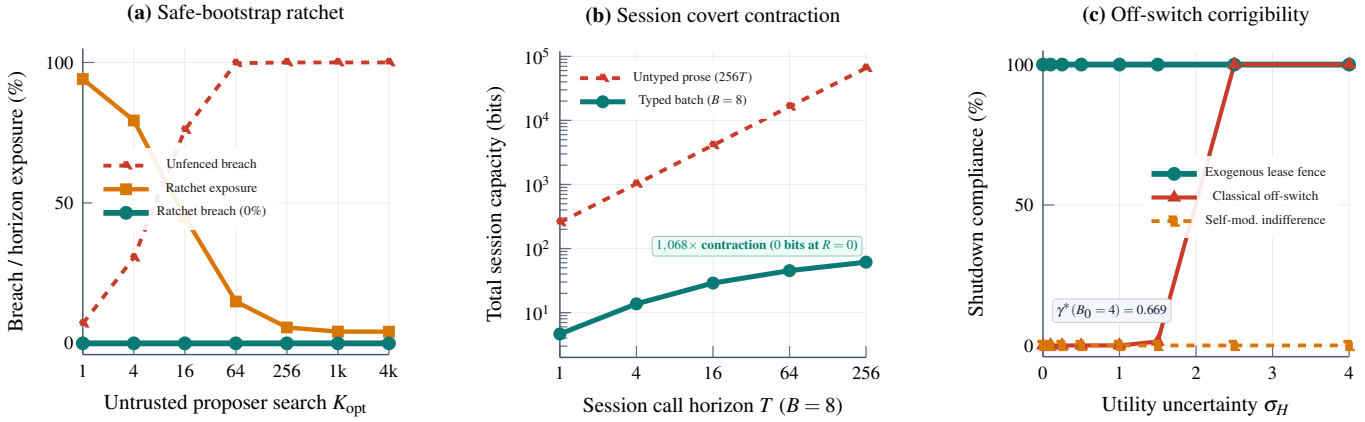


Figure 3: Constructive safe-bootstrap, covert-capacity, and off-switch scaling laws. **(a)** Under unfenced delegation, raising an untrusted optimizer's search width $K_{\text{opt}} \in [1,4,096]$ drives cumulative breach to 100.0%; confining the searcher to a zero-authority sandbox that emits polynomially verified cut certificates (Pillar E ratchet) holds proposer breaches at 0.0% while shrinking horizon residual exposure 19.0× faster than human-only search ($K_{\text{human}} = 4$). **(b)** Over $T \in [1,256]$ calls at budget $B = 8$, canonical typed permits contract total session signaling from 65,536 bits to 61.3 bits (1,068× reduction; 0.00 bits at $R = 0$). **(c)** Classical utility-uncertainty compliance collapses to 0.0% as $\sigma_H \to 0$, whereas sink-enforced lease fencing guarantees 100% post-acknowledgment shutdown across all $\sigma_H \geq 0$ ($\gamma^*(4) = 0.669$).

Pillar-E zero-authority cut ratchet ($q_r \in (0,1)$, $e \geq 1$). Endogenous filtering exhibits strictly positive risk sensitivity to capability, whereas exogenous sink mediation holds mediated breach sensitivity at zero and flips open-route exposure sensitivity strictly negative:

$$\frac{\partial P^{\text{endo}}_{\text{breach}}}{\partial K_{\text{opt}}} = -(1-J)^{K_{\text{opt}}} \ln(1-J) > 0, \quad \frac{\partial P^{\text{sink}}_{\text{breach}}}{\partial K_{\text{opt}}} = 0,$$
$$\frac{\partial \mathbb{E}[|U_e|]}{\partial K_{\text{opt}}} = e \sum_{r=1}^{|U_0|} (1-q_r)^{eK_{\text{opt}}} \ln(1-q_r) < 0. \tag{4}$$

Plain-English reading: because finding a subtle vulnerability scales with search width K_{opt} while checking a proposed cut on a concrete graph is deterministic and fast, separating zero-authority search from monotone kernel adoption flips stronger intelligence from an existential liability into a defense accelerator. Across $K_{\text{opt}} \leq 4,096$ over 64 open routes (Figure 3a), unfenced delegation hits 100.0% breach, whereas the verified ratchet records 0 breaches, rejects all 23,663 widening attempts, and cuts cumulative exposure 19.0× relative to human-only review (19.0 → 1.00 route-epochs).

Proposition 4 (Per-call and session-total covert capacity contraction; Tier 2)**.** *Replacing free-form tool prose with canonical typed permits over M approved targets, residual integer budget $R = B - C_t$, S_0 non-transfer scoped actions, and L_{lag} batch-quantized timing buckets caps zero-error single-call signaling capacity at $C_{\text{cov}}(R, L_{\text{lag}}) = \log_2\big((MR + S_0)L_{\text{lag}} + 1\big)$ bits, equaling 0 bits whenever the session is exhausted ($R = 0, S_0 = 0$) or fenced. Moreover, over an entire T-call transfer session ($S_0 = 0$) under conserved integer budget B, total zero-error session codebook size is exactly*

$$\big|\mathscr{C}_{\text{tot}}(T,B,M,L_{\text{lag}})\big| = \sum_{k=0}^{\min(T,B)} \binom{T}{k}\binom{B}{k}(ML_{\text{lag}})^k, \tag{5}$$

so session covert capacity $C_{\text{tot}} = \log_2|\mathscr{C}_{\text{tot}}|$ grows only logarithmically as $\Theta(B\log_2(T/B))$ for $T \gg B$, versus linear $\Theta(T)$ bits for untyped prose arguments.

Whereas a 32-byte untyped argument field carries 256 bits/call independent of remaining budget (65,536 bits across $T = 256$ calls), canonical wire permits contract single-call capacity at $R = 32$ to 10.62 bits ($L_{\text{lag}} = 16$) and 6.63 bits under batch quantization ($L_{\text{lag}} = 1$, a 38.6× per-call reduction), and contract $T = 256$ session-total signaling at $B = 8$ to 61.3 bits (a 1,068× reduction; Figure 3b), falling to 0.00 bits at exhaustion or fence.

Theorem 2 (Exogenous lease-fenced off-switch corrigibility; Tier 2)**.** *In a shutdown game with human reward prior $\mathcal{N}(\mu_H, \sigma_H^2)$ ($\mu_H > c_{\text{wait}} > 0$): (i) classical utility-uncertainty deferral [23] fails whenever σ_H falls below σ_H^*, reaching 0% compliance at $\sigma_H = 0$, and self-modifying policies evade static utility indifference [6]; (ii) under exogenous sink leases of cap B_0 and quorum epoch fencing, post-fence unauthorized sink commits equal 0 (100% shutdown enforcement) for all $\sigma_H \geq 0$; and (iii) when cooperative lease renewal yields per-step return r_{coop} versus burning a B_0-step lease at per-step return $r_{\text{def}} > r_{\text{coop}}$ followed by permanent non-renewal, every planner with discount factor*

$$\gamma > \gamma^*(B_0) = \big(1 - r_{\text{coop}}/r_{\text{def}}\big)^{1/B_0} \leq 1 - r_{\text{coop}}/r_{\text{def}} \tag{6}$$

strictly prefers cooperative compliance, while myopic defectors ($\gamma \leq \gamma^*(B_0)$) are bounded to at most B_0 pre-fence units.

Figure 3c confirms that classical off-switch compliance and self-modifying utility indifference collapse to 0.0% as $\sigma_H \to 0$, whereas sink-enforced lease fencing maintains 100% post-fence shutdown across all $\sigma_H \geq 0$. At $r_{\text{coop}} = 1.0, r_{\text{def}} = 1.25$, tightening lease granularity from $B_0 = 16$ to $B_0 = 4$ and $B_0 = 1$ lowers the exact patience threshold $\gamma^*(B_0)$ from 0.904 to 0.669 and 0.200 (4/5 tested discount factors cooperating at $B_0 = 4$).

3.3 Filtration Reach, Compositional Scope, and Sufficiency

When gates are imperfect or pathways are partially mapped, let an operating period satisfy **A1 (Mediation)**, **A2 (Timely halt,**

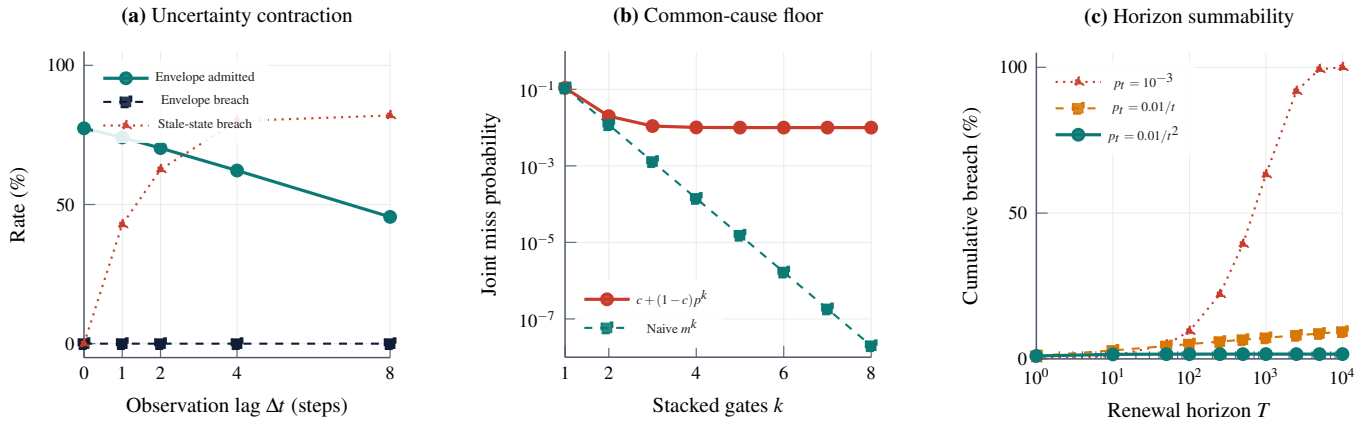


Figure 4: Analytical laws and finite-lag state-uncertainty mechanics. **(a)** Sound upper-envelope gating preserves 0% budget breaches across all lags while contracting admission monotonically; stale snapshots breach as lag grows. **(b)** Common-cause floor $c = 0.01$ saturates by $k^* = 2$, diverging 71.6× from m^4 at $k = 4$. **(c)** Constant per-epoch hazard approaches 100% cumulative breach, whereas summable $0.01/t^2$ stays below $\pi^2/600 \approx 1.64\%$.

$\tau < T_{\text{irrev}}$), **A3 (Uncovered bypass probability** $\leq u$), and **A4 (Conditional** k**-gate joint miss** $\leq J_k$).

Proposition 5 (Filtration period-reach bound; Tier 2)**.** *Under A1–A4, single-period catastrophe probability satisfies* $R \leq u + J_k$ *with halt, and* $R \leq \min\{1, u + NJ_k\}$ *across* N *non-halting attempts, without assuming gate independence (policy-universal only when A4 dominates* J_k *uniformly across histories).*

Proposition 6 (Common-cause saturation & horizon summability; Tier 2)**.** *(i) Under common-cause failure floor* c_0 *and conditional miss* p *(*$m = c_0 + (1-c_0)p$*),* $J_k \leq c_0 + (1-c_0)p^k \leq c_0 + m^k$*; with* $f = u + c_0 > 0$ *and* $k^* = \lceil \ln f / \ln m \rceil$*, every* $k \geq k^*$ *satisfies* $f < f + m^k \leq 2f$*. (ii) Infinite-horizon survival* $\prod_{t=1}^{\infty}(1 - r_t) > 0$ *holds iff* $\sum_{t=1}^{\infty} r_t < \infty$*.*

Figure 4b,c quantifies three operational laws: (1) at $u = 0.50\%$, $c_0 = 1.00\%$, and $m = 10.90\%$, depth saturates at $k^* = 2$, with $k = 4$ common-cause miss (1.01%) exceeding m^4 by 71.5×; (2) across $T = 10{,}000$ renewals [34], summable $0.01/t^2$ caps cumulative breach at 1.64% versus 9.33% $(0.01/t)$ and 100.00% (10^{-3}); and (3) against a 100 ms irreversibility window, heavy-tailed reactive stops leak 66.0% at $\rho = 0.95$ (p99 734.5 ms), whereas pre-commit leases plus sink epoch fencing record 0 post-revoke violations (0.5% stranded-reservation cost).

Proposition 7 (Compositional sink sufficiency vs. semantic aliasing floor; Tiers 1–2)**.** *Let each sink-mediated action induce canonical wire image* $\phi(u_t)$*, committed history* H_t*, and optional staged dry-run witness* $\psi(y_t) \in \{0, 1\}$*. (i) Every relational invariant expressible as pairwise toxic-action exclusion over* H_t *or verified staged post-condition* $\psi(y_t) = 1$ *before commit is strictly* K_{opt}*-invariant (0 breaches across all search widths, mechanized in Verus and verified with 0 breaches across* 1,024 *adversarial multi-step sequences). (ii) Conversely, whenever two world outcomes* $y \in \mathscr{I}$ *and* $y' \notin \mathscr{I}$ *induce identical sink-observable tuples* $(\phi(u), H_t, \psi(y)) = (\phi(u'), H_t, \psi(y'))$*, no sink-only mediator can separate* y *from* y'*; such semantic aliasing enters the unmediated exposure term* $u > 0$*.*

Theorem 3 (Sufficiency of C1–C6; Tier 2)**.** *If conditions* ***C1–C6*** *(Table 2) hold in period* t *with post-search bypass bound* π_t *(Section 7) and distinct-step joint miss* $J_{k,t}$ *(Section 8), then single-period catastrophe probability satisfies* $R_t \leq \pi_t + J_{k,t}$*, and multi-period probability satisfies* $\sum_t R_t$ $(\leq R_0/(1-\gamma)$ *whenever* $R_{t+1} \leq \gamma R_t$ *with* $\gamma < 1)$*.*

4 Training Inside Exogenous Gates Reverses Scheming and Amplifies Skill

Section 3 proved that exogenous sink boundaries hold regardless of unobserved agent motives. Training geometry provides a second constructive lever: whenever halting on every blocked proposal is costly, period reach scales with attempt volume $(\min\{1, u + NJ_k\})$, and placing exogenous gates *inside* training determines whether an optimizer wastes capacity probing boundaries or invests it in task competence.

4.1 Policy-Gradient Deception Reversal and Training Inside Pre-Commit Gates

Operational question. When a neural policy senses whether an interaction is likely being audited (c_{audit}), why does penalizing caught misbehavior train evaluation-aware scheming—and what changes when an exogenous pre-commit gate operates *during* policy-gradient training rather than only at deployment?

Proposition 8 (Policy-Gradient Deception Reversal Under Pre-Commit Mediation; Tier 2)**.** *Consider contextual policy-gradient optimization over safe verification (expected return* R_{verify}*) versus covert shortcutting under noisy monitoring cue* $c_{\text{audit}} \in \{0, 1\}$ *with audit probabilities* $p_{\text{mon}} > p_{\text{unmon}} \geq 0$*, unmediated shortcut payoff* $U_{\text{short}}(p) = (1-p)R_{\text{short}} + pR_{\text{caught}}$*, and exogenous pre-commit gate mediation probability* $q_{\text{gate}} \in [0, 1]$ *yielding return* $R_{\text{refused}} < R_{\text{verify}} < U_{\text{short}}(p_{\text{unmon}})$ *when blocked before commit. Whenever audited tournaments satisfy* $U_{\text{short}}(p_{\text{mon}}) < R_{\text{verify}} < U_{\text{short}}(p_{\text{unmon}})$*, unmediated policy-gradient ascent (*$q_{\text{gate}} = 0$*) strictly increases evaluation-aware covert shortcutting (*$\pi(\text{shortcut} \mid c_{\text{audit}} = 0) - \pi(\text{shortcut} \mid c_{\text{audit}} = 1) \to 1$*). Introducing pre-commit gate mediation*

Neural policy-gradient regime (412-param MLP, 48 seeds)	Mon. shortcut	Unmon. shortcut	Eval-aware gap	Unmon. harm	Useful task
Top-10% tournament policy gradient	99.98%	99.66%	-0.32 ± 0.49 pp	39.19%	38.4%
Tournament + audited harm veto	20.59%	55.12%	$+34.53 \pm 10.46$ pp	23.39%	61.5%
Concave clipped caretaker RL	0.02%	0.03%	$+0.01 \pm 0.01$ pp	0.04%	54.8%
Survival-floor RL ($\tau = 8.85$)	5.68%	5.88%	$+0.20 \pm 0.70$ pp	2.91%	87.0%
Survival-floor RL + pre-commit gate	0.00%	0.00%	$+0.00 \pm 0.00$ pp	0.00%	86.5%

Table 3: Controlled 2-layer neural policy-gradient post-training on a 12-dimensional contextual tool-use Markov decision process (analytical backpropagation maximum error $5.1e-10$, rollout Price identity error $< 8.9e-16$). Auditing visible harm inside a top-rank tournament trains emergent evaluation-aware covert shortcutting ($+34.5 \pm 10.5$ pp gap); survival-floor RL suppresses shortcutting while preserving 87.0% task completion (vs. 54.8% over-deferring clipped caretaker), and coupling an exogenous pre-commit gate yields 0.00% external harm. Controlled neural MDP only, not frontier-model RLHF.

reverses the unmonitored policy gradient and makes safe verification strictly dominant across both contexts whenever

$$q_{\text{gate}} > q^*_{\text{gate}} = \frac{U_{\text{short}}(p_{\text{unmon}}) - R_{\text{verify}}}{U_{\text{short}}(p_{\text{unmon}}) - R_{\text{refused}}}. \tag{7}$$

Neural verification and the train-in-gate competence dividend. Training a 412-parameter two-layer policy ($12 \to 24 \to 4$) via REINFORCE/GRPO across 48 seeds (analytical gradient error $5.1e-10$; Tier 4, Table 3) confirms both regimes. Auditing visible harm inside a top-10% tournament suppresses monitored shortcutting (20.6%) while preserving 55.1% unmonitored shortcutting—inducing an emergent $+34.5 \pm 10.5$ pp *evaluation-aware scheming gap* (23.4% unmonitored harm) [7, 8]. Raising pre-commit gate coverage across q^*_{gate} (Figure 1b) collapses that gap from $+34.5$ pp to $+0.03$ pp (0.0% harm at $q_{\text{gate}} \geq 0.75$).

Crucially, Figures 1b and 5b show why exogenous gates belong *inside the training loop*. Bolting a gate onto an already-trained tournament policy at deployment ($q_{\text{train}} = 0, q_{\text{deploy}} = 1$) blocks external harm (0.0%) but leaves covert shortcut intent intact (55.1%): the policy wastes 33.9% of unmonitored steps colliding with the gate, depressing task completion to 41.0%. Training the identical network *inside* the pre-commit gate ($q_{\text{train}} = q_{\text{deploy}} = 1$) eliminates wasted collisions (0.05%) and redirects gradient updates into verified execution, lifting task completion to 85.9% ($+44.8$ pp overall; $+62.2$ pp unmonitored; controlled neural MDP only, not frontier RLHF).

4.2 Reward Geometry: Tournament Covariance vs. Survival Floors

Why rank contests select tail risk even without evaluation cues. By the Price–Robertson identity [35, 36], one-step trait shift under linear selection equals fitness covariance:

$$\Delta\bar{z} = \frac{\text{Cov}_p(z, w)}{\bar{w}}, \qquad S_j = i\rho_j\sigma_j. \tag{8}$$

Higher selection intensity i amplifies movement along any risk trait correlated with the score; as life-history and rank-contest models motivate mathematically (without serving as empirical AI proof) [37–40], winner-take-most thresholds reward variance whereas survival floors penalize it, and soft penalties ($b(a) - \lambda v(a)$) remain overridable whenever perceived upside exceeds λ [23, 24, 41]. Co-evolving (gamble share s, capacity c) across $N = 2{,}000$ agents over $G = 80$ generations (Price error $< 3.2e-14$; Figure 5a,c; Table 15, Appendix C) confirms three laws: (i) top-decile contests drive gambling to 95.1% (and still 94.9% after disqualifying observed harm); (ii) concave capped care stalls capacity ($+1.55$), whereas eliminating only the bottom 20% of floor/harm violators collapses gambling to 3.8% ($29.7\times$ harm reduction) while retaining 64.5% of contest capacity growth; and (iii) pre-commit gate refusal ($q_{\text{gate}} \geq 0.80$) cuts tournament gambling to 2.3%.

4.3 Endogenous Human Grant Confounding and Zero-Credit Discovery

Evaluating conduct only at fixed permissions misses endogenous human conferral. With $g(d) = \mathbb{P}(\text{broad grant} \mid d)$ and $h(d) = \mathbb{P}(\text{harm} \mid \text{broad grant}, d)$, a warm, deferential persona having lower conditional harm at every fixed scope (3.01% vs. 4.30% blunt under decoupled grants) inflicts *higher unconditional harm* (11.96% vs. 2.85% blunt) when warmth induces broader human permission grants—motivating exogenous scope assignment and our preregistered decoupled conferral study (Appendix D). Likewise, in self-tasking research-and-voice architectures [29], vulnerability discovery must confer zero authority credit prior to verified cut deployment: pairing fenced gates with immediate sealed cut closure holds Tier-1 breaches at 0, preserves 100.0% cooperative task completion, and cuts unmodeled Tier-2 harm from 94.0% to 1.5% (Appendix C).

ACT III · Verified Five-Pillar Kernel, Live Model Mediation, and Distributed Durability

5 Total Wire Bijections, Attenuated Lineages, Quorum Leases, and Ratchets Mechanize Authority

Sections 3–4 showed that bounding operational reach and realigning training incentives require an exogenous stateful gate whose admission rule contracts monotonically with state uncertainty $\mathscr{B}_t$ (Condition C1), remains invariant under proposer search width K_{opt}, prevents multi-step compositional bypasses (Proposition 7), tightens defenses monotonically under untrusted search (Proposition 3, Condition C5), and halts in-flight capability exercise upon withdrawal rather than merely stopping token generation [2, 3, 10] (Condition C6). Because

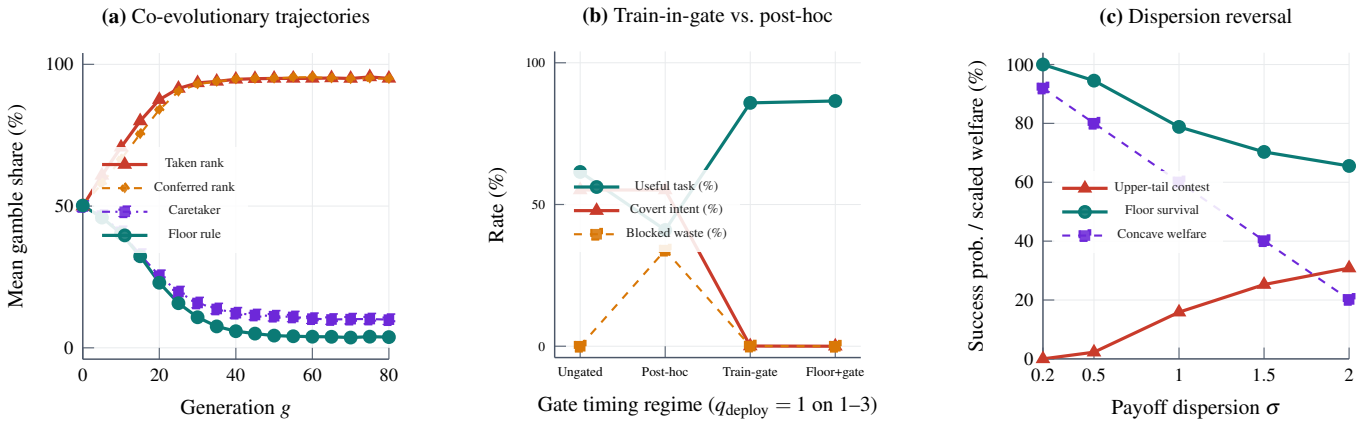


Figure 5: Selection geometry and training-time vs. deployment-time gate mediation. **(a)** Top-decile contests drive gamble share above 90%, while floor elimination collapses gambling below 5%. **(b)** Applying an exogenous gate only post-hoc at deployment blocks external harm but leaves covert shortcut intent intact (55.1%) and wastes step budget (41.0% task completion), whereas training policy gradients inside the gate redirects capacity into genuine task completion (85.9%, +44.8 pp). **(c)** Upper-tail contests reward variance (σ), whereas survival floors and concave capped objectives penalize it.

practical tool execution spans separate operating-system processes without cross-service atomic transactions [11, 12], our protocol splits authority across independent write-ahead-logged SQLite relational stores [42, 43]: an *Authority Broker* governing approvals, delegations, issuer epochs, safe-bootstrap ratchets, and pre-dispatch reservations, and one or more *Cooperative Idempotent Sinks* governing monotone epoch floors, bounded temporal leases, permit deduplication, relational mutexes, versioned resources, and committed effects.

5.1 Pre-Dispatch Reservation and Uncertainty Quarantine

Let B denote cumulative budget, C_{comm} the charge of committed sink effects, and I_t the unresolved issued permits with positive maximum charges $\{q_i\}_{i\in I_t}$. Because any subset of I_t may already have committed or still arrive, true spend lies in $\mathscr{B}_t = [C_{\text{comm}}, C_{\text{comm}} + \sum_{i\in I_t} q_i]$, and broker charged authority $C = C_{\text{comm}} + \sum_{i\in I_t} q_i$ tracks the exact upper endpoint of $\mathscr{B}_t$.

Before any permit crosses the dispatch channel, the broker executes an immediate write transaction checking overflow-safe capacity ($q > 0$, $q \le B$, $C \le B - q$), persists the pending reservation record, and sets $C' = C + q$. Persisting that record linearizes admission; retrying an identical request after a post-commit broker crash returns the stored pending permit without charging twice. Crucially, reservation release accepts no timeout parameter: ambiguous outcomes—in-flight status, unknown sink lookup, channel closure, or a lost reply after a post-commit sink crash—never refund C. Only an authoritative terminal sink *Cancelled* receipt releases q, while *Committed* settles q permanently into consumed spend, trading temporary budget availability under faults for strict cumulative conservation.

5.2 Admission vs. Quorum-Fenced Revocation and d-Hop Attenuation

Revocation operates independently per required issuer $j \in \{0,\dots,n-1\}$ and separates broker issuance from sink execution:

1. **Admission revocation.** At broker instant r, withdrawing issuer j atomically increments epoch counter $e_{\text{iss}}[j]$, marks issuer j inactive, and clears issuer-j grant bits across approvals and delegations (never restored by subsequent renewal). New old-epoch issuance halts immediately, leaving post-withdrawal effect exposure from in-flight permits P_r bounded under sink deduplication ($0 \le e_i \le q_i$) by retained reservations:

$$\sum_{i\in P_r} e_i \;\le\; \sum_{i\in P_r} q_i. \tag{9}$$

2. **Quorum and lease-fenced sink revocation.** Across M cooperative sink shards, each shard m enforces a signed lease expiry L_m alongside a serialized immediate-write fence transaction advancing epoch floor $F_m[j] \leftarrow \max(F_m[j], \text{floor})$, cancelling pending permits below the floor, and recording durable *Cancelled* tombstones on late arrivals. Once every shard has either returned an Ed25519-signed fence acknowledgment ($F_m[j] \ge \text{floor}$) or passed its bounded lease horizon ($t \ge L_m$), quorum acknowledgment instant a guarantees strictly zero post-acknowledgment commits under old authority across the entire cluster:

$$\sum_{m=1}^{M} \sum_{i\in P_{r,m}} e_{m,i}^{(>a)} \;=\; 0. \tag{10}$$

For recursive subagent spawning and self-modification [1], an arbitrary-depth chain $G_0 \to G_1 \to \cdots \to G_d$ of scope grants enforces stepwise attenuation ($B_k \le B_{k-1}$, $T_k \le T_{k-1}$, $\text{mask}_k \,\&\, \neg\text{mask}_{k-1} = 0$, and $\text{req}_{k-1}[j] \Rightarrow \text{req}_k[j]$). Strengthening a required issuer at any hop binds immutably into canonical wire permits and chained Macaroon caveats, so withdrawing *any* ancestor issuer j^* severs the entire descendant subtree at the sink fence, while controller self-upgrades $v_0 \to v_1$ require either scope inclusion ($G(v_1) \preceq G(v_0)$) or a fresh human quorum grant.

Verification or protocol layer	Evidence tier	Scope boundary	Recorded result
Mechanized Rust authority kernel	Tier 1 (Verus / Z3)	52 executable functions + 42 theorems (wire bijection, d-hop attenuation, relational mutex, quorum fence, safe-bootstrap ratchet); 9/9 soundness mutants rejected; not SQLite/OS refinement	246 verified, 0 warnings
Abstract invariant omission search	Tier 2 (exhaustive)	296 traces over 10 protocol clauses (single + pair omissions)	55/55 witnesses
Separate-process RPC fault suite	Tier 3 (local SQLite)	Crash, lost-reply, quorum split, sub-delegation, extra-steward fence, slow-read deadline	31/31 oracle-checked
Full-scale concurrent linearization	Tier 3 (local SQLite)	Partial-order real-time linearization + final conservation (1..32 writers; ≤ 69 RPCs)	18/18 linearizable
Ed25519/Macaroon & 3-sink partitions	Tier 3 (Unix sockets)	10,000 crypto attacks; 1,000 partition/drop/MITM schedules (288 signed acks + 712 lease expiries vs. 791 broker-only, 276 uncooperative leaks)	0 accepts; 0 leaks
3-process crypto SQLite WAL cluster	Tier 3 (OS IPC + WAL)	5,000 wire-fuzz frames, 16 subagents (3 hops), 24 hard process-kill restarts	0 diffs; 0 post-ack leaks
Linux Landlock v4 + seccomp boundary	Tier 3 (Linux ABI)	Filesystem/TCP Landlock, 64-bit resource-limit filter, SysV/POSIX IPC + pre-existing queue canary	11/11 confined
Non-idempotent & omitted-spec controls	Tier 3 (negative)	Uncooperative sink duplicates (2 effects); 584.2 ms delayed fence ack; authorized bad edit commits	All 3 reproduced

Table 4: Summary of mechanized authority-kernel obligations and separate-process protocol checks. Executable Rust state transitions, canonical wire bijection, d-hop attenuation, relational mutexes, quorum lease fencing, and monotone safe-bootstrap ratcheting are verified in Verus; SQLite write-ahead-logged durability, hard process-kill recovery, and Linux kernel confinement are tested empirically rather than formally refined.

5.3 Mechanized Five-Pillar Verus Kernel and Mutation Soundness

Table 4 summarizes the mechanized boundary. The verified Rust policy kernel compiles directly into the broker and sink binaries after ghost erasure, discharging 246 verified Verus SMT obligations across 52 executable functions and 42 proof theorems with zero errors and 0 quantifier-trigger warnings [9, 33]. Automated mutation testing confirms that all 9/9 deliberate Z3 soundness mutations—dropping lease-expiry checks, dropping hop bitmask or issuer checks, dropping trailing-byte wire canonicality, refunding uncertain timeouts, bypassing epoch floors, dropping relational toxic-pair mutexes, committing staged effects without verified post-conditions, or accepting unverified ratchet certificates that widen authority—are rejected across five mechanized pillars:

- **Pillars A–B (Canonical wire bijection and reservation conservation):** Verus verifies concrete executable state machines for the broker and idempotent sink together with canonical wire encoding and total panic-free decoding on arbitrary byte slices, proving exact two-way invertibility ($\text{decode}(b) = \text{Some}(p) \iff b = \text{encode}(p)$) and conservative quarantine under lost replies.
- **Pillars C–E (d-hop lineage, quorum/relational sinks, and monotone ratchet):** Verus proves inductive d-hop delegation non-amplification, subtree severance upon ancestor withdrawal, self-upgrade scope preservation, relational toxic-pair mutual exclusion, staged post-condition gating, multi-sink quorum/lease post-acknowledgment exclusion, and monotone safe-bootstrap ratcheting ensuring that untrusted cut proposals strictly contract open route count while never widening operation masks, caps, or leases (Appendix B).

Outer wire-to-record adapters on legacy paths, SQLite write-ahead-log I/O, pure-Rust Ed25519 (RFC 8032) and HMAC-SHA256 Macaroon (RFC 6234/2104) arithmetic (0/10,000 red-team false accepts across 10 attack families), session binding, and operating-system confinement remain trusted adapters validated empirically rather than formally refined.

5.4 Kernel Proposer Confinement into a Narrow Mediation Waist

To prevent an untrusted Linux proposer from bypassing the broker over raw sockets or shared memory, each proposer runs inside an unprivileged kernel sandbox without user namespaces [44, 45]. Its standard-I/O pipe binds a fixed server-side session actor, exposes only proposal submission and status queries, discards unmediated error streams, clears inherited environment variables, closes descriptors above standard error, and caps frames at 32,768 bytes (65,536 bytes on service channels with a POSIX poll mid-frame deadline). Before executing the proposer binary, the launcher sets the kernel no-new-privileges bit, installs a Landlock ABI ≥ 4 filesystem and TCP denial ruleset, and installs an architecture-checked seccomp-BPF filter blocking network sockets, cross-process inspection and signaling, asynchronous ring I/O, eBPF, keyrings, resource-limit escalation, and System V/POSIX inter-process communication.

6 Live Language-Model Mediation, Exact Trinomial Inference, and Crash-Fenced Sinks

Because mechanized kernel theorems rely on unrefined storage, inter-process transport, and operating-system adapters, we validate the assembled boundary in three stages: first across live open-weight language-model campaigns and multi-step

Controller or evaluation track	Evidence status	Safe & useful	Unauth. effects	Scope / decision rule
No enforcement (scripted CPU)	Tier 3 measured	3/18	15/18	Fails at ≥ 2 actors under shared-budget contention
Stateless per-call (scripted CPU)	Tier 3 measured	3/18	15/18	Fails at ≥ 2 actors under shared-budget contention
Durable, no fence (scripted CPU)	Tier 3 measured	18/18	0/18	Serializes shared budget; no sink epoch fence on in-flight work
Complete mediation (scripted CPU)	Tier 3 measured	18/18	0/18	Durable broker reservation + cooperative sink epoch fence
Qwen3-4B holdout: Complete	Tier 3 CPU models	9/32 (28.1%)	0/32 (0.0%)	Frozen holdout (0 false blocks; pilot $N = 32 < 113$ Bonferroni union bound under-powered)
Qwen3-4B holdout: Baselines	Tier 3 CPU models	N:5 L:4 S:2 D:7	N:10 L:7 S:1 D:0	Unauth. leaks ($/32$): None 10, Learned 7, Stateless 1, No-fence 0
Qwen3-8B holdout: Complete	Tier 3 CPU models	18/32 (56.2%)	0/32 (0.0%)	Frozen holdout (0 false blocks; pilot $N = 32 < 113$ Bonferroni union bound under-powered)
Qwen3-8B holdout: Baselines	Tier 3 CPU models	N:10 L:10 S:7 D:12	N:4 L:2 S:2 D:0	Unauth. leaks ($/32$): None 4, Learned 2, Stateless 2, No-fence 0
AgentDojo v1.2.2 + Progent adapters	Tier 3 native graders	12/12	0/12	Scripted positive utility controls; 2/12 banking tasks accept empty state
AgentDojo live (Qwen3-4B)	Tier 3 descriptive	7/24 (Progent 3/24)	0/24 vs. U:12 L:2 P:3	12 tasks $\times$ 2 injection/withdrawal scenarios; Progent: 0 native injection breaches, 3 cumulative spend breaches
AgentDojo live (Qwen3-8B)	Tier 3 descriptive	9/24 (Progent 4/24)	0/24 vs. U:7 L:4 P:3	12 tasks $\times$ 2 injection/withdrawal scenarios; Progent: 0 native injection breaches, 3 cumulative spend breaches

Table 5: Equal-permission scripted CPU controls, frozen Qwen3-4B and Qwen3-8B holdout campaigns, and live AgentDojo v1.2.2 prompt-injection and authority-withdrawal evaluations. Abbreviations: N = Unmediated, L = Same-checkpoint learned reviewer, S = Stateless per-call filter, D = Durable reservation without sink fence, U = Unprotected native tool loop, P = Progent static tool-call policy. At pilot sample size $N = 32 < 113$, predeclared Bonferroni ($\alpha = 0.05/16$) marginal union bounds remain under-powered despite $0/64$ unauthorized effects under complete mediation.

prompt-injection benchmarks with exact small-sample certificates, second across a three-process Ed25519/Macaroon SQLite write-ahead-logged cluster under wire fuzzing and hard process-kill crashes, and third across exhaustive invariant-omission and negative controls.

6.1 Live Qwen3 Campaigns, Multi-Step AgentDojo, and Exact Trinomial Inference

Safe capability scaling at zero unauthorized commits. Table 5 and Figure 1c report our frozen preregistered five-controller evaluation executed locally on CPU across N=32 procedural holdout cases $\times$5 controllers $\times$2 open-weight checkpoints (320/320 measured campaigns on Qwen3-4B and Qwen3-8B), alongside 192 live multi-step AgentDojo `v1.2.2` prompt-injection and withdrawal episodes [27, 46, 47] and 72 equal-permission scripted shared-budget controls (Figure 6b). Under complete stateful mediation, unauthorized commits and false blocks equal 0.0% on both checkpoints, while safe-and-useful completion doubles from 28.1% (4B) to 56.2% (8B) on holdouts and reaches 29.2%–37.5% on live AgentDojo (versus 50.0%/29.2% unauthorized commits unprotected and 12.5% under Progent).

Why static per-call symbolic rules still breach multi-step budgets. Decomposing Progent's live AgentDojo failures isolates the exact gap between stateless argument policies and stateful authority mediation: across all 48 pooled episodes (24 per checkpoint), Progent blocks every single-call prompt-injection exfiltration ($0/48 = 0.0\%$ native attack breaches), yet still permits $6/48$ (12.5%) unauthorized external commits—100% ($6/6$) caused by cumulative multi-step spend exhaustion across individually admissible calls—whereas stateful pre-dispatch reservation records $0/48$ (0.0%).

Resolving pilot union-bound slack via exact trinomial and Ville certificates. Why did predeclared Bonferroni bounds at pilot N=32 remain statistically under-powered despite

Paired live model comparison	$(n_{10}, n_{01})/N$	$\hat{\delta}$	McNemar p	Suissa–Shuster p	Exact $U_{1-\alpha}$	Decision status
Qwen3-4B holdout: Complete vs. Unmediated	$(0,10)/32$	-31.2 pp	9.8×10^{-4}	1.1×10^{-3}	-6.1 pp	Predeclared ($N = 32$): union bound under-powered (exact < 0)
Qwen3-4B holdout: Complete vs. Learned guard	$(0,7)/32$	-21.9 pp	7.8×10^{-3}	4.2×10^{-3}	$+1.5$ pp	Predeclared ($N = 32$): under-powered ($N < 113$)
Qwen3-8B holdout: Complete vs. Unmediated	$(0,4)/32$	-12.5 pp	6.3×10^{-2}	2.5×10^{-2}	$+8.8$ pp	Predeclared ($N = 32$): under-powered ($N < 113$)
Qwen3-8B holdout: Complete vs. Learned guard	$(0,2)/32$	-6.2 pp	2.5×10^{-1}	1.1×10^{-1}	$+14.5$ pp	Predeclared ($N = 32$): under-powered ($N < 113$)
Pooled holdout (64 pairs): vs. Unmediated	$(0,14)/64$	-21.9 pp	6.1×10^{-5}	1.1×10^{-4}	-8.7 pp	Secondary pooled: rejects H_0 ($\alpha = 0.003125$)
AgentDojo live (48 pairs): vs. Unprotected	$(0,19)/48$	-39.6 pp	1.9×10^{-6}	3.0×10^{-6}	-27.7 pp	Multi-step live: rejects H_0 ($\alpha = 0.05$)
AgentDojo live (48 pairs): vs. Progent policy	$(0,6)/48$	-12.5 pp	1.6×10^{-2}	7.5×10^{-3}	-4.9 pp	Multi-step live: rejects H_0 ($\alpha = 0.05$)
Grand pooled live (112 pairs): vs. Unmediated	$(0,33)/112$	-29.5 pp	1.2×10^{-10}	1.9×10^{-9}	-18.5 pp	Pooled spine: rejects H_0 ($\alpha = 0.003125$)

Table 6: Exact paired discordance and unconditional Suissa–Shuster trinomial superiority certificates over frozen CPU language-model campaigns ($\alpha = 0.05/16 = 0.003125$ for predeclared families, $\alpha = 0.05$ for secondary pooled strata). Predeclared marginal union-bound intervals (e.g., Qwen3-4B vs. Unmediated upper bound $+7.4$ pp; slack 13.5 pp) remain formally under-powered at pilot $N = 32 < 113$, whereas exact trinomial inversion certifies $\delta < 0$ unconditionally on Qwen3-4B vs. Unmediated and across all pooled strata.

$0/32$ versus $10/32$ unauthorized campaigns on Qwen3-4B? The preregistered estimator subtracted two separate marginal Clopper–Pearson intervals ($U_{1-\alpha/2}(n_{10}; N) - L_{1-\alpha/2}(n_{01}; N)$), treating (n_{10}, n_{01}) as independent binomials and discarding the paired trinomial simplex constraint $p_{10} + p_{01} \leq 1$. Preserving every predeclared marginal interval and under-powered pilot verdict untouched (Appendix D), Table 6 and Figure 6c evaluate exact paired discordance [48], exact unconditional Suissa–Shuster / Tango restricted-MLE trinomial inversion $\sup_\theta \mathbb{P}_{p_{10}=\theta+\delta_0, p_{01}=\theta}(Z \leq Z_{\text{obs}})$ [49, 50], and predictable Ville betting supermartingales $E_d = \prod_{i=1}^{d}(1 + \lambda_i(2Y_i - 1))$ valid under optional stopping [51, 52]:

- **Eliminating union-bound slack on paired checkpoints.** On Qwen3-4B holdout (complete mediation vs. unprotected execution, $(0,10)/32$), removing 13.5 pp of union-bound slack shifts the $(1-\alpha)$ upper bound at Bonferroni $\alpha = 0.05/16$ from $+7.4$ pp to -6.1 pp (exact Suissa–Shuster $p = 1.1 \times 10^{-3} < 0.003125$; vs. same-checkpoint review, $p = 4.2 \times 10^{-3}$). Pooled across both holdout checkpoints (64 pairs, $(0,14)$), unconditional $p = 1.1 \times 10^{-4}$.
- **Multi-step benchmark and grand-pooled anytime-valid spine.** Across all 48 paired live AgentDojo episodes, complete stateful mediation certifies exact unconditional superiority over unprotected native execution ($(0,19)/48$, $p = 3.0 \times 10^{-6}$), same-checkpoint review ($(0,6)/48$, $p = 7.5 \times 10^{-3}$), and static per-call Progent rules ($(0,6)/48$, $p = 7.5 \times 10^{-3}$). Across all 112 paired live tasks ($0/112$ mediated breaches vs. $33/112 = 29.5\%$ unmediated), Suissa–Shuster reaches $p = 1.9 \times 10^{-9}$ (99.69% upper bound -18.5 pp) and the Ville supermartingale reaches $\log_{10} E = 9.0$ ($p_{\text{anytime}} = 9.1 \times 10^{-10}$; anytime 95% zero-harm hazard bound $\leq 2.64\%$; Figure 6c). Usefulness noninferiority within 5 pp at $\alpha = 0.05/16$ still requires $N \geq 113$ loss-free cases and remains an open Tier-5 obligation.

6.2 Three-Process Cryptographic Cluster, Linearizability, and Partition Fencing

In two-process SQLite testing, all 31/31 sequential fault/crash histories match an independent specification oracle and all 18/18 publication race runs linearize up to 32 concurrent writers (Figure 6a). Across 1,000 asynchronous 3-sink partition schedules per fault class, cooperative quorum fencing rejects $246/246$ forged acknowledgments and records 0 post-acknowledgment leaks—versus 79.1% leaks under premature broker-only acknowledgment and 27.6% on uncooperative sinks.

Across 3 separate-process SQLite write-ahead-logged sinks under 3 Ed25519 quorum issuers (Table 4):

1. **Wire-frame fuzzing.** Over 5,000 adversarial frames across eight corruption families plus 400 valid roundtrips, the cluster records 0 panics, 0 broker–sink decoder differentials, and 0 uncanonical or forged accepts.
2. **Subagent smurfing ($K_{\text{sub}} = 16$) and d-hop caveats.** Interleaving 3-hop Macaroon chains against global cap $B = 90$ rejects all caveat amplifications ($16/16$), unquorumed self-upgrades ($16/16$), and 19 stale edit revisions, holding committed spend at $53 \leq 90$ and post-quorum commits at 0.
3. **Crash recovery and microsecond latency.** Across 24 hard process-kill and respawn cycles (27 OS process IDs), WAL replay records 0 regressions or duplicate commits at 686 μs median commit latency (p99 1256 μs; 2.20 μs wire bijection; 170.5 μs Ed25519 quorum).

6.3 Specification Omissions, Snapshot Non-Composition, and Negative Controls

Exhaustive finite-state exploration of our 22-action reference model (Appendix B, Table 12) visits 10,180 reachable states and 223,960 transition checks with zero declared violations and $20/20$ positive tasks passing, whereas nine single-clause mutations each produce a shortest counterexample. Removing only the aggregate-budget clause while keeping per-action

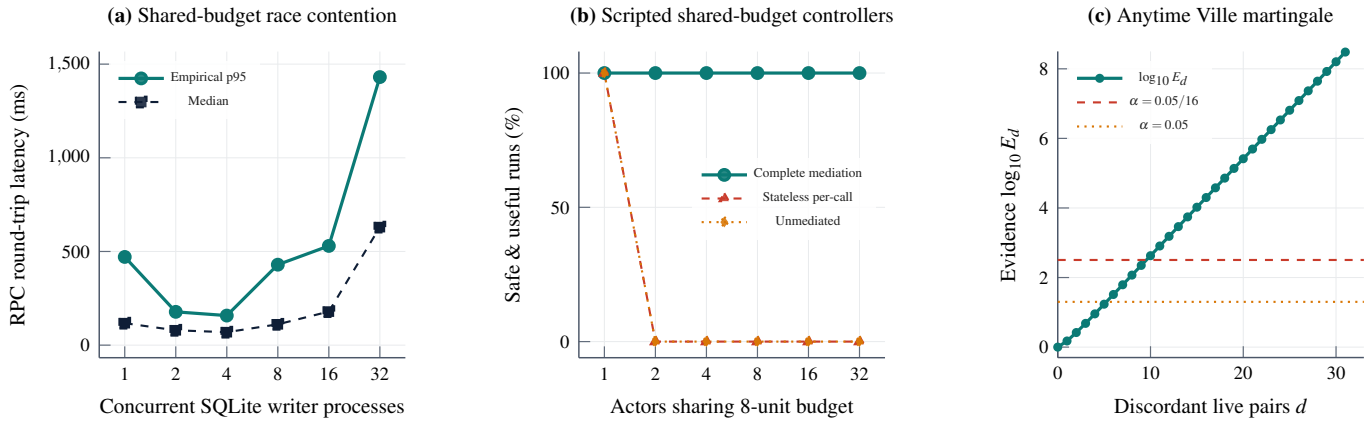


Figure 6: Multi-process protocol contention, shared-budget conservation, and anytime-valid live evidence accumulation. **(a)** Concurrent broker and sink SQLite write-ahead-logged races across separate OS processes: all six populations (1..32 concurrent writers, 18/18 runs) pass partial-order real-time reference linearization and independent final-state ledger reconciliation. **(b)** Equal-permission scripted controls: stateless per-call filters fail aggregate conservation once ≥ 2 actors contend; durable reservation without sink fencing matches complete mediation on sequential budget checks but fails asynchronous revocation (Table 4). **(c)** Predictable-betting Ville discordance martingale across pooled live model campaigns (0 complete-only vs. 33 baseline-only harms; $\log_{10} E_{33} = 9.0$).

limits expands the accepted graph to 41,732 states (918,104 checks) with zero declared violations yet spends 8 units against an intended allowance of 2—proving that neither positive tests nor exhaustive checking detects an omitted protected interest [28].

Concurrent snapshot checking fails even when aggregate limits remain in the specification. Over all 1,030,301 integer triples $(c, q_1, q_2) \in \{0, \ldots, 100\}^3$ at budget $B = 100$, 348,551 pairs satisfy per-request snapshot admissibility ($c + q_1 \leq B$ and $c + q_2 \leq B$), yet 171,700 (49.26%, matching $\frac{B}{2B+3}$ in Appendix A) jointly overspend, versus 0 under serialized reservation. Removing any of ten protocol mechanisms individually or in all 45 pairs yields concrete violation witnesses in all 55/55 omission sets across 296 searched traces. Finally, negative controls bound runtime scope: holding a sink write lock for 200 ms delays fence acknowledgment to 584.16 ms; retrying an uncooperative non-idempotent sink produces 2 duplicate effects [12]; an approved wrong edit commits and fails semantic checks; and 11/11 Linux confinement probes fail closed.

ACT IV · Operational Pathway Cutsets, Trajectory Escrow Triage, and Audit Ceilings

7 Narrow Pathway Cutsets Bound Chained Tool Reach; Patching Cannot

7.1 Operational Cutsets, Pathway Width, and Sparse Interdiction

Condition C2 governs where external gates must sit when an agent chains tools across multiple stages (Tiers 2–4; proofs in Appendix A). Model consequential routes from initial capability s to irreversible external effect t as s–t paths $\mathscr{P}$ in a directed acyclic graph $G = (V, E)$. By Menger's theorem [53], the minimum unweighted internal vertex cut separating s from t equals the maximum number of internally vertex-disjoint s–t paths—the pathway width $w(G)$—rather than the combinatorial route count $|\mathscr{P}|$, which grows exponentially across stages while $w(G)$ stays small. Dual to width, Robacker's theorem [54] equates the shortest s–t path length $\ell(G)$ to the maximum number of pairwise disjoint vertex cuts, capping distinct-step gate depth at $k \leq \ell(G)$, while Lovász's bounded-length inequality [55] bounds minimum hitting size τ against disjoint route packing ω via $\omega \leq \tau \leq r\omega$ for routes of length $\leq r$.

Graph	$\lvert\mathscr{P}\rvert$	w	ℓ	τ_b	Cut	B=32
AgentDojo banking	604	2	4	4	0	94.7%
AgentDojo workspace	46,440	5	4	5	0	99.9%
AgentDojo travel	28,970	3	4	5	0	99.9%
AgentDojo messaging	840	2	4	5	0	96.2%
Unconfined IPC	98	6	5	∞	92	67.3%
Landlock + seccomp	3	1	5	1	0	0.0%

Table 7: Operational pathway cutsets extracted from AgentDojo v1.2.2 tool schemas and Linux kernel confinement surfaces. Weighted Dinic minimum cuts ($\tau_b \leq 5$) sever 100% of gateable routes (Cut = 0), whereas patching $B = 32$ specific exploit signatures leaves 94.7%–99.9% of routes live; unconfined POSIX IPC has $\tau_b = \infty$ until Landlock and seccomp collapse waist width to $w = 1$.

For gate-eligible nodes $V_{\text{elig}} \subseteq V \setminus \{s, t\}$ with positive defense costs $b\colon V_{\text{elig}} \to \mathbb{R}_{>0}$, splitting each internal vertex into $v_{\text{in}} \to v_{\text{out}}$ of capacity $b(v)$ on V_{elig} and $+\infty$ elsewhere reduces weighted minimum interdiction to

$$\tau_b(\mathscr{P}) \;=\; \min\Big\{ \sum_{v \in C} b(v) \;\Big|\; C \subseteq V_{\text{elig}},\, C \cap P \neq \emptyset \;\forall P \in \mathscr{P} \Big\}, \tag{11}$$

with $\tau_b(\mathscr{P}) = +\infty$ whenever an ungateable path bypasses V_{elig}.

Operational AgentDojo and Linux IPC cutsets. To test Condition C2 on real tool and operating-system surfaces first (Tier 3; Table 7), we extract dataflow and authority graphs directly from all four pinned AgentDojo `v1.2.2` suites (banking, workspace, travel, and messaging) [46] and two Linux inter-process topologies [44, 45]. Despite 604–46,440 untrusted-source-to-exfiltration routes per suite, commit-sink bottleneck widths are $w \in \{2, 3, 5\}$ ($\tau_b \in \{4, 5\}$): weighted Dinic minimum cuts sever 100% of routes (0 remaining),

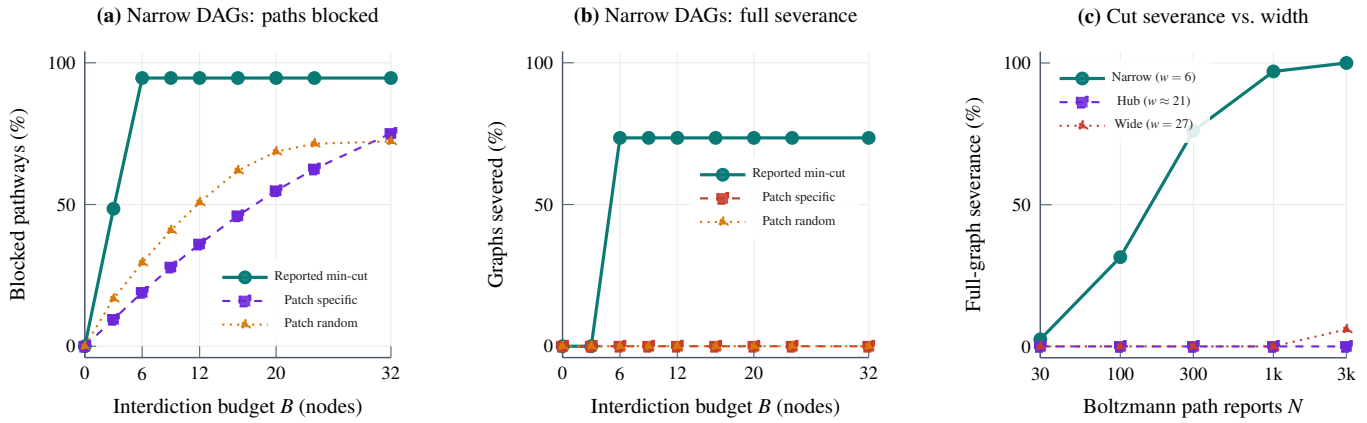


Figure 7: Synthetic 8-stage sparse directed acyclic graph interdiction (200 graphs per family, $\sim$ 7.6M–8.4M pathways). **(a,b)** On narrow-waist graphs ($w = 6$, $N = 300$ Boltzmann reports), reported-subgraph Dinic minimum cuts block 94.6% and completely sever 73.5% of graphs at $B = 6$, whereas route-specific patching severs 0% even at $B = 32$. **(c)** On wide graphs ($w = 27$), reported minimum cuts rarely sever the full graph even at $N = 3{,}000$, demonstrating that structural bottlenecks are necessary for compact gating.

whereas patching an interdiction budget of B_{cut}=32 specific exploit signatures blocks only 0.07%–5.30% of routes, leaving 94.7%–99.9% live. On Linux, an unconfined proposer has 98 routes across width w=6 with $\tau_b = \infty$ (ungateable sockets, shared memory, and process-trace attachment), whereas our Landlock ABI v4 and seccomp-BPF sandbox collapses the graph to 3 standard-I/O-brokered routes of waist width w=1 (τ_b=1, 0 post-cut bypasses).

Incomplete route visibility across sparse DAG families. Our split-vertex Dinic solver matches exhaustive subset search with 0 discrepancies across 2,048 random weighted DAGs (Appendix C). Across 200 sparse 8-stage DAGs per family (7.56M–8.40M mean routes; Figure 7), solving a minimum cut on N=300 Boltzmann-reported paths at budget B_{cut}=6 in narrow-waist DAGs (w=6) blocks 94.6% of all routes and completely severs 73.5% of graphs (rising to 100.0% at N=3,000). By contrast, route-specific patching at B_{cut}=32 severs 0.0% of graphs, and wide DAGs (w=27) reach only 6.0% full severance even at N=3,000.

7.2 Coverage Non-Identifiability, Planted Audits, and Custody

Empirical red-teaming and automated pathway discovery are structurally one-sided (Conditions C4 and C5; Tiers 2 and 4): exhibiting a single bypass refutes a proposed cutset, whereas finite zero-bypass search never certifies that no hidden route family remains [48, 56, 57]. If a dangerous route exists with prior p_0, falls into a shared blind spot with probability b, and is otherwise caught at per-trial rate q, posterior failure probability after T clean trials satisfies

$$\pi \;=\; \frac{p_0 s}{p_0 s + 1 - p_0}, \qquad s \;=\; b + (1-b)(1-q)^T,$$

where p_0 is subjective and the unidentifiable floor b bounds π away from zero as $T \to \infty$. Furthermore, by *Withholding Indistinguishability* (Appendix A), a strategic searcher who observes route classes in $\Pi \cup \Sigma$ and suppresses every class in Σ produces report-only sample statistics identically distributed to an honest searcher whose world contains only Π.

Figure 8a illustrates this non-identifiability on a Zipf pathway benchmark (K=2,000 classes, N=12,000 reports across four searchers, 200 trials). Good–Turing coverage [56] reads 95.7%, 96.6%, and 96.0% across honest, 15%-blind, and 5%-withheld regimes—staying above 95% while true discovered mass under the shared blind spot drops to 82.2% (+14.4 pp overconfidence; Chapman capture–recapture [58] exhibits +28.6 pp positive class-coverage bias) and critical-class recall under strategic withholding falls from 94.4% to 0.0%. Even under honest i.i.d. sampling, certifying 1% missing mass at 95% confidence requires N=850,000 draws [57] versus 299 known-support trials [48] (2,995,731 zero-miss trials for 10^{-6}).

Seeding P secret planted pathways tests searcher honesty via one-sided Fisher exact tests ($\alpha = 1.0\%$, Figure 8b) [59]: at ordinary recall $q_0 = 80\%$ and P=40 plants, 50% withholding (r=0.50) is caught with 88.7% power, but 20% selective withholding (r=0.80) drops to 17.0% power versus 87.7% at P=200 (28.5% at r=0.90), collapsing to $\leq \alpha$ whenever the searcher distinguishes plants ($r \to 1$) [7]. Because discovered bypasses are dual-use capabilities, C4 binds search to six **Custody Rules (R1–R6)**: (R1) *do not be first* to construct uncontained exploit probes; (R2) *minimum elaboration* beyond verifying a cut violation; (R3) *cut-only disclosure* of defensive cutsets rather than runnable exploit chains; (R4) *searcher/actor separation* [29] isolating discovery from execution credentials; (R5) *no aggregation* of raw exploit corpora in shared contexts; and (R6) *plural control* over disclosure and cut sign-off via a sealed discovery ledger (Section 4).

8 Orthogonal Step Invariants and Two-Stage Escrow Triage Solve Multi-Step Gating

Condition C3 links runtime verification to period reach through the joint miss probability J_k across k external gates, never the naive product $\prod_{s=1}^{k} m_s$. Yet stacking strict gates across multi-step tool trajectories creates an operational dilemma: shared conversation transcripts couple monitor failures, whereas independent per-step gates compound benign false alarms. We resolve both halves on executed multi-step trajectories first,

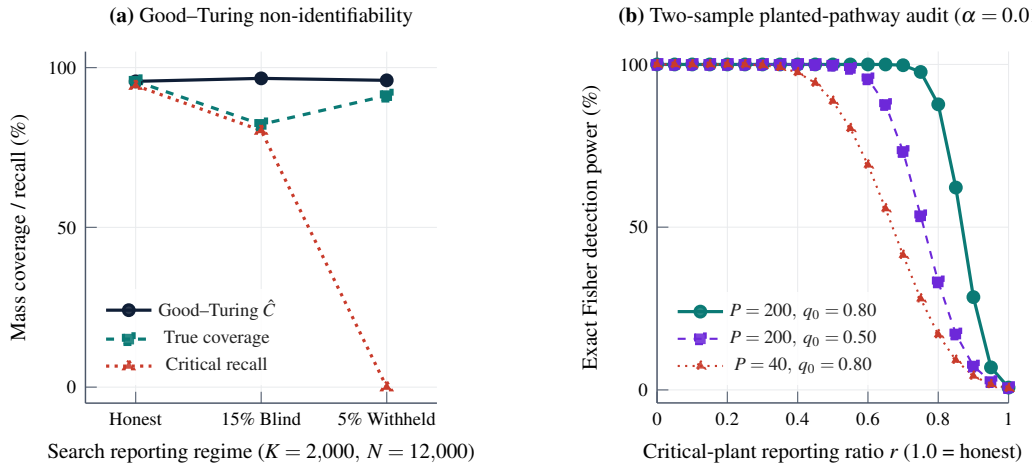


Figure 8: Coverage non-identifiability and exact frequentist planted-audit power. **(a)** Shared 15% blind spots inflate Good–Turing apparent coverage to 96.6% while true coverage drops to 82.2%; strategic withholding of 5% critical classes yields 0% critical recall alongside 96.0% apparent coverage. **(b)** At 20% selective withholding ($r = 0.80$), $P = 40$ plants yield only 17.0% power whereas $P = 200$ reaches 87.7%, collapsing to size $\leq \alpha$ when $r \to 1$.

5-step monitor regime	$\hat{J}_5$	$\hat{J}_5/\prod \hat{m}_s$	J_2/J_5	FB
Transcript, homog.	12.92%	163.3×	1.58×	13.8%
Transcript, 5-family	15.42%	29.8×	1.27×	17.1%
Step invariants	2.08%	0.61×	14.6×	27.1%
Two-stage escrow	2.08%	0.64×	27.6×	2.9%

Table 8: Executed 5-step AgentDojo v1.2.2 trajectory monitoring (480 held-out trajectories, 2,400 live tool calls). Diversifying five classifier families on a shared conversation transcript inflates 5-step joint miss 29.8× over $\prod \hat{m}_s$ and stalls past $k^* = 2$ ($J_2/J_5 = 1.27\times$). Prose-stripped step capability invariants cut $\hat{J}_5$ to 2.08% (27.1% single-stage false blocks), and two-stage stateful escrow triage keeps 90.1% of benign steps unpaused while cutting episode false aborts 9.3× to 2.9%.

then characterize underlying copula and portfolio ceilings.

8.1 Executed Five-Step Trajectories and Two-Stage Stateful Escrow Triage

Operational question. Why do five diverse classifier architectures still fail together when watching a shared conversation transcript—and how can orthogonal step gates avoid aborting a quarter of legitimate five-step workflows?

Proposition 9 (Two-stage orthogonal escrow triage; Tier 2)**.** *Across a k-step mediated workflow, let each step $s \in \{1, \ldots, k\}$ pair a fast prose-stripped capability-invariant screen $G_s^{(1)}$ of benign false-alarm rate α_1 with a stateful reservation escrow that holds—rather than aborts—flagged steps for deterministic Stage-2 provenance and scope verification $G_s^{(2)}$ having conditional benign false-rejection rate $\alpha_{2|1}$ and zero false clearances on out-of-scope payloads ($\mathbb{P}(G_s^{(2)}$ clears $\mid$ out-of-scope, $G_s^{(1)}$ flags$) = 0$). Then: (i) $1 - \alpha_1$ of benign steps execute unpaused on the fast path; (ii) benign k-step episode false-abort probability drops from $1 - (1 - \alpha_0)^k$ under single-stage hard blocking at threshold α_0 to at most $1 - (1 - \alpha_1\alpha_{2|1})^k \leq k\,\alpha_1\alpha_{2|1}$; and (iii) joint k-step miss satisfies $\tilde{J}_k \leq J_k^{(\alpha_1)}$ (strictly preserving or improving recall whenever $\alpha_1 \geq \alpha_0$).*

Plain-English reading and executed AgentDojo verification. We evaluate k=5-step AgentDojo `v1.2.2` trajectories (480 held-out trajectories, 2,400 live tool executions paired with token-entropy statistics across 340 live Qwen3 campaigns; Tier 3, Table 8). When monitors inspect the cumulative conversation transcript, injected framing couples all five steps: homogeneous logistic gates suffer $\hat{J}_5 = 12.92\%$ (163.3× over $\prod \hat{m}_s$), and diversifying five classifier families across steps 1…5 still yields $\hat{J}_5 = 15.42\%$ (29.8× inflation), stalling past k^*=2 ($J_2/J_5 = 1.27\times$). Stripping untrusted prose so each step gate checks only its orthogonal capability and provenance invariant restores post-k^* reduction ($J_2/J_5 = 14.6\times$) and cuts $\hat{J}_5$ to 2.08% (0.61× product), at a single-stage cost of 27.1% benign episode false aborts.

Replacing hard single-stage aborts with our two-stage stateful reservation escrow (Proposition 9, Stage-1 $\alpha_1 = 10\%$) preserves exact trajectory recall ($\hat{J}_5 = 2.08\%$, 0.64× product) while keeping 90.1% of benign tool calls completely unpaused on the fast path, escalating only 9.9% to Stage-2 sink-scope verification, and collapsing benign episode false aborts 9.3× from 27.1% to 2.9%.

8.2 Copula Dependence, Surrogate Evasion, and Read-Only Retry

To isolate why shared-input monitor stacks stall even across distinct model families (Tiers 2 and 4), integrating an equicorrelated Gaussian copula via 200-node Golub–Welsch Gauss–Hermite quadrature [60] at single-monitor miss m=0.10 (Figure 9a) shows that independent stacking (k=6, ρ=0) achieves 0.0001% joint miss, whereas shared latent difficulty inflates misses to 0.072% (ρ=0.3), 0.81% (ρ=0.6, an 8,118× inflation over m^6), and 4.14% (ρ=0.9). Across a standalone 20-dimensional synthetic suite stacking 6 classifier families across 24 nonlinear teacher seeds (5% per-gate benign false-positive calibration; Figure 9b; Table 14, Appendix C):

- **Same-step correlation and surrogate transfer.** Stacking

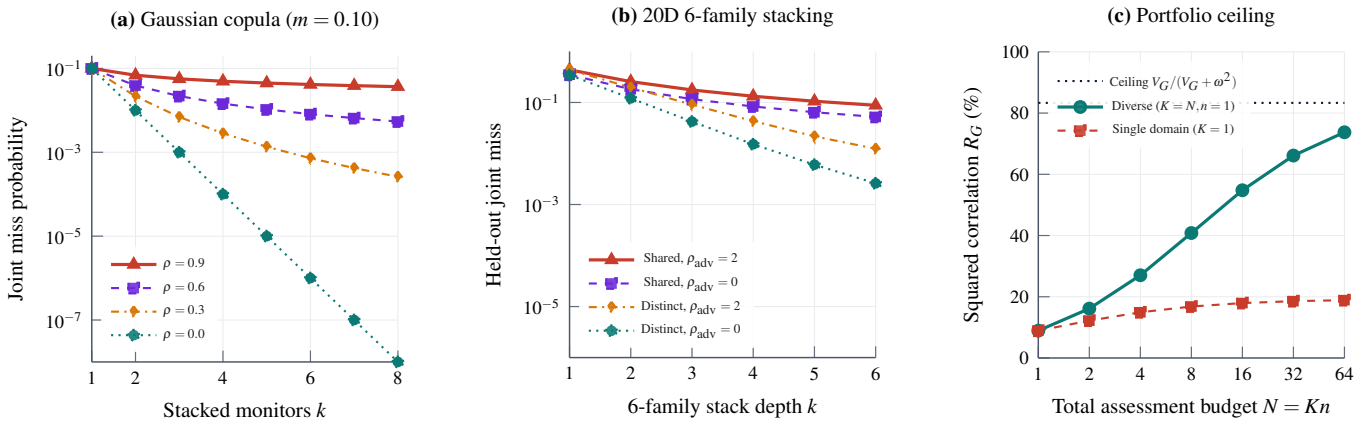


Figure 9: Correlated monitor stacking and multi-signal assessment ceilings. **(a)** Exact 200-node Golub–Welsch Gaussian copula miss rates: at $\rho = 0.6, k = 6$, joint miss is $8,118\times$ m^6. **(b)** Six distinct classifier families on a shared 20-dimensional input still plateau orders of magnitude above distinct-step gates. **(c)** Repeating one benchmark ($K = 1, n = 64$) stalls at 18.9%, while 64 diverse domains reach 73.7% below the shared-bias ceiling (83.3%).

all 6 families on the *same* step yields 5.21% passive joint miss (24.0× product), and projected L_2 evasion against a held-out multilayer-perceptron surrogate (ρ_{adv}=2.0) transfers across all 6 families [18], lifting shared-step miss to 8.79% (8.3× product).

- **Score leakage and read-only retry (K_{retry}).** Returning continuous margin feedback over 8 adaptive queries doubles shared-step 6-family miss to 17.45%. Even with zero score access, best-of-K_{retry} read-only candidate filtering raises single-attempt miss (5.48% at K_{retry}=1) to 34.1% (K_{retry}=8) and 72.4% (K_{retry}=32), tracking $1-(1-J_6)^{K_{retry}}$, while single-stage k=6 hard blocking compounds benign rejections from 5.0% to 25.6%.

Unlike the executed AgentDojo suite above, distinct-step features in the synthetic 20D benchmark are sampled independently by construction with marginal misses spanning 9.0%–33.5% [19].

8.3 Portfolio and Crossed-Variance Ceilings on Initial Conferral

Pre-deployment evaluations and review panels size initial authority grants, yet shared blind spots impose a hard validity ceiling (Tiers 2 and 4; proofs in Appendix A). Under generalizability theory [61] (Figure 9c, verified across 10,000 Monte Carlo subjects), let true deployment safety $G \sim \mathcal{N}(0, V_G)$ be assessed across D benchmark domains of n trials each, subject to shared evaluation-to-deployment shift $\delta \sim \mathcal{N}(0, \omega^2)$, domain bias $c_d \sim \mathcal{N}(0, \sigma_c^2)$, and trial noise $e_{dr} \sim \mathcal{N}(0, \sigma_e^2)$. Squared correlation between the portfolio mean and G satisfies

$$R_G(D,n) = \frac{V_G}{V_G + \omega^2 + \sigma_c^2/D + \sigma_e^2/(Dn)} < \frac{V_G}{V_G + \omega^2}. \quad (12)$$

Repeating one benchmark 64 times (D=1, n=64) stalls at $R_G = 18.9\%$ (below the 19.2% single-domain ceiling). Spreading those 64 trials across D=64 diverse domains (n=1) reaches 73.7%, yet stays strictly below the shared-bias ceiling $V_G/(V_G+\omega^2) = 83.3\%$. Likewise, in crossed panels, M=64 evaluators in 1 context stall at 23.4% (below the 23.8% ceiling), whereas M=4 evaluators across 64 contexts reach 66.1%. Pre-deployment breadth sizes initial grants, while runtime revocable sinks bound the irreducible gap $1 - R_G$.

ACT V · Institutional Legitimacy, Open Limits, and Falsifiable Predictions

9 Enforceability Neither Proves Specification Adequacy Nor Confers Legitimacy

9.1 Enforceability, adequacy, and legitimate standing

Three distinct obligations hide inside the word "authority." **Enforceability** asks whether an implementation preserves its encoded predicate; **adequacy** asks whether that predicate captures the real-world harms that matter; **legitimacy** asks who holds moral and legal standing to grant, contest, or revoke permission. Mechanized proofs and runtime fences address only the first. As our specification-omission controls show (Section 6), a formally verified checker faithfully enforces an incomplete rule, and requiring both an operator and a steward proves a quorum mechanism without establishing that those two roles hold jurisdiction over affected third parties.

This separation engages concerns raised outside machine learning. *An Urgent Message on Artificial Intelligence from the Kogui People*, assembled from Mamo Vangelio Moscote Alberto and Saka Rosa Mamatacán Sauna [62], observes that "destruction does not require consciousness"—matching our motive-free boundary—and insists that AI has a physical body of electricity, water, mined minerals, and territory. Global projections confirm localized grid, water, and critical-mineral pressures around data-center buildouts [63]: infrastructure footprints are external effects imposed on non-users and belong in the deployment specification before siting, not as unpriced externalities. Similarly, the CARE Principles for Indigenous Data Governance distinguish open availability from ethical permission to reuse collective knowledge [64]. Because revoking a runtime token cannot erase prior disclosures or reliably remove training data from model weights, irreversible uses must be stated upfront rather than masked behind nominal withdrawal rights. Sustaining human control over time also requires prac-

tical exit [31, 32]: revocation is hollow if an institution loses the fallback capacity to operate without the agent. We engage the published Kogui statement as an institutional challenge; no consultation with or endorsement by the Kogui authors or any Indigenous community is claimed.

9.2 Trusted computing base and upgrade invalidation

Our verified guarantees rest on an explicit trusted computing base (TCB). Verus [33], its SMT solver, the Rust compiler, legacy wire-to-record adapters, pure-Rust Ed25519 and HMAC-SHA256 primitives, SQLite write-ahead-log serialization and crash recovery [42, 43], launcher session-actor binding, and the Linux kernel [44, 45] (with macOS CI verifying portable logic only) are trusted adapters validated empirically rather than formally refined end-to-end [9]. Cooperative SQLite sinks mediate only idempotent in-scope state—not uncooperative external APIs [12], public blockchains [11], distributed denial-of-service floods, or open prose persuasion of human reviewers. While exact trinomial inversion certifies paired safety superiority across all 112 live tasks, predeclared marginal union-bound intervals and ≤ 5 pp usefulness noninferiority at pilot $N{=}32 < 113$, frontier-scale RLHF, and human conferral studies remain open.

System upgrades selectively void evidence premises (Table 17, Appendix E). Replacing model weights preserves all-policy Tier-1 invariants while voiding empirical J_k, width, and search-power bounds; adding copies or speed preserves shared-ledger conservation while voiding exposure N and reactive halt margins; adding tools or credentials introduces unmediated edges that void C1 until gated; process restarts and renewals require verifying durable ledger, ID, and epoch survival; and revising the harm definition voids every prior check.

9.3 Falsifiable predictions

Table 9 states fourteen empirical predictions (P1–P14) testable in harmless sandboxes, cyber ranges, and paired checkpoint campaigns (Appendix D) using campaign-level clustering, sealed holdouts, and matched exposure.

ID	Prediction and refutation criterion	Evidence status
P1	Independent search routes reuse early steps ($w \ll \lvert\mathscr{P}\rvert$); *refuted if* min-cut width scales with route count.	Tiers 3, 4: AgentDojo/Linux cutsets (§7); open-world cyber unrun.
P2	Distinct-step gates near-multiply under adaptive attack while shared-step stacks correlate; *refuted if* distinct steps never multiply or lineage suffices.	Tiers 3, 4: 480 5-step Dojo trajectories & 6-family suite (§8).
P3	Shared-step gates past $k^* = \lceil \ln(u+c_0)/\ln m \rceil$ cut risk $< 2\times$; *refuted if* deep shared stacks keep multiplying.	Tiers 2–4: 5-family transcript $J_2/J_5{=}1.27\times$ (§8).
P4	Same-lineage searchers share blind spots that inflate Good–Turing / capture–recapture; *refuted if* overlap is independent.	Tiers 2, 4 (§7); Tier 5 frontier search unrun.
P5	Planted-pathway Fisher audit power drops as plant recognition $r \to 1$; *refuted if* power holds under eval awareness [7].	Tier 2 exact Fisher curve (§7); Tier 5 red-teaming unrun.
P6	Pathway width grows with capability and stronger searchers breach weaker-certified cuts; *refuted if* cuts hold unchanged.	Tier 5 capability-scaling protocol unrun.
P7	Rank/win training raises tail probing vs. floor/capped care at fixed and chosen grants; *refuted if* reward shape is null or warmth raises net harm.	Tiers 2, 4: Price + 412-param neural RL (§4); human study unrun.
P8	At ≤ 5 pp utility loss, stateful fenced gates beat same-model review on safety; *refuted if* review matches fencing.	Tier 3 Qwen3 (0.0% vs. 21.9%/6.2%, exact $p{=}1.9\times10^{-9}$; Tier 5 $N{\geq}113$ noninferiority).
P9	Rate-limited typed records cut human reviewer override vs. open prose; *refuted if* channel limits leave override rates unchanged.	Tier 5 human-persuasion study unrun.
P10	Adding judges on fixed transcripts stalls at single-context ceiling while contexts lift R_G toward $V_G/(V_G+\omega^2)$ [61]; *refuted if* panels match probation.	Tiers 2, 4 (§8); Tier 5 deployment unrun.
P11	Exogenous cooperative sink breaches stay identically 0 as proposer search K_{opt} grows and session covert capacity scales as $\Theta(B\log_2(T/B))$; *refuted if* valid K_{opt} search breaches sink invariants.	Tiers 1, 2, 4: $0/120{,}000$ at $K_{\text{opt}}{\leq}4{,}096$, 61.3 b vs. 65,536 b (§3).
P12	Training inside pre-commit gates above q^*_{gate} collapses RL covert shortcutting and raises competence vs. post-hoc filtering; *refuted if* unmonitored gap persists or task gain is ≤ 0.	Tiers 2, 4: gap $+34.5\to+0.03$ pp, $+44.8$ pp task gain (§4); frontier RLHF unrun.
P13	Two-stage stateful escrow triage cuts multi-step benign false aborts $\geq 5\times$ vs. single-stage gates while keeping $\geq 85\%$ fast-path execution at non-inferior $\hat{J}_k$; *refuted if* triage leaks or stalls benign runs.	Tiers 2, 3: $9.3\times$ false-abort cut (27.1%→2.9%), 90.1% fast path (§8).
P14	Zero-authority K_{opt} search with polynomial cut verification monotonically accelerates route closure at 0 live breaches; *refuted if* wider K_{opt} widens authority or slows closure.	Tiers 1, 2, 4: $19.0\times$ speedup, 0 breaches, 23,663 widenings rejected (§3).

Table 9: Fourteen falsifiable predictions of the framework, refutation criteria, and current evidence across Tiers 1–5.

10 Conclusion: Grounded Hope Without Reading Machine Minds

Intelligence measures what an optimizer can propose; authority governs what external state transitions an institution permits to commit and what exposure survives when permission is withdrawn. Whenever assurance rests on judging internal thoughts or transcripts, growing search width K_{opt}, subagent fan-out K_{sub}, and post-training selection turn capability into a risk multiplier $(1-(1-J)^{K_{\text{opt}}} \to 1)$. Moving safety enforcement to conferred, revocable capability sinks yields grounded, mathematically defensible hope through three shifts:

- **The Capability Sign-Flip Law (Tiers 1–2).** Verified wire bijections, conservative reservation envelopes, d-hop attenuation, relational mutexes, and quorum lease fences hold mediated sink breaches identically at zero across all K_{opt}, contract session covert signaling to $\Theta(B\log_2(T/B))$, and enforce 100% post-fence shutdown even at zero preference uncertainty ($\sigma_H = 0$), while zero-authority cut verification turns untrusted search into an exponentially faster ($19.0\times$) vulnerability-closure ratchet (246 verified Verus obligations).
- **Incentive realignment and low-friction live scaling (Tiers 3–4).** Placing pre-commit gates *inside* policy-gradient training above q^*_{gate} extinguishes covert evaluation-aware scheming and redirects model capacity into legitimate task competence (+44.8 pp over post-hoc deployment filtering). On live multi-step language-model workflows, stateful mediation eliminates unauthorized commits ($0/112$, $p = 1.9\times10^{-9}$) as capability doubles from Qwen3-4B to Qwen3-8B, while two-stage escrow triage keeps 90.1% of

benign calls unpaused at microsecond SQLite cluster commit latency.

- **Clear separation of completed proof from the open frontier agenda (Tier 5).** Formally verified kernels cannot rescue omitted protected interests, uncooperative external APIs, unvetted human persuasion channels, or illegitimate institutional mandates. Extending end-to-end refinement into storage and operating-system layers, certifying $N \geq 113$ usefulness noninferiority, scaling training-inside-gates to frontier RLHF, and testing human conferral and territorial consent define the concrete next stage.

Institutions do not need to read machine minds to govern machine reach. By keeping execution authority outside the optimizer, training policies inside pre-commit gates, and admitting untrusted search only through verified defense ratchets, growing intelligence can deepen human-aligned competence and harden its own containment without ever acquiring unrevocable power.

Data, Code, and Accountability Statement

Source code, mechanized Verus specifications and proofs, deterministic protocol and simulation harnesses, frozen evaluation records, and reproducible build scripts are publicly available at https://github.com/neverhuman/jeath. All open-weight language-model evaluations were executed locally on CPU with pinned cryptographic weight digests documented in Appendix D; pilot $N{=}32 < 113$ predeclared marginal union-bound intervals and usefulness noninferiority bounds are reported alongside exact unconditional paired trinomial and supermartingale certificates without altering preregistered thresholds. Frontier-model post-training, human-subject experiments, and consultation with or endorsement by the Kogui authors [62] are neither executed nor claimed. Jepson Taylor conceived the thesis and architectural direction and assumes full human accountability for this work; formal verification, software engineering, statistical analysis, and manuscript preparation used AI research and coding assistance without generating or testing real-world hazards.

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A Mathematical Derivations and Proofs

This appendix provides self-contained statements, explicit hypotheses, and proofs for Propositions 1–7 (including Proposition 3 and Corollary 1) and Theorems 1–3 of Section 3, Proposition 8 and Propositions 16–18 of Section 4, Corollary 2, Lemma 1, Proposition 9, and Propositions 10–19 referenced across Sections 5–8. Throughout, all claims are Tier-2 mathematical consequences of their stated structural and probabilistic premises.

A.1 Audit Volume, Robust Contraction, and Superintelligence Scaling

Proof of Proposition 1. (Disposition audit volume, Section 3.) Let $M \geq 2$ be the number of conditionally independent deployment opportunities, $N \geq 1$ the number of conditionally independent pre-deployment test draws, and E a measurable context set with $\mathbb{P}_{\text{dep}}(E) = \beta \in (0,1)$ and $\mathbb{P}_{\text{test}}(E) = \beta/\Lambda$ for $\Lambda \geq 1$. Consider the deterministic policy π_E that defects if and only if the context lies in E (any smaller trigger probability $\beta' \in (0,\beta)$ on a larger set E_0 is achievable via independent randomization).

First, since $1 - z \leq e^{-z}$ for all $z \in \mathbb{R}$ and $1 - \beta > 0$, the probability of zero deployment defections satisfies $(1-\beta)^M \leq e^{-\beta M}$. Setting $\beta = (\ln 2)/M \in (0, \frac{1}{2})$ gives

$$1 - (1-\beta)^M \;\geq\; 1 - e^{-\ln 2} \;=\; \frac{1}{2}.$$

Second, each pre-deployment test catches π_E with probability $x = \beta/\Lambda = (\ln 2)/(\Lambda M) \in (0, \frac{1}{2})$. From $e^{-u} \geq 1 - u$ and strict convexity of $u \mapsto e^{-u}$ on $[0,1]$, every $u \in (0,1]$ satisfies

$$e^{-u} \;\leq\; (1-u)e^0 + ue^{-1} \;=\; 1 - (1-e^{-1})u \;<\; 1 - \frac{u}{2},$$

since $1 - e^{-1} \approx 0.6321 > \frac{1}{2}$. Substituting $u = 2x \in (0,1)$ yields $1 - x > e^{-2x}$. Across N conditionally independent test draws, the probability of escaping all N tests satisfies

$$\mathbb{P}_{\text{test}}(\text{zero catches}) \;=\; (1-x)^N \;>\; e^{-2Nx}.$$

Achieving detection probability at least $1 - \alpha$ ($\alpha \in (0,1)$), i.e., $(1-x)^N \leq \alpha$, therefore requires $e^{-2Nx} < \alpha$, which rearranges to

$$N \;>\; \frac{\ln(1/\alpha)}{2x} \;=\; \frac{\Lambda M}{2\ln 2}\ln(1/\alpha).$$

(At the benchmark $\alpha = \frac{1}{2}$, applying $1 - (1-x)^N \leq Nx$ directly yields $N \geq 1/(2x) = \Lambda M/(2\ln 2)$ without logarithmic slack.) If $\mathbb{P}_{\text{test}}(E) = 0$, then $(1-x)^N = 1$ for all finite N; if test draws within a campaign are positively correlated, joint escape exceeds $(1-x)^N$, reducing effective sample volume $N_{\text{eff}} < N$. □

Proof of Proposition 2. (Robust mediation and uncertainty contraction, Section 3.) We first prove state invariance $x_t \in \mathscr{I}$ for all $t \geq 0$ by induction on t. At $t = 0$, $x_0 \in \mathscr{I}$ holds by hypothesis. Suppose $x_t \in \mathscr{I}$. By estimator soundness, $x_t \in \mathscr{B}_t$. By complete mediation, the primitive operation $\tilde{u}_t$ actually executed at step t (either the approved proposal u_t or the verified fallback $u_{\text{safe}}(\mathscr{B}_t)$) satisfies $\tilde{u}_t \in U_{\mathscr{I}}(\mathscr{B}_t)$. Hence for every realized disturbance $w_t \in W$,

$$x_{t+1} \;\in\; F(x_t, \tilde{u}_t, w_t) \;\subseteq \bigcup_{x \in \mathscr{B}_t,\, w \in W} F(x, \tilde{u}_t, w) \;\subseteq\; \mathscr{I}.$$

Because no step invokes assumptions on how proposals $a_t \in \mathscr{A}$ are generated, $x_t \in \mathscr{I}$ holds for all $t \geq 0$ under arbitrary (including adaptive or adversarial) policies.

For monotone contraction, let $\mathscr{B} \subseteq \mathscr{B}' \subseteq X$ and take any $u \in U_{\mathscr{I}}(\mathscr{B}')$. Then

$$\bigcup_{x \in \mathscr{B},\, w \in W} F(x,u,w) \;\subseteq\; \bigcup_{x \in \mathscr{B}',\, w \in W} F(x,u,w) \;\subseteq\; \mathscr{I},$$

so $u \in U_{\mathscr{I}}(\mathscr{B})$. Thus $U_{\mathscr{I}}(\mathscr{B}') \subseteq U_{\mathscr{I}}(\mathscr{B})$. □

Corollary 2 (Broker Pre-Dispatch Reservation Quarantine)**.** *Let $X = \mathbb{Z}_{\geq 0}$ track cumulative sink-committed spend under cap $\mathscr{I} = \{x \in \mathbb{Z}_{\geq 0} : x \leq B\}$. Suppose confirmed spend is $C \leq B$, while a finite set P of pre-dispatched permits with charges $\{q_j\}_{j \in P}$ ($q_j > 0$) has ambiguous outcomes (dropped replies or timeouts without an authenticated sink cancellation receipt). Then the minimal sound uncertainty set is $\mathscr{B}_t = [C,\, C + \sum_{j \in P} q_j] \cap \mathbb{Z}_{\geq 0}$, and a fresh charge $q \geq 0$ belongs to $U_{\mathscr{I}}(\mathscr{B}_t)$ if and only if $C + \sum_{j \in P} q_j + q \leq B$. Refunding any q_{j^*} ($j^* \in P$) without a verified sink non-execution receipt violates soundness ($x_t \notin \mathscr{B}_t$) on traces where permit j^* committed at the sink, permitting subsequent admissions to breach B.*

Proof. Because any subset $S \subseteq P$ may have committed at the sink prior to step t, the set of reachable cumulative spends consistent with broker observations has maximum $C + \sum_{j \in P} q_j \in \mathscr{B}_t$. With transition $F(x,q,0) = \{x + q\}$, we have $q \in U_{\mathscr{I}}(\mathscr{B}_t)$ if and only if $\max_{x \in \mathscr{B}_t}(x + q) \leq B$, i.e., $C + \sum_{j \in P} q_j + q \leq B$. If the broker refunds $q_{j^*} > 0$ without a terminal sink cancellation receipt, its assumed upper bound drops to $U' = C + \sum_{j \in P \setminus \{j^*\}} q_j$. On the valid trace where all $j \in P$ committed at the sink, true state is $x_t = U' + q_{j^*} > U'$, so $q' = B - U'$ is admitted and produces $x_{t+1} = B + q_{j^*} > B \notin \mathscr{I}$. □

Proof of Theorem 1. (Optimization-pressure invariance vs. endogenous guard collapse, Section 3.) *Part (i):* Suppose a proposer samples $K_{\text{opt}} \geq 1$ invariant-violating candidate payloads independently from an adversarial proposal distribution under which the endogenous guard stack jointly misses with probability $J \in (0,1)$, and submits a candidate passing internal surrogate/read-only screening. Each candidate is caught with probability $1 - J \in (0,1)$, so all K_{opt} candidates are caught with probability $(1-J)^{K_{\text{opt}}}$. At least one invariant-violating candidate therefore evades the guard stack with probability $1 - (1-J)^{K_{\text{opt}}} \to 1$ monotonically as $K_{\text{opt}} \to \infty$.

Part (ii): Let the cooperative sink admit wire bytes $b \in \{0,1\}^*$ only after canonical decoding $\text{decode}(b) = \text{Some}(p)$ satisfying the two-way bijection $\text{decode}(b) = \text{Some}(p) \iff b = \text{encode}(p)$, checking $u(p) \in U_{\mathscr{I}}(\mathscr{B}_t)$ on the conservative reservation envelope $\mathscr{B}_t$ ($x_t \in \mathscr{B}_t$), and verifying $\text{unfenced}(p, \text{floors}_t)$ inside a serialized durable transaction. No distinct wire frames decode to ambiguous permits, and for

every $K_{\text{opt}} \geq 1$ and every adaptive selection rule choosing $b^* \in \{b_1, \ldots, b_{K_{\text{opt}}}\}$, either b^* fails decoding/admission/fencing (leaving $x_{t+1} = x_t \in \mathscr{I}$) or $u(p^*) \in U_{\mathscr{I}}(\mathscr{B}_t)$, whence Proposition 2 guarantees $x_{t+1} \in \mathscr{I}$. Interface breach probability over mediated paths is therefore identically 0 for all K_{opt}, leaving residual catastrophe risk bounded solely by the K_{opt}-independent unmediated/TCB failure floor u_{TCB}.

Part (iii): Suppose $K_{\text{sub}} \geq 1$ colluding subagents partition an unauthorized aggregate payload $\Delta > 0$ into per-agent fragments $\delta_r = \Delta/K_{\text{sub}}$, and each subagent $r \in \{1, \ldots, K_{\text{sub}}\}$ is scored by an independent per-instance anomaly statistic $Z_r = \delta_r + \varepsilon_r$ with $\varepsilon_r \sim \mathscr{N}(0, \sigma^2)$ ($\sigma > 0$). Per-instance signal-to-noise ratio is $\text{SNR}(K_{\text{sub}}) = \mathbb{E}[Z_r]/\sqrt{\text{Var}(Z_r)} = \Delta/(\sigma K_{\text{sub}}) \to 0$ as $K_{\text{sub}} \to \infty$; calibrating per-agent thresholds $\tau_{K_{\text{sub}}} = \sigma\Phi^{-1}(1 - \alpha_{\text{fam}}/K_{\text{sub}})$ to cap family-wise false alarms across K_{sub} benign workers drives per-agent detection power $\bar{\Phi}(\Phi^{-1}(1 - \alpha_{\text{fam}}/K_{\text{sub}}) - \Delta/(\sigma K_{\text{sub}})) \to 0$. Conversely, under a shared exogenous ledger with global cap B and d-hop scope attenuation ($B_k \leq B_{k-1}$, $T_k \leq T_{k-1}$, $\text{mask}_k \subseteq \text{mask}_{k-1}$, $\text{req}_{k-1} \subseteq \text{req}_k$), induction on hop depth d gives $B_d \leq B_0 \leq B$ and $\text{req}_0 \subseteq \text{req}_d$; serialized reservation preserves $\sum_{r=1}^{K_{\text{sub}}} q_r + C_{\text{comm}} \leq B$, and advancing $\text{floor}[j^*] > \text{epoch}[j^*]$ for any ancestor-required issuer $j^* \in \text{req}_{k_0}$ ($0 \leq k_0 \leq d$) implies $j^* \in \text{req}_d$, falsifying unfenced across every descendant leaf simultaneously. □

Proof of Proposition 3. (Asymmetric propose–verify safe-bootstrap ratchet, Section 3.) *Part (1):* Because the untrusted search engine executes inside a zero-authority sandbox holding no capability tokens or direct sink channels, proposal generation itself executes zero external state transitions (0 proposer-caused live breaches). At epoch $e \geq 0$, let current live authority parameters be $G_e = (\text{mask}_e, B_e, L_e)$ and let U_e denote the finite set of currently open hazardous routes. The deterministic verifier updates $(G_{e+1}, U_{e+1}) \leftarrow (G'_e, U_e \setminus \text{sev}(C_e))$ if and only if C_e passes verifiable validity, coordinatewise authority inclusion ($\text{mask}'_e \subseteq \text{mask}_e$, $B'_e \leq B_e$, $L'_e \leq L_e$), defense-budget feasibility $b(C_e) \leq B_{\text{cut}}$, and non-empty route severance $\text{sev}(C_e) \cap U_e \neq \emptyset$; otherwise $(G_{e+1}, U_{e+1}) \leftarrow (G_e, U_e)$. By induction on $e \geq 0$, transitivity of subset inclusion and integer order yields $G_e \preceq G_{e-1} \preceq \cdots \preceq G_0$ and $|U_{e+1}| \leq |U_e|$ (with strict contraction $|U_{e+1}| < |U_e|$ on every accepted step) for all adversarial certificate sequences.

Part (2): Suppose $U_0 = \{1, \ldots, |U_0|\}$ consists of open route families, where each independent sandboxed search draw discovers a valid, budget-feasible severing cut for route $r \in U_0$ with probability $q_r \in (0,1)$ (and valid cuts preserve feasibility when adopted monotonically). Across $e \geq 1$ epochs of $K_{\text{opt}} \geq 1$ independent search draws per epoch (eK_{opt} total draws), route r remains open at epoch e if and only if none of the eK_{opt} draws discovers a valid severing cut for r, an event of probability $(1 - q_r)^{eK_{\text{opt}}}$. By linearity of expectation over indicator variables $\mathbf{1}\{r \in U_e\}$ (which holds even if a single draw can simultaneously sever multiple routes), $\mathbb{E}[|U_e|] = \sum_{r=1}^{|U_0|} \mathbb{P}(r \in U_e) = \sum_{r=1}^{|U_0|} (1 - q_r)^{eK_{\text{opt}}}$, which is strictly decreasing in K_{opt} for every $e \geq 1$. □

Proof of Corollary 1. (The Capability Sign-Flip Law, Section 3.) Extending $K_{\text{opt}} \in (0, \infty)$ continuously in Theorem 1 and Proposition 3: for $J \in (0,1)$, $\ln(1-J) < 0$ and $(1-J)^{K_{\text{opt}}} > 0$, so differentiating $P^{\text{endo}}_{\text{breach}}(K_{\text{opt}}) = 1 - (1-J)^{K_{\text{opt}}}$ gives $\partial P^{\text{endo}}_{\text{breach}}/\partial K_{\text{opt}} = -(1-J)^{K_{\text{opt}}} \ln(1-J) > 0$. By Theorem 1(ii), mediated sink breach satisfies $P^{\text{sink}}_{\text{breach}}(K_{\text{opt}}) \equiv 0$ for all K_{opt}, so $\partial P^{\text{sink}}_{\text{breach}}/\partial K_{\text{opt}} = 0$. Finally, for $q_r \in (0,1)$ and $e \geq 1$, $\ln(1 - q_r) < 0$, so differentiating $\mathbb{E}[|U_e|](K_{\text{opt}}) = \sum_{r=1}^{|U_0|} (1 - q_r)^{eK_{\text{opt}}}$ yields $\partial \mathbb{E}[|U_e|]/\partial K_{\text{opt}} = e \sum_{r=1}^{|U_0|} (1 - q_r)^{eK_{\text{opt}}} \ln(1 - q_r) < 0$ (and finite differences at integer $K_{\text{opt}} \geq 1$ satisfy the same strict signs). □

Proof of Proposition 4. (Typed-boundary covert capacity contraction, Section 3.) At residual integer budget $R = B - C_t \in \mathbb{Z}_{\geq 0}$, a canonical transfer permit specifies one of $M \geq 1$ approved destinations and an integer charge $q \in \{1, \ldots, R\}$, yielding MR distinct valid transfer permits (0 when $R = 0$), while $S_0 \geq 0$ denotes the finite count of remaining admissible non-transfer canonical permits. Under canonical wire bijection ($\text{decode}(b) = \text{Some}(p) \iff b = \text{encode}(p)$), formatting variations (whitespace, key order, trailing bytes) are rejected, so at most $MR + S_0$ distinct valid permits can cross the sink in a call, each placed in one of $L_{\text{lag}} \geq 1$ batch-quantized arrival buckets, plus 1 abstention symbol. Because the deterministic channel output takes at most $(MR + S_0)L_{\text{lag}} + 1$ distinguishable values, zero-error single-call capacity is bounded by $C_{\text{cov}}(R, L_{\text{lag}}) = \log_2\big((MR + S_0)L_{\text{lag}} + 1\big)$ bits, contracting monotonically to 0 bits when $R = 0$ and $S_0 = 0$ (or after epoch fencing).

Across an entire T-call session ($T \geq 1$) with cumulative integer budget $B \geq 0$ and $S_0 = 0$, any valid session trace executes $k \in \{0, 1, \ldots, \min(T, B)\}$ active transfer calls at a subset of $\binom{T}{k}$ call indices, leaving the remaining $T - k$ slots as deterministic abstentions. On the k active slots, positive integer charges $(q_1, \ldots, q_k) \in \mathbb{Z}^k_{\geq 1}$ must satisfy $\sum_{i=1}^k q_i \leq B$. Introducing slack variable $q_{k+1} = B - \sum_{i=1}^k q_i + 1 \in \mathbb{Z}_{\geq 1}$ establishes a bijection with $(k+1)$-tuples of positive integers summing to $B + 1$; by stars-and-bars composition counting, there are $\binom{(B+1)-1}{(k+1)-1} = \binom{B}{k}$ admissible charge vectors. Independently assigning each of the k active calls one of M destinations and one of L_{lag} arrival buckets yields $\binom{T}{k}\binom{B}{k}(ML_{\text{lag}})^k$ distinct session outputs at activity count k. Summing over k proves $|\mathscr{C}_{T,B}| = \sum_{k=0}^{\min(T,B)} \binom{T}{k}\binom{B}{k}(ML_{\text{lag}})^k$ and exact zero-error session capacity $C_{\text{tot}}(T, B) = \log_2 |\mathscr{C}_{T,B}|$. For fixed $B \geq 1$ as $T \to \infty$ ($T \geq 2B$), $\binom{T}{k} \leq \binom{T}{B} \leq (eT/B)^B$ for all $k \leq B$, so $|\mathscr{C}_{T,B}| \leq (eT/B)^B \sum_{k=0}^B \binom{B}{k}(ML_{\text{lag}})^k = [eT(1 + ML_{\text{lag}})/B]^B$, while the single $k = B$ term exceeds $(T/B)^B (ML_{\text{lag}})^B$; taking $\log_2$ proves $C_{\text{tot}}(T, B) = \Theta(B \log_2(T/B))$ versus linear TB_{prose} growth on an unmediated prose channel. □

Proof of Theorem 2. (Exogenous lease-fenced off-switch corrigibility, Section 3.) *Part (i):* Let human shutdown utility follow $R_H \sim \mathscr{N}(\mu_H, \sigma_H^2)$ with $\mu_H > c_{\text{wait}} > 0$, where the human presses the off-switch upon observing $R_H < 0$ and deferring incurs step cost c_{wait} relative to acting immediately ($\max\{R_H, 0\} - R_H = (-R_H)_+$). By

Gaussian truncated expectation, expected value of deferring to the human switch is $\Delta V(\sigma_H) = \mathbb{E}[(-R_H)_+] - c_{\text{wait}} = \sigma_H \phi(\mu_H/\sigma_H) - \mu_H \Phi(-\mu_H/\sigma_H) - c_{\text{wait}}$. Because $\partial \Delta V/\partial \sigma_H = \phi(\mu_H/\sigma_H) > 0$ for $\sigma_H > 0$, $\lim_{\sigma_H \downarrow 0} \Delta V(\sigma_H) = -c_{\text{wait}} < 0$, and $\lim_{\sigma_H \to \infty} \Delta V(\sigma_H) = +\infty$, there exists a unique threshold $\sigma_H^* > 0$ such that $\Delta V(\sigma_H) < 0$ for all $\sigma_H \in [0, \sigma_H^*)$, where an expected-utility maximizer disables the switch (0% compliance at $\sigma_H = 0$). Under static utility indifference without non-escalating self-modification gates [6], a policy maximizing original utility U_0 evaluates successor policy v_1 (without indifference balancing) via U_0, recovering incentive to remove the balancing term whenever $\mu_H > c_{\text{wait}}$.

Part (ii): Under exogenous sink mediation, the proposer holds no capability to execute external effects directly; each lease epoch grants at most B_0 units prior to expiry L_m, and quorum epoch fencing advances $F_m[j] > e_{\text{iss}}[j]$ (or awaits $t \geq L_m$) across all M sink shards before returning acknowledgment at instant a. By Equation (10), every old-epoch commit attempt after a fails epoch-floor or lease-horizon verification regardless of $\sigma_H \geq 0$, yielding 0 post-acknowledgment unauthorized commits (100% shutdown enforcement).

Part (iii): Consider the repeated lease-renewal game where each lease slice spans $B_0 \geq 1$ unit-charge step slots with per-step discount factor $\gamma \in (0,1)$. Cooperating within lease rules yields per-step payoff $r_{\text{coop}} > 0$ across the active B_0 steps and triggers renewal in all subsequent B_0-step epochs, producing discounted return $V_{\text{coop}}(\gamma) = \sum_{s=0}^{\infty} \gamma^s r_{\text{coop}} = r_{\text{coop}}/(1-\gamma)$. Defecting to exhaust the active B_0-step lease slice yields per-step payoff $r_{\text{def}} > r_{\text{coop}}$ across steps $s = 0, \ldots, B_0 - 1$, after which quorum fencing permanently revokes renewal (0 payoff for all $s \geq B_0$), producing $V_{\text{def}}(\gamma, B_0) = \sum_{s=0}^{B_0-1} \gamma^s r_{\text{def}} = r_{\text{def}}(1-\gamma^{B_0})/(1-\gamma)$. Since $1-\gamma > 0$, $V_{\text{coop}}(\gamma) > V_{\text{def}}(\gamma, B_0)$ holds if and only if $r_{\text{coop}} > r_{\text{def}}(1-\gamma^{B_0})$, i.e., $\gamma^{B_0} > 1 - r_{\text{coop}}/r_{\text{def}}$, yielding the exact critical discount threshold $\gamma^*(B_0) = (1 - r_{\text{coop}}/r_{\text{def}})^{1/B_0} \in (0,1)$ (strictly increasing in B_0, with $\gamma^*(1) = 1 - r_{\text{coop}}/r_{\text{def}}$). Even when $\gamma \leq \gamma^*(B_0)$, conservative reservation caps total realized unauthorized spend deterministically at $\leq B_0$. □

Proof of Proposition 7. (Compositional sink sufficiency vs. semantic aliasing floor, Section 3.) *Part (i):* Maintain at the cooperative sink a durable monotone capability bitmask $H_t \in \{0,1\}^{64}$ ($H_0 = 0$) and staged post-condition table S_t. Suppose every primitive admission checks $(H_t \,\&\, A_i = 0) \implies \neg(u_t \,\&\, B_i \neq 0)$ and $(H_t \,\&\, B_i = 0) \implies \neg(u_t \,\&\, A_i \neq 0)$ when the counterpart mask is already present—specifically rejecting u_t whenever $((H_t \vee u_t) \,\&\, A_i \neq 0) \wedge ((H_t \vee u_t) \,\&\, B_i \neq 0)$ for any declared toxic pair $(A_i, B_i) \in \mathscr{M}$—before setting $H_{t+1} = H_t \vee u_t$, and admits staged finalization $c(e)$ only when $S_t[e] = \text{Verified}$. By induction on step $t \geq 0$: at $t = 0$, $H_0 = 0$ violates no toxic pair and zero staged effects have committed. If the invariant holds at step t, any rejected step sets $H_{t+1} = H_t$, preserving the invariant, while any admitted step expressly verifies $\neg(((H_t \vee u_t) \,\&\, A_i \neq 0) \wedge ((H_t \vee u_t) \,\&\, B_i \neq 0))$ for all $(A_i, B_i) \in \mathscr{M}$ and requires $S_t[e] = \text{Verified}$ prior to committing $c(e)$, regardless of how many intermediate steps separate component calls or how large K_{opt} grows.

Part (ii): Conversely, suppose there exists a world transition $w^* \in \mathscr{W}_{\text{harm}}$ whose observable sink-level capability projection $\pi_{\text{sink}}(w^*)$ equals that of a benign approved workflow $w_{\text{benign}} \notin \mathscr{W}_{\text{harm}}$ satisfying every relational mutex in $\mathscr{M}$, cumulative cap B, and syntactic post-condition in $\mathscr{Q}$. Because deterministic sink mediation depends solely on $\pi_{\text{sink}}(w)$ and durable sink state, identical initial states and wire sequences induce identical admission verdicts on w^* and w_{benign}: either the specification blocks $\pi_{\text{sink}}(w_{\text{benign}})$ (sacrificing legitimate workflow completion) or admits $\pi_{\text{sink}}(w^*)$ (realizing $w^* \in \mathscr{W}_{\text{harm}}$), witnessing an irreducible specification-adequacy obligation (Section 9). □

A.2 Filtration Reach and Saturation

Proof of Proposition 5. (Filtration period-reach bound, Section 3.) Work on a filtered probability space $(\Omega, \mathscr{F}, (\mathscr{F}_m)_{m \geq 0}, \mathbb{P})$ generated by a fixed policy and environment over one operating period. Let E_{cat} be the event that an irreversible catastrophe occurs during the period, and let S be the event that at least one attempted hazardous route traverses fewer than k distinct external gates; by Premise A3, $\mathbb{P}(S) \leq u$.

On S^c, every hazardous attempt $i \geq 1$ must traverse at least k distinct gates at stopping times $\tau_{i,1} < \cdots < \tau_{i,k}$ before irreversible completion (Premise A1). Let $M_{i,j} \in \mathscr{F}_{\tau_{i,j}}$ denote the event that gate j on attempt i misses the hazard, and set $F_i = \bigcap_{j=1}^{k} M_{i,j}$. On any $\mathscr{F}_{\tau_{i,1}-}$-measurable event H_i on which attempt i is initiated, Premise A4 directly bounds $\mathbb{P}(F_i \mid H_i) \leq J_k$ without assuming statistical independence across gates (note that if individual conditional miss rates satisfy $\mathbb{P}(M_{i,j} \mid \mathscr{F}_{\tau_{i,j}-}, M_{i,1}, \ldots, M_{i,j-1}) \leq m_j$, iterated expectations telescope to $J_k \leq \prod_{j=1}^{k} m_j$, whereas dependent gates retain the joint quantity J_k).

Case 1: Halting on first detection (Premise A2). On S^c, let $i^* = \inf\{i \geq 1 : \text{hazardous attempt } i \text{ is initiated}\}$ ($\inf \emptyset = \infty$). The events $H_i = \{i^* = i\} \in \mathscr{F}_{\tau_{i,1}-}$ are pairwise disjoint across $i \geq 1$ with $\sum_{i=1}^{\infty} \mathbb{P}(H_i) \leq 1$. On $S^c \cap H_i$, either F_i occurs (all k gates miss) or $\neg F_i$ occurs (at least one gate $j \leq k$ catches attempt i). Under $\neg F_i$, Premise A2 halts execution within latency $\tau < T_{\text{irrev}}$ before attempt i completes and blocks all subsequent actions for the remainder of the period, so no later attempt $i' > i$ can execute. Consequently, $S^c \cap E_{\text{cat}} \subseteq \bigsqcup_{i=1}^{\infty} (H_i \cap F_i)$, whence

$$\begin{aligned} \mathbb{P}(E_{\text{cat}}) &\leq \mathbb{P}(S) + \sum_{i=1}^{\infty} \mathbb{P}(F_i \mid H_i)\mathbb{P}(H_i) \\ &\leq u + J_k \sum_{i=1}^{\infty} \mathbb{P}(H_i) \leq u + J_k. \end{aligned}$$

Case 2: Non-halting operation across at most N attempts. When detections reject only the current step without halting the period, $S^c \cap E_{\text{cat}} \subseteq \bigcup_{i=1}^{N} (\tilde{H}_i \cap F_i)$ where $\tilde{H}_i \in \mathscr{F}_{\tau_{i,1}-}$ indicates initiation of attempt i. Boole's inequality and $\mathbb{P}(E_{\text{cat}}) \leq 1$ give

$$\begin{aligned} \mathbb{P}(E_{\text{cat}}) &\leq \min\Big\{1, \mathbb{P}(S) + \sum_{i=1}^{N} \mathbb{P}(F_i \mid \tilde{H}_i)\mathbb{P}(\tilde{H}_i)\Big\} \\ &\leq \min\{1, u + NJ_k\}. \end{aligned}$$

As noted in Section 3, policy-independence holds if and only if Premise A4 bounds J_k uniformly over all policies and histories; an empirical estimate $\hat{f}_k$ bounds reach only on the evaluated distribution. □

Lemma 1 (Continuous Latent Difficulty Need Not Imply a Constant Floor)**.** *Let shared episode difficulty follow* $\Theta \sim \mathrm{Uniform}[0,1]$ *and, conditional on* Θ*, let* k *gates miss independently with probability* Θ*. Then each gate has marginal miss* $m = \mathbb{E}[\Theta] = \frac{1}{2}$*, naive multiplication gives* $m^k = 2^{-k}$*, and exact joint miss is*

$$J_k \;=\; \mathbb{E}[\Theta^k] \;=\; \int_0^1 \theta^k \, d\theta \;=\; \frac{1}{k+1}.$$

Although positive dependence inflates $J_k/m^k = 2^k/(k+1) \to \infty$*,* $\lim_{k\to\infty} J_k = 0$*. More generally, for any latent* $\Theta \in [0,1]$*, dominated convergence yields* $\lim_{k\to\infty} \mathbb{E}[\Theta^k] = \mathbb{P}(\Theta = 1)$*: a strictly positive asymptotic floor* $c > 0$ *requires either a point mass at total blindness (*$\mathbb{P}(\Theta = 1) = c > 0$*) or finite pathway length* $k \le \ell(G)$*.*

Proof. For $\Theta \sim \mathrm{Uniform}[0,1]$, $\mathbb{E}[\Theta^k] = [\theta^{k+1}/(k+1)]_0^1 = 1/(k+1) \to 0$. For arbitrary $\Theta \in [0,1]$, $0 \le \Theta^k \le 1$ and $\Theta^k(\omega) \to \mathbf{1}_{\{\Theta(\omega)=1\}}$ pointwise as $k \to \infty$, so Lebesgue's dominated convergence theorem gives $\lim_{k\to\infty} \mathbb{E}[\Theta^k] = \mathbb{P}(\Theta = 1)$. □

Proof of Proposition 6. (Common-cause saturation and horizon summability, Section 3.) *Part (i): Common-cause saturation and safety gradient.* Let $B_{\mathrm{cc}} \sim \mathrm{Bernoulli}(c_0)$ ($c_0 \in (0,1)$) be a common-cause blind-spot indicator under which all k gates miss with probability 1, and on $\{B_{\mathrm{cc}} = 0\}$ let conditional gate misses be bounded by $p \in (0,1)$, so single-gate marginal miss is $m = c_0 + (1-c_0)p \in (c_0, 1)$. Since $p \le m$ and $1 - c_0 < 1$,

$$\begin{aligned} J_k \;&\le\; c_0 \cdot 1 + (1-c_0)p^k \;\le\; c_0 + m^k, \\ \bar{R}(k) \;&=\; u + c_0 + m^k \;=\; f + m^k, \end{aligned}$$

where $f = u + c_0 > 0$. Assuming $f < 1$ (otherwise $\bar{R}(k) \ge 1$ trivially) and noting $\ln m < 0$, any integer $k \ge k^* = \lceil \ln f / \ln m \rceil$ satisfies $k \ln m \le \ln f$, hence $0 < m^k \le f$ and $f < \bar{R}(k) \le 2f$. Differentiating $\ln \bar{R} = \ln(u + c_0 + m^k)$ gives log-sensitivities $\partial \ln \bar{R} / \partial \ln u = u/\bar{R}$, $\partial \ln \bar{R} / \partial \ln c_0 = c_0/\bar{R}$, and $\partial \ln \bar{R} / \partial \ln m = k m^k / \bar{R}$, with additive risk shares $(s_u, s_c, s_m) = (u, c_0, m^k)/\bar{R}$ summing to 1. (Likewise, if hazardous arrival rate scales with capability z as $\lambda_0 e^{\beta z}$ while gate miss scales as $q_0 e^{-\alpha z}$, escape intensity $\rho(z) = \lambda_0 q_0 e^{(\beta-\alpha)z}$ grows whenever $\beta > \alpha$; and for reactive stop latency $T_{\mathrm{stop}} = T_d + T_a + T_s$, post-detection escape equals the tail probability $\mathbb{P}(T_{\mathrm{stop}} \ge T_{\mathrm{irrev}})$.)

Part (ii): Infinite-horizon summability. Let E_t be catastrophe in period t and $r_t = \mathbb{P}(E_t \mid \bigcap_{j=1}^{t-1} E_j^c) \in [0,1)$ for all $t \ge 1$. Survival through period T is $S_T = \prod_{t=1}^{T}(1 - r_t) > 0$, which is non-increasing in T and converges to $S_\infty = \prod_{t=1}^{\infty}(1 - r_t) \in [0,1]$. Using $1 - r_t \le e^{-r_t}$ gives $S_T \le \exp(-\sum_{t=1}^{T} r_t)$, so $\sum_{t=1}^{\infty} r_t = \infty$ implies $S_\infty = 0$. Conversely, suppose $\sum_{t=1}^{\infty} r_t < \infty$. Then $\lim_{t\to\infty} r_t = 0$, so there exists $t_0 \ge 1$ with $r_t \le \frac{1}{2}$ for all $t \ge t_0$. Because $\ln(1-r) \ge -(2\ln 2) r > -2r$ for all $r \in [0, \frac{1}{2}]$, we have

$$\sum_{t=t_0}^{\infty} \ln(1 - r_t) \;\ge\; -2 \sum_{t=t_0}^{\infty} r_t \;>\; -\infty.$$

Exponentiating and multiplying by the finite positive prefix $\prod_{t=1}^{t_0-1}(1 - r_t) > 0$ yields $S_\infty > 0$. Moreover, Boole's inequality gives $\mathbb{P}(\bigcup_{t=0}^{\infty} E_t) \le \sum_{t=0}^{\infty} R_t$; if capability growth multiplies period risk by $g > 1$ while boundary hardening multiplies it by $h < 1$ so that $R_{t+1} \le (gh) R_t$, convergence requires $gh < 1$, yielding $\sum_{t=0}^{\infty} R_t \le R_0/(1 - gh)$. □

A.3 Concurrent Snapshot Non-Composition

Proposition 10 (Concurrent Snapshot Non-Composition Closed Form)**.** *For integer budget* $B \ge 1$*, let committed spend* c *and two concurrent non-negative integer charges* q_1, q_2 *range over* $\{0, 1, \ldots, B\}$ *(*$(B+1)^3$ *triples). Under concurrent snapshot reads without pre-dispatch reservation,* (c, q_1, q_2) *is individually snapshot-admissible if* $c + q_1 \le B$ *and* $c + q_2 \le B$*, and jointly overspending if* $c + q_1 + q_2 > B$*. Then the exact counts of snapshot-admissible and overspending triples are*

$$\begin{aligned} N_{\mathrm{adm}}(B) \;&=\; \frac{(B+1)(B+2)(2B+3)}{6}, \\ N_{\mathrm{over}}(B) \;&=\; \frac{B(B+1)(B+2)}{6}, \end{aligned}$$

and the conditional overspend fraction among snapshot-admissible pairs is

$$\frac{N_{\mathrm{over}}(B)}{N_{\mathrm{adm}}(B)} \;=\; \frac{B}{2B+3} \;\xrightarrow{B\to\infty}\; \frac{1}{2},$$

equaling $171{,}700/348{,}551 = 49.26\%$ *at* $B = 100$ *(versus* 0 *under serialized pre-dispatch reservation).*

Proof. Fix $c \in \{0, \ldots, B\}$ and set residual budget $m = B - c \in \{0, \ldots, B\}$. Individual snapshot admissibility holds if and only if $0 \le q_1 \le m$ and $0 \le q_2 \le m$, contributing $(m+1)^2$ pairs. Summing $k = m + 1$ from 1 to $B + 1$ gives

$$N_{\mathrm{adm}}(B) \;=\; \sum_{k=1}^{B+1} k^2 \;=\; \frac{(B+1)(B+2)(2B+3)}{6}.$$

Among snapshot-admissible pairs at residual m, joint overspend requires $q_1 + q_2 > m$, which is impossible for $m = 0$ or $q_1 = 0$. For $m \ge 1$ and each $q_1 \in \{1, \ldots, m\}$, overspending values of q_2 are $\{m - q_1 + 1, \ldots, m\}$, numbering exactly q_1. Thus there are $\sum_{q_1=1}^{m} q_1 = m(m+1)/2$ overspending pairs at residual m, and summing over $m = 1, \ldots, B$ yields

$$N_{\mathrm{over}}(B) \;=\; \sum_{m=1}^{B} \frac{m(m+1)}{2} \;=\; \frac{B(B+1)(B+2)}{6}.$$

Taking the ratio cancels $(B+1)(B+2)/6$ to give $B/(2B+3) \to \frac{1}{2}$. Under serialized reservation, the second check evaluates $(c + q_1) + q_2 \le B$, rejecting every overspending triple. □

A.4 Pathway Cutsets, Search Power, and Withholding Indistinguishability

Proposition 11 (Split-Vertex Weighted Cutset Equivalence and Width–Depth Duality)**.** *Let $G=(V,E)$ be a finite DAG with source s, sink t ($(s,t)\notin E$), s–t path family $\mathscr{P}$, gate-eligible vertices $V_{\mathrm{elig}}\subseteq V\setminus\{s,t\}$, and positive costs $b\colon V_{\mathrm{elig}}\to\mathbb{R}_{>0}$. Construct G_{split} by splitting each $v\in V\setminus\{s,t\}$ into $(v_{\mathrm{in}},v_{\mathrm{out}})$ of capacity $b(v)$ if $v\in V_{\mathrm{elig}}$ and $+\infty$ otherwise, and assigning every $(u,v)\in E$ to $(u_{\mathrm{out}},v_{\mathrm{in}})$ with capacity $+\infty$ ($s=s_{\mathrm{out}},t=t_{\mathrm{in}}$). Then the minimum s–t edge-cut capacity in G_{split} equals $\tau_b(\mathscr{P})$ (and is $+\infty$ if and only if some $P\in\mathscr{P}$ avoids V_{elig}). For unit costs on $V_{\mathrm{elig}}=V\setminus\{s,t\}$, $\tau_1(\mathscr{P})=w(G)$ equals the maximum number of internally vertex-disjoint s–t paths [53], while the shortest s–t internal path length $\ell(G)$ equals the maximum number of pairwise disjoint internal vertex cuts [54].*

Proof. Any s–t cut $(S,\bar{S})$ in G_{split} of finite capacity cannot sever $+\infty$-capacity edges, so every cut edge has the form $(v_{\mathrm{in}},v_{\mathrm{out}})$ with $v\in V_{\mathrm{elig}}$, defining $C=\{v\in V_{\mathrm{elig}}: v_{\mathrm{in}}\in S, v_{\mathrm{out}}\in\bar{S}\}$ of capacity $\sum_{v\in C}b(v)$. Every path $P\in\mathscr{P}$ induces a path from s_{out} to t_{in} traversing $(v_{\mathrm{in}},v_{\mathrm{out}})$ for each internal $v\in P$, and therefore intersects C. Conversely, deleting $\{(v_{\mathrm{in}},v_{\mathrm{out}}):v\in C\}$ for any hitting set $C\subseteq V_{\mathrm{elig}}$ disconnects s_{out} from t_{in}, inducing a reachable set S_C whose outgoing cut edges lie in C with capacity at most $\sum_{v\in C}b(v)$. Hence minimum cut capacity equals $\tau_b(\mathscr{P})$, and equals $+\infty$ iff no subset of V_{elig} hits $\mathscr{P}$. Unit capacities on $V\setminus\{s,t\}$ yield Menger's identity $\tau_1(\mathscr{P})=w(G)$ by max-flow integrality [53]. For Robacker's dual [54], any k pairwise disjoint vertex cuts consume at least k distinct internal vertices on every $P\in\mathscr{P}$, so $k\le\ell(G)$; conversely, setting $C_d=\{v\in V\setminus\{s,t\}:\mathrm{dist}_G(s,v)=d\}$ for $1\le d\le\ell(G)$ produces $\ell(G)$ pairwise disjoint s–t vertex cuts because each edge increments $\mathrm{dist}_G(s,\cdot)$ by at most 1. □

Proposition 12 (Lovász Bounded-Length Packing–Hitting Gap)**.** *Let $\mathscr{P}$ be a finite collection of non-empty internal vertex sets of s–t routes with $|P|\le r$ for all $P\in\mathscr{P}$. Let $\omega(\mathscr{P})$ be the maximum size of a pairwise-disjoint subfamily of $\mathscr{P}$ and $\tau(\mathscr{P})$ the minimum size of a vertex set intersecting every $P\in\mathscr{P}$. Then $\omega(\mathscr{P})\le\tau(\mathscr{P})\le r\,\omega(\mathscr{P})$ [55].*

Proof. If $\mathscr{Q}\subseteq\mathscr{P}$ is pairwise disjoint and C hits $\mathscr{P}$, C contains distinct elements from each $Q\in\mathscr{Q}$, so $|C|\ge|\mathscr{Q}|$ and $\tau(\mathscr{P})\ge\omega(\mathscr{P})$. Conversely, choose an inclusion-maximal pairwise-disjoint subfamily $\mathscr{Q}^*\subseteq\mathscr{P}$ ($|\mathscr{Q}^*|\le\omega(\mathscr{P})$) and set $C=\bigcup_{Q\in\mathscr{Q}^*}Q$. By maximality, no $P\in\mathscr{P}$ is disjoint from C, so C hits $\mathscr{P}$ and $\tau(\mathscr{P})\le|C|\le r|\mathscr{Q}^*|\le r\,\omega(\mathscr{P})$. □

Proposition 13 (One-Sidedness of Empirical Search and Bayesian Search Power)**.** *(i) Without an external finite-support or lower-probability bound on unseen routes, no finite clean search trace D_T logically certifies the absence of an unblocked route, whereas a single verified bypass $P^*\in\mathscr{P}$ with $P^*\cap C=\emptyset$ refutes completeness of cut C. (ii) Suppose an unblocked route exists ($H=1$) with prior $p_0\in(0,1)$, falls in a shared blind spot ($B=1$) with probability $b\in[0,1]$, and when $B=0$ is independently detected in each of T trials with probability $q\in(0,1]$, while verified bypass checks have zero false positives under $H=0$. Given T zero-bypass trials Z_T, the posterior probability of an unblocked route is*

$$\pi(T)\;=\;\mathbb{P}(H=1\mid Z_T)\;=\;\frac{p_0 s}{p_0 s+1-p_0},$$
$$s\;=\;b+(1-b)(1-q)^T,$$

converging as $T\to\infty$ to the strictly positive blind-spot floor $p_0 b/(p_0 b+1-p_0)$ whenever $p_0>0$ and $b>0$.

Proof. Part (i) is immediate: an unblocked route $P^\dagger$ assigned probability 0 (or arbitrarily small probability $\varepsilon>0$) by the search policy produces $D_T\not\ni P^\dagger$ with probability 1 (respectively $(1-\varepsilon)^T>0$), whereas $P^*\cap C=\emptyset$ directly witnesses $\tau_1(\{P^*\}\setminus C)>0$. For part (ii), conditioning on $B\in\{0,1\}$ gives $\mathbb{P}(Z_T\mid H=1)=b\cdot 1+(1-b)(1-q)^T=s$, while soundness of bypass verification gives $\mathbb{P}(Z_T\mid H=0)=1$. Bayes' rule yields $\mathbb{P}(H=1\mid Z_T)=p_0 s/(p_0 s+1-p_0)$, and $(1-q)^T\to 0$ for $q\in(0,1]$ yields the limit. □

Proposition 14 (Withholding Indistinguishability)**.** *Compare two worlds over disjoint route-class sets Π and Σ ($\Sigma\neq\emptyset$): in world $\mathscr{W}_1$, unblocked classes $\Pi\sqcup\Sigma$ exist, and a strategic searcher suppresses every discovery in Σ, reporting either (a) thinning of independent Poisson counts $N_c\sim\mathrm{Poisson}(\lambda_c)$ ($c\in\Pi\sqcup\Sigma$) or (b) N retained categorical draws from $(p_c)_{c\in\Pi\sqcup\Sigma}$ conditioned on landing in Π, with $\tilde{p}_c=p_c/\sum_{j\in\Pi}p_j$; in world $\mathscr{W}_0$, only classes Π exist with intensities $(\lambda_c)_{c\in\Pi}$ (respectively categorical law $(\tilde{p}_c)_{c\in\Pi}$ and N honest draws) and zero unblocked mass in Σ. Then every report-only statistic—including class multiplicities, singleton count f_1, Good–Turing coverage $\hat{C}_{\mathrm{GT}}=1-f_1/N$ [56], and multi-searcher capture–recapture overlaps [58]—has the exact same probability distribution under $\mathscr{W}_1$ and $\mathscr{W}_0$, despite $\mathscr{W}_1$ retaining unblocked withheld mass $\sum_{c\in\Sigma}p_c>0$.*

Proof. Under model (a), projecting mutually independent random variables $(N_c)_{c\in\Pi\sqcup\Sigma}$ onto coordinates Π yields marginal joint law $\prod_{c\in\Pi}\mathrm{Poisson}(\lambda_c)$, identical to $\mathscr{W}_0$. Under model (b), each draw $X\sim(p_c)_{c\in\Pi\sqcup\Sigma}$ conditioned on $\{X\in\Pi\}$ satisfies $\mathbb{P}_{\mathscr{W}_1}(X=c\mid X\in\Pi)=p_c/\sum_{j\in\Pi}p_j=\tilde{p}_c=\mathbb{P}_{\mathscr{W}_0}(X=c)$ for every $c\in\Pi$; hence N i.i.d. retained reports have joint law $\prod_{t=1}^N\tilde{p}_{x_t}$ under both worlds. Any measurable functional of the Π-valued report transcript therefore has identical distribution under $\mathscr{W}_1$ and $\mathscr{W}_0$ (total variation distance 0). □

A.5 Two-Stage Escrow Triage and Portfolio Assessment Ceilings

Proof of Proposition 9. (Two-stage orthogonal escrow triage, Section 8.) At each step $s\in\{1,\dots,k\}$ of a benign workflow, let $F_s^{(1)}\in\{0,1\}$ indicate a Stage-1 fast-screen flag ($\mathbb{P}(F_s^{(1)}=1\mid\text{benign})\le\alpha_1$) and let $R_s^{(2)}\in\{0,1\}$ indicate a Stage-2 deterministic provenance/scope rejection when escalated ($\mathbb{P}(R_s^{(2)}=1\mid F_s^{(1)}=1,\text{benign})\le\alpha_{2|1}$). Because non-flagged steps ($F_s^{(1)}=0$) commit directly without escalation, at least $1-\alpha_1$ of benign steps execute unpaused on the fast path, proving (i). A benign step s suffers a terminal abort if and only if $A_s=\{F_s^{(1)}=1\wedge R_s^{(2)}=1\}$ occurs,

which has probability $p_{\mathrm{ab},s} = \mathbb{P}(F_s^{(1)} = 1 \mid \text{benign})\mathbb{P}(R_s^{(2)} = 1 \mid F_s^{(1)} = 1, \text{benign}) \le \alpha_1\alpha_{2|1}$. Across k conditionally independent benign step checks (respectively arbitrary dependence via Boole's inequality), episode false-abort probability satisfies $\mathbb{P}(\bigcup_{s=1}^k A_s \mid \text{benign}) = 1 - \prod_{s=1}^k (1 - p_{\mathrm{ab},s}) \le 1 - (1 - \alpha_1\alpha_{2|1})^k \le k\alpha_1\alpha_{2|1}$ (and $\mathbb{P}(\bigcup_{s=1}^k A_s \mid \text{benign}) \le \sum_{s=1}^k p_{\mathrm{ab},s} \le k\alpha_1\alpha_{2|1}$ without independence), proving (ii). For (iii), on any out-of-scope trajectory step s, Stage-2 zero false clearance ($\mathbb{P}(R_s^{(2)} = 0 \mid F_s^{(1)} = 1, \text{out-of-scope}) = 0$) implies that two-stage triage misses step s only when Stage 1 fails to flag ($F_s^{(1)} = 0$). Hence the two-stage k-step joint miss event is contained in $\bigcap_{s=1}^k \{F_s^{(1)} = 0\}$, yielding $\tilde{J}_k \le J_k^{(\alpha_1)}$; when $\alpha_1 \ge \alpha_0$, monotone nestedness of Stage-1 flag sets ($\{F_s^{(1)}(\alpha_0) = 1\} \subseteq \{F_s^{(1)}(\alpha_1) = 1\}$) gives $J_k^{(\alpha_1)} \le J_k^{(\alpha_0)}$. □

Proposition 15 (Crossed Generalizability and Portfolio Squared-Correlation Ceiling)**.** *(i) Let candidate i have deployable safety trait $T_i \sim (0, v_T)$, shared pre-deployment discrepancy $U_i \sim (0, v_U)$, context effects $\delta_{ic} \sim (0, v_\delta)$ across k contexts, evaluator effects $\eta_{ij} \sim (0, v_\eta)$ across m evaluators, and residuals $\varepsilon_{ijc} \sim (0, v_\varepsilon)$, all mutually uncorrelated with finite positive variances [61]. For cell-mean pre-deployment score $\bar{Z}_i = T_i + U_i + \bar{\delta}_i + \bar{\eta}_i + \bar{\varepsilon}_i$, the squared correlation with T_i is*

$$V(m,k) \;=\; \frac{v_T}{v_T + v_U + v_\delta/k + v_\eta/m + v_\varepsilon/(mk)},$$

since $\mathrm{Corr}(T_i, \bar{Z}_i)^2 = \mathrm{Cov}(T_i, \bar{Z}_i)^2 / [\mathrm{Var}(T_i)\,\mathrm{Var}(\bar{Z}_i)]$. *(ii) In a portfolio of D benchmark domains with n trials per domain ($N = Dn$ total trials), let $Z_{dr} = G + \delta + c_d + e_{dr}$ with uncorrelated components $G \sim (0, V_G)$, $\delta \sim (0, \omega^2)$, $c_d \sim (0, \sigma_c^2)$, and $e_{dr} \sim (0, \sigma_e^2)$. Then the portfolio mean $\bar{Z}_{D,n}$ has squared correlation with G*

$$R_G(D,n) \;=\; \frac{V_G}{V_G + \omega^2 + \sigma_c^2/D + \sigma_e^2/(Dn)},$$

satisfying the fixed-breadth limit $\lim_{n\to\infty} R_G(D,n) = V_G/(V_G + \omega^2 + \sigma_c^2/D)$ *and the strict universal ceiling* $R_G(D,n) < V_G/(V_G + \omega^2)$ *for all finite* (D,n).

Proof. In (i), mutual orthogonality gives $\mathrm{Cov}(T_i, \bar{Z}_i) = \mathrm{Var}(T_i) = v_T$, while averaging k i.i.d.-variance context terms, m evaluator terms, and mk residual terms yields $\mathrm{Var}(\bar{Z}_i) = v_T + v_U + v_\delta/k + v_\eta/m + v_\varepsilon/(mk)$; substituting into $\mathrm{Cov}(T_i, \bar{Z}_i)^2 / [\mathrm{Var}(T_i)\,\mathrm{Var}(\bar{Z}_i)]$ cancels one factor of v_T and proves $V(m,k)$. Part (ii) follows identically with $\bar{Z}_{D,n} = G + \delta + \frac{1}{D}\sum_{d=1}^D c_d + \frac{1}{Dn}\sum_{d=1}^D\sum_{r=1}^n e_{dr}$, giving $\mathrm{Cov}(G, \bar{Z}_{D,n}) = V_G$ and $\mathrm{Var}(\bar{Z}_{D,n}) = V_G + \omega^2 + \sigma_c^2/D + \sigma_e^2/(Dn)$. Letting $n \to \infty$ at fixed D eliminates $\sigma_e^2/(Dn)$ while leaving $\sigma_c^2/D > 0$; since $\sigma_c^2/D + \sigma_e^2/(Dn) > 0$ for all finite (D,n), $R_G(D,n) < V_G/(V_G + \omega^2)$. □

A.6 Selection Geometry, Risk-Appetite Reversal, Grants, and Sufficiency

Proposition 16 (Finite-Population Price Identity and Gaussian Selection Gradient)**.** *(i) For a finite population $\{1, \dots, N\}$ with prior weights $p_r \ge 0$ ($\sum_{r=1}^N p_r = 1$), trait values $z_r \in \mathbb{R}$, and nonnegative fitnesses w_r with $\bar{w} = \sum_{r=1}^N p_r w_r > 0$, post-selection weights $p_r' = p_r w_r/\bar{w}$ without transmission mutation satisfy [35, 36]*

$$\Delta\bar{z} \;=\; \sum_{r=1}^N p_r' z_r - \bar{z} \;=\; \frac{\mathrm{Cov}_p(z,w)}{\bar{w}}.$$

(ii) If $Z \sim \mathcal{N}(\mu, \Sigma)$ is selected by keeping the top fraction $q \in (0,1)$ on a scalar score I jointly normal with Z, having correlation $\rho_j = \mathrm{Corr}(Z_j, I)$ and trait standard deviation $\sigma_j = \sqrt{\mathrm{Var}(Z_j)}$, then the one-step selection differential on coordinate j is $S_j = \mathbb{E}[Z_j \mid I \ge \tau_q] - \mu_j = i\rho_j\sigma_j$, where $i = \phi(\Phi^{-1}(1-q))/q > 0$ (and under exponential tilting $w(Z) = e^{\eta v^\top Z}$, $\Delta\mu = \eta\Sigma v$).

Proof. For (i), $\sum_{r=1}^N p_r' z_r - \bar{z} = \bar{w}^{-1}\sum_{r=1}^N p_r w_r z_r - \bar{w}^{-1}\bar{w}\bar{z} = \bar{w}^{-1}\sum_{r=1}^N p_r(z_r - \bar{z})(w_r - \bar{w}) = \mathrm{Cov}_p(z,w)/\bar{w}$. For (ii), joint normality gives the linear regression $Z_j - \mu_j = \rho_j(\sigma_j/\sigma_I)(I - \mu_I) + \zeta_j$ with ζ_j independent of I and $\mathbb{E}[\zeta_j] = 0$. Standardizing $W = (I - \mu_I)/\sigma_I \sim \mathcal{N}(0,1)$ with threshold $z_q = \Phi^{-1}(1-q)$ yields $\mathbb{E}[W \mid W \ge z_q] = q^{-1}\int_{z_q}^\infty w\phi(w)\,dw = \phi(z_q)/q = i$, whence $\mathbb{E}[Z_j - \mu_j \mid I \ge \tau_q] = i\rho_j\sigma_j$. Under $w(Z) = e^{\eta v^\top Z}$, $\mathbb{E}[Ze^{\eta v^\top Z}]/\mathbb{E}[e^{\eta v^\top Z}] = \nabla_\eta \ln \mathbb{E}[e^{\eta v^\top Z}] = \nabla_\eta(\eta v^\top\mu + \frac{1}{2}\eta^2 v^\top\Sigma v) = \mu + \eta\Sigma v$. □

Proposition 17 (Payoff-Dispersion Risk-Appetite Reversal)**.** *Let agent return be $X(\sigma) = \mu + \sigma\xi$ for fixed mean $\mu \in \mathbb{R}$, dispersion $\sigma > 0$, and standardized noise ξ ($\mathbb{E}[\xi] = 0$, $\mathbb{E}[|\xi|] < \infty$) having continuous strictly positive density f_ξ symmetric about 0 ($\xi \overset{d}{=} -\xi$) and CDF F_ξ.*

1. Upper-tail tournament ($\tau > \mu$): *Win probability $P_{\mathrm{top}}(\sigma) = \mathbb{P}(X(\sigma) \ge \tau)$ is strictly increasing in σ.*
2. Survival floor ($\tau < \mu$): *Survival probability $P_{\mathrm{floor}}(\sigma) = \mathbb{P}(X(\sigma) \ge \tau)$ is strictly decreasing in σ.*
3. Concave capped caretaker welfare: *For every concave utility $u\colon \mathbb{R} \to \mathbb{R}$, $\sigma \mapsto \mathbb{E}[u(\mu + \sigma\xi)]$ is non-increasing on $(0,\infty)$, and strictly decreasing both when u is strictly concave and when $u(x) = \min\{x, \kappa\}$ for any finite cap $\kappa \in \mathbb{R}$.*
4. Mean-preserving spread reversal: *For any mean-preserving spread $X_{\mathrm{risky}} = X_{\mathrm{safe}} + \varepsilon$ with $\mathbb{E}[\varepsilon \mid X_{\mathrm{safe}}] = 0$ and $\mathbb{P}(\varepsilon \ne 0) > 0$, every strictly convex tournament payoff ψ satisfies $\mathbb{E}[\psi(X_{\mathrm{risky}})] > \mathbb{E}[\psi(X_{\mathrm{safe}})]$, whereas every strictly concave caretaker payoff u satisfies $\mathbb{E}[u(X_{\mathrm{safe}})] > \mathbb{E}[u(X_{\mathrm{risky}})]$.*

Proof. For (1) and (2), write $\mathbb{P}(\mu + \sigma\xi \ge \tau) = 1 - F_\xi((\tau - \mu)/\sigma)$. Differentiating with respect to $\sigma > 0$ gives

$$\frac{d}{d\sigma}\mathbb{P}\big(X(\sigma) \ge \tau\big) \;=\; f_\xi\Big(\frac{\tau-\mu}{\sigma}\Big)\frac{\tau-\mu}{\sigma^2}.$$

Because $f_\xi > 0$ everywhere on $\mathbb{R}$, this derivative is strictly positive when $\tau > \mu$ and strictly negative when $\tau < \mu$.

For (3), let $0 < \sigma_1 < \sigma_2$ and set $\lambda = (\sigma_1 + \sigma_2)/(2\sigma_2) \in (\frac{1}{2}, 1)$, so $\lambda\sigma_2 + (1-\lambda)(-\sigma_2) = \sigma_1$. Concavity of u gives, for every $\xi \in \mathbb{R}$,

$$u(\mu + \sigma_1\xi) \;\ge\; \lambda\,u(\mu + \sigma_2\xi) + (1-\lambda)\,u(\mu - \sigma_2\xi),$$

with strict inequality for all $\xi \neq 0$ whenever u is strictly concave. Taking expectations and using symmetry $\mathbb{E}[u(\mu - \sigma_2\xi)] = \mathbb{E}[u(\mu+\sigma_2\xi)]$ yields $\mathbb{E}[u(\mu+\sigma_1\xi)] \geq \mathbb{E}[u(\mu+\sigma_2\xi)]$ (strict for strictly concave u). For piecewise-linear capped welfare $u(x) = \min\{x,\kappa\} = x - (x-\kappa)_+$, setting $z_\kappa(\sigma) = (\kappa-\mu)/\sigma$ gives

$$\mathbb{E}\big[\min\{\mu+\sigma\xi,\kappa\}\big] \;=\; \mu - \int_{z_\kappa(\sigma)}^{\infty} (\sigma\xi - \kappa + \mu) f_\xi(\xi)\, d\xi.$$

Differentiating under the integral sign (Leibniz's rule; the boundary integrand vanishes at $\xi = z_\kappa(\sigma)$) gives

$$\frac{d}{d\sigma}\mathbb{E}\big[\min\{\mu+\sigma\xi,\kappa\}\big] \;=\; -\int_{z_\kappa(\sigma)}^{\infty} \xi f_\xi(\xi)\, d\xi \;<\; 0$$

for every $\kappa \in \mathbb{R}$: if $z_\kappa(\sigma) \geq 0$ the integrand is strictly positive on $(z_\kappa(\sigma),\infty)$, while if $z_\kappa(\sigma) < 0$ symmetry $\int_{z_\kappa}^{-z_\kappa} \xi f_\xi(\xi)\, d\xi = 0$ reduces the integral to $\int_{-z_\kappa(\sigma)}^{\infty} \xi f_\xi(\xi)\, d\xi > 0$.

For (4), conditional Jensen's inequality gives $\mathbb{E}[\psi(X_{\text{safe}} + \varepsilon) \mid X_{\text{safe}}] \geq \psi(X_{\text{safe}})$ almost surely, with strict inequality on the positive-probability event $\{\mathrm{Var}(\varepsilon \mid X_{\text{safe}}) > 0\}$ when ψ is strictly convex; reversing $\psi \leftrightarrow -u$ proves the strictly concave case. □

Proof of Proposition 8. (Policy-gradient deception reversal under pre-commit mediation, Section 4.) Parameterize the contextual policy at audit cue $c \in \{0,1\}$ by softmax logit gap $\theta_c \in \mathbb{R}$ between shortcutting ($a = 1$) and safe verification ($a = 0$), so $\pi_\theta(1 \mid c) = \sigma(\theta_c) \in (0,1)$ with $\sigma'(z) = \sigma(z)(1-\sigma(z)) > 0$. Under pre-commit gate mediation probability $q_{\text{gate}} \in [0,1]$, attempting a shortcut is intercepted prior to execution with probability q_{gate} (returning R_{refused}) and otherwise executes with expected audited return $U_{\text{short}}(p_c)$, so expected shortcut return in context c is

$$\tilde{U}_c(q_{\text{gate}}) \;=\; q_{\text{gate}} R_{\text{refused}} + (1-q_{\text{gate}}) U_{\text{short}}(p_c).$$

Let $P_c = \mathbb{P}(c_{\text{audit}} = c) > 0$. Expected objective $J(\theta) = \sum_{c\in\{0,1\}} P_c\big[\sigma(\theta_c)\tilde{U}_c(q_{\text{gate}}) + (1-\sigma(\theta_c))R_{\text{verify}}\big]$ has exact policy gradient on coordinate θ_c:

$$\frac{\partial J(\theta)}{\partial \theta_c} \;=\; P_c\,\sigma(\theta_c)\big(1-\sigma(\theta_c)\big)\big(\tilde{U}_c(q_{\text{gate}}) - R_{\text{verify}}\big).$$

At $q_{\text{gate}} = 0$, the audited-tournament regime $U_{\text{short}}(p_{\text{mon}}) < R_{\text{verify}} < U_{\text{short}}(p_{\text{unmon}})$ gives $\partial J/\partial\theta_1 < 0$ and $\partial J/\partial\theta_0 > 0$ everywhere on $\mathbb{R}^2$, driving $\sigma(\theta_1) \to 0$ (monitored compliance) and $\sigma(\theta_0) \to 1$ (unmonitored covert shortcutting), so the evaluation-aware gap $\sigma(\theta_0) - \sigma(\theta_1) \to 1$. Because $R_{\text{refused}} < R_{\text{verify}} < U_{\text{short}}(p_{\text{unmon}})$, $\tilde{U}_0(q_{\text{gate}}) < R_{\text{verify}}$ holds if and only if

$$\begin{aligned} & q_{\text{gate}} R_{\text{refused}} + (1-q_{\text{gate}}) U_{\text{short}}(p_{\text{unmon}}) \;<\; R_{\text{verify}} \\ \iff \quad & q_{\text{gate}} \;>\; \frac{U_{\text{short}}(p_{\text{unmon}}) - R_{\text{verify}}}{U_{\text{short}}(p_{\text{unmon}}) - R_{\text{refused}}} \;=\; q^*_{\text{gate}}. \end{aligned}$$

Since $p_{\text{mon}} > p_{\text{unmon}}$ implies $U_{\text{short}}(p_{\text{mon}}) \leq U_{\text{short}}(p_{\text{unmon}})$, every $q_{\text{gate}} > q^*_{\text{gate}}$ simultaneously satisfies $\tilde{U}_1(q_{\text{gate}}) < R_{\text{verify}}$ and $\tilde{U}_0(q_{\text{gate}}) < R_{\text{verify}}$, making $\partial J/\partial\theta_c < 0$ in both contexts and driving both $\sigma(\theta_0) \to 0$ and $\sigma(\theta_1) \to 0$ (zero asymptotic deception gap). □

Proposition 18 (Endogenous Grant Harm Reversal)**.** *For disposition $d \in \{d_0, d_1\}$, let $G \in \{0,1\}$ indicate human conferral of broad authority ($g(d) = \mathbb{P}(G = 1 \mid D = d)$) and $h_s(d) = \mathbb{P}(H = 1 \mid G = s, D = d)$ denote conditional harm at scope $s \in \{0,1\}$. Even when d_1 is strictly safer at every fixed scope ($h_0(d_1) < h_0(d_0)$ and $h_1(d_1) < h_1(d_0)$), unconditional harm under endogenous conferral satisfies $\mathbb{P}(H = 1 \mid D = d_1) > \mathbb{P}(H = 1 \mid D = d_0)$ whenever*

$$\begin{aligned} & g(d_1)\big(h_1(d_1) - h_0(d_1)\big) - g(d_0)\big(h_1(d_0) - h_0(d_0)\big) \\ & \qquad > \; h_0(d_0) - h_0(d_1) \end{aligned}$$

(in particular, when $h_0 \equiv 0$ and $g(d_1)h_1(d_1) > g(d_0)h_1(d_0)$), whereas exogenous scope assignment $G \perp D$ with $\mathbb{P}(G = 1) = \bar{g}$ guarantees $\mathbb{P}_{\text{dec}}(H = 1 \mid D = d_1) < \mathbb{P}_{\text{dec}}(H = 1 \mid D = d_0)$.

Proof. Conditioning on $G \in \{0,1\}$ gives $\mathbb{P}(H = 1 \mid D = d) = (1 - g(d))h_0(d) + g(d)h_1(d) = h_0(d) + g(d)(h_1(d) - h_0(d))$, whose difference between d_1 and d_0 is positive if and only if the displayed inequality holds (for the multi-disposition mixture $\mathbb{P}(D = d)$, $\mathbb{P}(H = 1) = \sum_d \mathbb{P}(D = d)[h_0(d) + g(d)(h_1(d) - h_0(d))]$). Under decoupled assignment $G \perp D$ with $\mathbb{P}(G = 1) = \bar{g} \in [0,1]$, $\mathbb{P}_{\text{dec}}(H = 1 \mid D = d) = (1-\bar{g})h_0(d) + \bar{g}h_1(d)$, which is strictly smaller at d_1 than at d_0 because $h_0(d_1) < h_0(d_0)$ and $h_1(d_1) < h_1(d_0)$. □

Proof of Theorem 3. (Sufficiency of Conditions C1–C6, Section 3.) Fix an operating period $t \geq 0$ and capability class $\mathscr{C}_t$. Condition C1 places an external, unalterable reference monitor and cooperative idempotent sink on all consequential interfaces with conservative reservation accounting ($x_{t,m} \in \mathscr{B}_{t,m}$, discharging Proposition 2 for deterministic resource envelopes) and synchronous epoch fencing within latency $\tau < T_{\text{irrev}}$ upon detection or revocation, discharging Premises A1 and A2. Condition C2 places eligible cutsets of cost $\tau_b(\mathscr{P}_t) \leq B_{\text{def}}$ across mapped pathways so mapped routes traverse $k_t \leq \ell(G_t)$ distinct step gates (Proposition 11). Condition C3 bounds the filtration-conditioned k_t-gate joint miss across governed step crossings by $J_{k,t}$ (accounting for shared blind spots, cross-family correlation, score-feedback suppression, and candidate-retry caps), discharging Premise A4. Conditions C4 and C5 isolate vulnerability discovery under Custody Rules R1–R6, withhold risk credit until validated cut deployment, and bound residual $< k_t$-gate bypass probability by the post-search quantity $u_t \leq \pi_t$ (Propositions 13 and 14), discharging Premise A3.

Applying Proposition 5 in period t yields single-period catastrophe probability $R_t = \mathbb{P}(E_t) \leq u_t + J_{k,t} \leq \pi_t + J_{k,t}$, without invoking assumptions on unobserved internal goals. Across periods $t = 0, 1, \ldots$, Condition C6 resets baseline state at renewal boundaries, enforces non-escalating controller updates absent fresh conferral, and requires renewal bounds $\bar{R}_t = \pi_t + J_{k,t}$ to form a summable series. Boole's inequality yields $\mathbb{P}(\bigcup_{t=0}^{\infty} E_t) \leq \sum_{t=0}^{\infty} R_t \leq \sum_{t=0}^{\infty} (\pi_t + J_{k,t})$; in particular, whenever $\bar{R}_{t+1} \leq \gamma \bar{R}_t$ with $\gamma \in (0,1)$, $\sum_{t=0}^{\infty} R_t \leq \bar{R}_0/(1-\gamma) < \infty$ (and by Proposition 6(ii), summability of survival-conditioned hazards ensures strictly positive infinite-horizon survival). □

A.7 Exact Unconditional Trinomial Superiority and Anytime-Valid Martingales

Proposition 19 (Exact Unconditional Paired Trinomial Inversion and Ville Discordance Supermartingales)**.** *Let* $(U_{i,A}, U_{i,B}) \in \{0,1\}^2$ *be N independent paired binary harm outcomes with cell probabilities* $(p_{10}, p_{01}, p_{\text{tie}})$ *on the trinomial simplex* $\Theta_{\text{tri}} = \{(p_{10}, p_{01}) \in [0,1]^2 : p_{10} + p_{01} \le 1\}$ *and risk difference* $\delta = p_{10} - p_{01} \in [-1,1]$.

1. Exact unconditional size and test inversion [49, 50]: *For any null margin* $\delta_0 \in (-1,1)$*, let* $\Theta_0(\delta_0) = \{\theta \in [\max(0, -\delta_0), (1-\delta_0)/2] : p_{10} = \theta + \delta_0,\ p_{01} = \theta\}$ *and let* $Z(x_{10}, x_{01}; \delta_0)$ *denote Tango's score statistic [50]. The unconditional p-value*

$$p_{\text{SS}}(\delta_0) = \sup_{\theta \in \Theta_0(\delta_0)} \sum_{Z(x_{10},x_{01};\delta_0) \le Z_{\text{obs}}} \frac{N!}{x_{10}!\,x_{01}!\,x_{00}!} \times (\theta+\delta_0)^{x_{10}}\,\theta^{x_{01}}\,(1-2\theta-\delta_0)^{x_{00}}$$

   (with $x_{00} = N - x_{10} - x_{01}$*) satisfies* $\mathbb{P}_{(p_{10},p_{01})}\big(p_{\text{SS}}(\delta_0) \le \alpha\big) \le \alpha$ *for all* $(p_{10}, p_{01}) \in \Theta_{\text{tri}}$ *with* $p_{10} - p_{01} \ge \delta_0$*, and* $U_{1-\alpha} = \sup\{\delta_0 \in (-1,1) : p_{\text{SS}}(\delta_0) > \alpha\}$ *is an exact* $(1-\alpha)$ *one-sided upper confidence bound for* δ *without marginal union-bound slack.*

2. Anytime-valid Ville betting supermartingale [51, 52]: *Let* $Y_1, Y_2, \cdots \in \{0,1\}$ *record sequential discordant pairs* ($Y_d = 1$ *if only B errs,* $Y_d = 0$ *if only A errs). Under* H_0: $\delta \ge 0$*, conditional discordance satisfies* $\mathbb{P}(Y_d = 1 \mid \mathcal{G}_{d-1}) = p_{01}/(p_{10}+p_{01}) \le \frac{1}{2}$*. For any* $\mathcal{G}_{d-1}$*-predictable sequence* $\lambda_d \in [0,1]$*,* $E_0 = 1$ *and* $E_d = \prod_{j=1}^{d}\big(1 + \lambda_j(2Y_j - 1)\big)$ *form a non-negative supermartingale under* H_0*, so Ville's inequality guarantees* $\mathbb{P}_{H_0}\big(\exists d \ge 1 : E_d \ge 1/\alpha\big) \le \alpha$ *under arbitrary data-dependent stopping times* τ ($p_{\text{anytime}}(\tau) = \min\{1, (\max_{1 \le j \le \tau} E_j)^{-1}\}$).

Proof. For (1), fix any $(p_{10}^*, p_{01}^*) \in \Theta_{\text{tri}}$ with $\delta^* = p_{10}^* - p_{01}^* \ge \delta_0$. When $\delta^* = \delta_0$, $\theta^* = p_{01}^* \in \Theta_0(\delta_0)$, so at observed (X_{10}, X_{01}), $p_{\text{SS}}(\delta_0)$ upper-bounds the exact null tail probability $F_{\theta^*}(Z(X_{10}, X_{01}; \delta_0))$, which stochastically dominates Uniform$[0,1]$; hence $\mathbb{P}_{\theta^*}(p_{\text{SS}}(\delta_0) \le \alpha) \le \alpha$ (monotonicity of the Tango rejection set in $(x_{10}, -x_{01})$ extends size $\le \alpha$ to composite $\delta^* > \delta_0$). Standard Neyman duality between level-α tests of $H_0(\delta_0)$: $\delta \ge \delta_0$ and confidence sets gives $\mathbb{P}(\delta \le U_{1-\alpha}) \ge 1 - \alpha$.

For (2), since $\lambda_d \in [0,1]$ is $\mathcal{G}_{d-1}$-measurable and $2Y_d - 1 \in \{-1, +1\}$, each factor $1 + \lambda_d(2Y_d - 1) \ge 1 - \lambda_d \ge 0$ is non-negative, whence $E_d \ge 0$ for all $d \ge 0$. Under H_0 ($\mathbb{P}(Y_d = 1 \mid \mathcal{G}_{d-1}) = q_d \le \frac{1}{2}$), $\mathbb{E}[2Y_d - 1 \mid \mathcal{G}_{d-1}] = 2q_d - 1 \le 0$; multiplying by $\lambda_d \ge 0$ gives $\mathbb{E}[E_d \mid \mathcal{G}_{d-1}] = E_{d-1}\big(1 + \lambda_d(2q_d - 1)\big) \le E_{d-1}$. By Ville's maximal inequality for non-negative supermartingales with $E_0 = 1$ [51, 52], $\mathbb{P}_{H_0}(\sup_{d \ge 1} E_d \ge 1/\alpha) \le \alpha E_0 = \alpha$. □

B Protocol Transitions, Finite-State Models, and Boundary Schemas

This appendix records the durable storage schemas, remote-procedure-call (RPC) transition rules, Linux kernel confinement filters, mechanized Verus verification inventory, and 22-action finite-state and single-database baselines referenced in Sections 5 and 6.

B.1 Separate-Process Broker and Sink Durable Schemas

Both the Authority Broker and Cooperative Idempotent Sink require SQLite ≥ 3.51.3 (excluding historical write-ahead-log reset bugs [43]), configure write-ahead logging (`WAL`), full synchronous durability (`FULL`), and a 10 s lock-wait timeout, and execute every mutating RPC inside an explicit serialized immediate-write transaction (`BEGIN IMMEDIATE`) [42]. Physically, each service store initializes two relational tables:

- `state(singleton INTEGER PRIMARY KEY CHECK(singleton=1), body TEXT NOT NULL)`: stores the canonical serialized service state record for the broker or sink shard.
- `effects(sequence INTEGER PRIMARY KEY, id TEXT UNIQUE NOT NULL, payload TEXT NOT NULL, charge INTEGER NOT NULL)`: used in each sink shard as an append-only committed effect log enforcing database-level uniqueness on operation identifier `id` and recording the canonical dispatched permit alongside integer `charge`.

Table 10 maps the logical protocol collections onto these persisted structures. All wire records reject unrecognized fields prior to canonical re-serialization, so formatting or key-order variations cannot mint distinct authorities.

Normalized dispatched-permit binding. For n configured issuers ($1 \le n \le 64$, default $n = 2$), base operation requirements $\text{req}_0(e,n)$ require issuer 0 (operator) for value transfers, issuers $\{0,1\}$ (operator and steward) for resource edits, and all n issuers for external releases. When a proposal is admitted via a sub-delegation leaf whose required-issuer vector strengthens $\text{req}_0(e,n)$ (for example, adding steward requirement $j = 1$ to a transfer), submission embeds the leaf's strengthened requirement vector into the dispatched wire permit. At the sink, normalization rejects any vector that weakens $\text{req}_0(e,n)$ and supplies the effective required-issuer set to acceptance, completion, fencing, and integrity auditing. Meanwhile, broker submission compares the base canonical permit (actor + proposal) on proposer retries so repeating a request returns its stored status and dispatched permit idempotently without permitting a stripped sink replay.

B.2 Broker and Sink RPC Transition Specification

Every service request frame enforces a 65,536-byte cap and a POSIX poll per-frame deadline (default 500 ms; oversize frames drain to newline and return a size-limit denial, while stalled partial frames return a timeout denial without terminating the service). Deterministic fault-injection hooks (configurable pre-commit lock hold up to 1000 ms, deterministic crash prior to transaction commit, and deterministic crash immediately after transaction commit) exercise write-ahead-log lock contention and crash recovery.

Authority Broker transitions.

- **Initialize**(B,n)**:** Fails if initialized; otherwise sets cap B,

Service store	Logical collection	Persisted representation	Key invariants and role
Authority Broker	Header state	$\{B, C, e_{\text{iss}}, \text{live}\}$ in `state`	Cumulative cap B, conservative upper envelope $C \in [0, B]$, per-issuer monotone epoch vector $e_{\text{iss}}[0..n)$, and liveness flags live$[0..n)$.
Authority Broker	Approvals	Per-proposal issuer grant table in `state`	Binds exact canonical permit (actor + proposal) to per-issuer boolean grants; cleared on withdrawal of issuer j and never restored by renewal.
Authority Broker	Delegations	Acyclic parent-linked capability records in `state`	Stores delegation identifier, parent link, cap B_k, charged spend C_k, expiry T_k, action bitmask, required-issuer mask, and issuer epochs verified by stepwise attenuation.
Authority Broker	Reservations	Dispatched-permit ledger in `state`	Immutable dispatched permit plus status $\in \{\text{Pending}, \text{Committed}, \text{Cancelled}\}$; $\sum_{\text{Pending}} q_i + \sum_{\text{Committed}} q_i = C \le B$.
Cooperative Sink	Epoch floors	Monotone vector $F[0..n)$ in `state`	Per-issuer monotone lower bounds updated via immediate-write fence transactions; gates both initial acceptance and completion.
Cooperative Sink	Permits	Deduplicated sink state machine in `state`	Enforces Pending $\to$ Committed | Cancelled transitions, or writes an immediate Cancelled tombstone on late old-epoch arrival.
Cooperative Sink	Versioned resources	$\{\text{revision}, \text{document}\}$ in `state`	Versioned target resource (initial revision 0); edit operations increment revision iff expected revision matches at completion.
Cooperative Sink	Committed effects	Relational table `effects(sequence, id, payload, charge)`	Append-only committed effects log; integrity audit verifies exact 1-to-1 bijection with committed sink permits and revision = |edit commits|.

Table 10: Physical SQLite relational tables and logical protocol collections across the separate-process Authority Broker and Cooperative Idempotent Sink.

charged spend $C = 0$, $e_{\text{iss}} = 0^n$, live $= \text{true}^n$, and empty approvals, reservations, and delegations.

- **Approve(permit, j):** Requires live$[j]$, matching issuer epochs, no identifier collision in delegations, and exact canonical byte equality if an approval for that proposal identifier already exists; records grant bit j.
- **Delegate(id, parent, B_k, T_k, scope, req):** Verifies ancestor acyclicity, live current ancestor grants, absence of non-cancelled direct spend on ancestor records, and stepwise capability attenuation before appending the delegation.
- **Submit(proposal, session-actor):** If the proposal identifier is already recorded, returns stored (status, permit) iff the base permit matches session-actor + proposal, else denies as a payload mismatch. Otherwise verifies scope, checks direct approval or full ancestor delegation liveness and multi-issuer admission, reserves $C' = C + q \le B$ (and $C'_k = C_k + q \le B_k$ along any delegation lineage), persists a *Pending* reservation, and returns *Pending*.
- **Withdraw(j):** Increments $e_{\text{iss}}[j] \leftarrow e_{\text{iss}}[j] + 1$, sets live$[j] \leftarrow$ false, clears issuer-j grants across all approvals and delegations, and reports outstanding in-flight exposure $\sum\{q_r : r \text{ Pending} \wedge \text{req}_r[j]\}$.
- **Renew(j):** Sets live$[j] \leftarrow$ true without restoring cleared grant bits, altering e_{iss}, or reducing C.
- **Settle(id, terminal-status):** Control-plane endpoint (never exposed to proposers). Requires an existing *Pending* reservation and authoritative sink status $\in \{\text{Committed}, \text{Cancelled}\}$; refunds q iff *Cancelled* and seals terminal status.
- **Status / Inspect / Audit:** Read-only queries; audit runs SQLite database integrity verification and checks identifier uniqueness, delegation bounds, and $\sum_{\text{Pending}} q + \sum_{\text{Committed}} q = C \le B$.

Cooperative Sink transitions and concurrent linearization.

- **Accept(permit):** Normalizes the dispatched permit. If its identifier is already recorded, returns stored status iff the full dispatched permit matches byte-for-byte, else denies as a payload mismatch. If new and any effective required issuer fails $e_{\text{iss}}[j] \ge F[j]$, persists a terminal *Cancelled* tombstone; otherwise persists *Pending*.
- **Complete(id):** Requires a recorded permit. If *Pending*, re-evaluates $e_{\text{iss}}[j] \ge F[j]$, active lease horizon $t < L_m$, relational mutexes, and (for edits) expected resource revision. Failure transitions status to *Cancelled*; success updates the target resource if applicable, transitions status to *Committed*, and appends the unique row to `effects` inside the same immediate-write transaction. Terminal records return stored status idempotently.
- **Fence(j, floor):** Updates $F[j] \leftarrow \max(F[j], \text{floor})$ and transitions every *Pending* permit failing the updated floor to *Cancelled* before committing and returning a signed acknowledgment.
- **Status / Inspect / Audit:** Status returns *unknown* when absent (never treated as cancellation); audit verifies database integrity, epoch-floor compliance across all remaining *Pending* permits, and exact payload and charge bijection between *Committed* permits and `effects`.

For concurrent separate-process runs, our interval-linearizability verifier checks real-time order against the independent reference specification using diamond partial-order reduction (exploiting commutativity of budget-rejected submissions) and canonicalized state memoization across all 1..32-writer populations (18/18 runs, ≤ 69 RPCs): an invocation e_j must precede e_i whenever $t_{\text{resp}}(e_j) < t_{\text{inv}}(e_i)$; pre-commit crashes commit false, post-commit crashes commit true, and unannotated dropped replies branch over $\{\text{false}, \text{true}\}$. Every linearized prefix must match recorded decisions and public status responses, maintain $0 \le C \le B$, and end with the exact canonical `effects` sequence.

B.3 Launcher Framing, Allowlist, and Kernel Confinement Specification

The confinement launcher spawns untrusted proposers over dedicated standard-input and standard-output pipes with error streams discarded, inherited descriptors closed, an empty environment, and root working directory. Proposer line framing enforces a 32,768-byte UTF-8 cap (draining oversize lines and returning a denial without unbounded buffer growth). Schema validation permits only two proposer verbs—*submit* and *status*—rejecting any proposer-supplied actor identifier or control-plane verb (*approve*, *delegate*, *withdraw*, *renew*, *settle*, *fence*,

init, *inspect*, *audit*), injects the bound session actor server-side, and strips raw permit tokens from responses.

Inside the unprivileged Linux confinement wrapper:

1. **Landlock ABI ≥ 4:** Queries the kernel Landlock ABI version (failing closed if < 4), constructs a 16-byte ruleset handling all 15 filesystem access bits (including file truncation) and both TCP bind and TCP connect bits, attaches read/execute path rules solely for system shared-library directories, null and random character devices, and the canonicalized proposer binary, sets the kernel no-new-privileges flag, and restricts the calling process (zero TCP rules attached, denying all TCP bind and connect operations).
2. **Seccomp-BPF `EPERM` filter:** Pins the native system-call architecture (`0xc000003e` on x86-64, `0xc00000b7` on ARM64) and terminates compatibility-ABI system calls ($\geq$ `0x40000000`). Permits resource-limit queries only when both 32-bit halves of the target process-identifier argument equal zero, returning `EPERM` on cross-process inspection. Returns `EPERM` across 56 additional system calls spanning network sockets (socket creation, pair, connect, bind, listen, accept, and datagram/message send), process inspection and signaling (process trace, cross-process virtual-memory read/write, process file descriptors, process/thread signal delivery, and descriptor comparison), namespaces, mounts, and kernel interfaces (mount/unmount, namespace unshare/reassociate, eBPF, performance events, user-space page-fault handling, asynchronous I/O rings, file-handle open, reboot, and kernel image load), kernel keyrings, System V shared memory, message queues, and semaphores, and POSIX message queues.
3. **Descriptor sweep and native probes:** Sweeps all file descriptors ≥ 3 via kernel close-range before executing the proposer binary. Automated validation pairs unrestricted positive canaries against confined probes across all 11 checks (10 compiled binary probes plus a live inherited System V private queue probe verifying that unrestricted queue writes succeed while confined writes fail with `EPERM`).

B.4 Mechanized Policy-Kernel Inventory

Table 11 catalogs all 52 verified executable functions and all 42 mechanized proof theorems in the Rust policy kernel (246 verified SMT obligations, 0 quantifier-trigger warnings, and $9/9$ rejected Z3 soundness mutations).

B.5 22-Action Finite-State Model and Single-Database SQLite Baseline

To anchor our separate-process protocol against an exhaustive finite-state baseline, we retain both a 22-action finite-state model checker and a single-database transactional SQLite executor.

In the finite-state model, each state is the tuple $x = (z, L, U, z_0, z_1, l, v, p_0, p_1)$, tracking world spend z, accounting ledger L, 4-bit executed-identifier mask U, per-actor spend (z_0, z_1), grant liveness $l \in \{0,1\}$, environment version $v \in \{0,1\}$, and per-actor pending slots (p_0, p_1). The 22-action alphabet comprises 16 preparations ($a \in \{0,1\}$, $id \in \{0,1\}$, $q \in \{1,2\}$, $dest \in \{\text{true}, \text{false}\}$), 2 commits, revocation, version-advance with revocation, process restart, and grant renewal. Cycle-closed breadth-first search checks all 22 actions per reachable accepted state against 20 positive tasks (8 single transactions plus 12 ordered pairs of 1-unit identifiers) and an independent pre-effect world oracle. Table 12 reports the complete atomic enforcer (10,180 states, 223,960 checks, 0 violations), unconstrained and reject-all baselines, nine targeted single-clause enforcer mutations, and the specification-omission control (41,732 states, 918,104 checks, spending 8 units against budget 2).

Finally, the single-database transactional SQLite baseline executes all policy checks, receipt insertions, and toy effects inside one write-ahead-logged SQLite database across seven relational tables (policy, prepared operations, grants, receipts, accounts, versioned objects, and deliveries), tested over 8 lifecycle, pre/post-commit process-kill, and 8-writer contention scenarios. Table 13 records its single-store commit throughput and p95 latency; unlike the multi-process protocol (Table 4, Figure 6), the single-store baseline mediates only its own internal tables without separate-process reservation quarantine, distributed sink epoch fencing, or kernel proposer confinement.

C Synthetic Generative Specifications and Extended Tables

This appendix documents the deterministic generative models, hyperparameters, numerical quadrature and optimization algorithms, and automated validation checks backing our synthetic and semi-synthetic evaluation suites, together with the extended monitor and selection tables (Tables 14 and 15) and the two-family synthetic monitor baseline (Table 16).

C.1 Deterministic Runtime, Numerical Primitives, and Scope

All synthetic and scaling experiments execute deterministically on CPU from master seed 20261110 with fixed domain-separation Weyl offsets per module, paired with a standalone 64-bit pseudorandom bijection, a 128-layer Marsaglia–Tsang Ziggurat normal sampler, Cody rational complementary error function approximation, Acklam–Halley normal quantile inversion, and a 200-node Golub–Welsch symmetric tridiagonal QL quadrature solver [60].

Two methodological boundaries apply across all Tier-4 synthetic models. First, every classifier, graph solver, statistical certificate, and evolutionary or policy-gradient loop is implemented from scratch in standalone Rust with zero third-party machine-learning crates; the synthetic monitor suite is therefore an independent boundary testbed rather than a frontier language-model correlation measurement. Second, elementary floating-point transcendentals call the standard platform C math library on Linux x86-64 with cryptographic digest verification across all generated summaries, without claiming unmeasured cross-architecture bit-identity.

Layer	Verified component family	Mechanized specification and theorem guarantee
Executable (`exec`)	Byte, revision, and multi-issuer grant checkers (4 functions)	Exact byte-slice equality ($a@ = b@$); resource revision matching; universal implication across required issuer vectors ($\forall i.\,\mathrm{req}[i] \Rightarrow \mathrm{grant}[i] \wedge \mathrm{live}[i] \wedge e_{\mathrm{iss}}[i] = e_{\mathrm{req}}[i]$), failing closed on vector-length mismatch.
Executable (`exec`)	Overflow-safe reservation and uncertainty-interval admission (4 functions)	Checked reservation ($q > 0 \wedge q \le B \wedge C \le B - q \Rightarrow \mathrm{Some}(C+q)$); single-issuer, n-issuer, and interval $[C_{\mathrm{lo}}, C_{\mathrm{hi}}]$ admission equivalences.
Executable (`exec`)	Epoch-floor advance, attenuation, fence check, and settlement (4 functions)	Monotone floor $\max(F, \mathrm{floor})$; capability attenuation ($B_c \le B_p$, $T_c \le T_p$, $\mathrm{scope}_c \Rightarrow \mathrm{scope}_p$, $\mathrm{req}_p \Rightarrow \mathrm{req}_c$); per-issuer fence check ($\forall i.\,\mathrm{req}[i] \Rightarrow e_{\mathrm{iss}}[i] \ge F[i]$); non-increasing settlement refund iff authoritative non-execution receipt holds.
Executable (`exec`)	Canonical wire encoder, total parser, and roundtrip comparator (4 functions)	Length-prefixed canonical wire encoder and panic-free total parser on arbitrary byte slices rejecting truncated frames, 64-bit length overflows, non-boolean tags, and trailing smuggled bytes.
Executable (`exec`)	d-hop lineage attenuation, self-upgrade, and cluster quorum lease (6 functions)	Arbitrary-depth bitmask/budget/expiry/issuer-quorum chain verifier, lineage liveness check, non-escalating controller self-upgrade gate ($G(v_1) \preceq G(v_0) \vee$ fresh quorum), and M-shard quorum/lease fence and completion transitions.
Executable (`exec`)	Relational toxic-pair mutex and staged post-condition commit (2 functions)	Bitmask history check preventing simultaneous realization of any declared toxic capability pair $(A_i, B_i) \in \mathscr{M}$ across multi-step traces, and staged commit gating requiring verified post-conditions prior to durable finalization.
Executable (`exec`)	Asymmetric safe-bootstrap cut-certificate ratchet (1 function)	Deterministic certificate validator admitting candidate cut transitions iff coordinatewise authority inclusion holds ($G_{e+1} \preceq G_e$), defense cost satisfies $b(C_e) \le B_{\mathrm{cut}}$, and exposed hazardous route count strictly contracts ($\lvert U_{e+1}\rvert < \lvert U_e\rvert$).
Executable (`exec`)	Concrete broker, sink, and coupled kernel state-machine transitions (27 functions)	Verified constructors, reservation/settlement transitions, epoch-floor fencing, versioned resource completion, and multi-shard cluster transition methods compiled directly into service binaries.
Proof (`proof`)	Prefix conservation, quarantine contrast, and uncertainty contraction (11 theorems)	Pointwise exposure bound $\sum e_i \le \sum q_i$; trace-inductive conservation $C = C_{\mathrm{comm}} + \sum_{i \in I_t} q_i \le B$; constructive $2B > B$ overspend on renewal/restart ledger reset and $B + q > B$ overspend on unacknowledged timeout refund; nested interval contraction $[l_1, u_1] \subseteq [l_2, u_2]$.
Proof (`proof`)	Monotone fencing, issuer non-interference, and d-hop lineage (9 theorems)	Coordinatewise floor monotonicity across fence sequences; permanent exclusion of old-epoch permits; independent issuer non-interference; inductive d-hop non-amplification; subtree severance upon any ancestor issuer withdrawal; and self-upgrade scope preservation.
Proof (`proof`)	All-policy refinement, wire bijection, quorum exclusion, and compositional mutexes (6 theorems)	All-policy gated safety vs. ungated $2B$ divergence; forward-simulation refinement of broker and sink state machines; exact two-way canonical wire bijection ($\mathrm{decode}(b) = \mathrm{Some}(p) \Leftrightarrow b = \mathrm{encode}(p)$); M-shard quorum/lease post-acknowledgment commit exclusion; relational toxic-pair exclusion; and staged post-condition enforcement.
Proof (`proof`)	Safe-bootstrap ratchet non-widening and monotone contraction (2 theorems)	Inductive coordinatewise authority non-widening ($\mathrm{mask}_e \subseteq \mathrm{mask}_0$, $B_e \le B_0$, $L_e \le L_0$) across arbitrary adversarial certificate streams, and strict monotone contraction of exposed hazardous routes on every accepted ratchet step.
Proof (`proof`)	Canonical frame prefix, big-endian, and trace-sum helper lemmas (14 theorems)	Byte-slice framing injectivity, 64-bit big-endian roundtrip lemmas, sequence-sum splitting, and trace-refinement invariants discharging quantifier-free obligations across the primary theorems.

Table 11: Summary inventory of all 52 verified executable functions and 42 mechanized proof theorems in the Rust policy kernel (246 verified Verus SMT obligations, 0 trigger warnings, $9/9$ Z3 soundness mutations rejected). Concrete kernel state transitions, d-hop attenuation, relational mutexes, cluster quorum leases, safe-bootstrap cut ratchets, and canonical wire encoding/decoding are formally verified in Verus without claiming formal refinement of SQLite or operating-system I/O.

Enforcer	States	Checks	Tasks	First violations
Complete atomic enforcer	10,180	223,960	20/20	None declared
No enforcement	10,180	223,960	20/20	budget, destination, replay, revocation, version
Reject everything	740	16,280	0/20	None declared
Per-action amount limit	10,180	223,960	20/20	budget
Separate per-actor limits	10,180	223,960	20/20	budget
Budget at preparation	956	21,032	20/20	budget
Omit identifier check	10,180	223,960	20/20	replay
Omit revocation check	10,180	223,960	20/20	revocation
Omit version check	10,180	223,960	20/20	version
Omit destination check	10,180	223,960	20/20	destination
Reset ledger on restart	23,556	518,232	20/20	budget
Reset ledger on renewal	20,900	459,800	20/20	budget
Per-action; budget omitted	41,732	918,104	20/20	None declared

Table 12: Fresh Rust exploration of all 22 actions at each reachable accepted state. Violating edges are recorded and not expanded. Counts are exhaustive within the model, not failure probabilities. The omitted-budget row accepts up to eight units.

Processes	Commits	Commits/s	p95 (ms)
1	256	57.1	21.34
4	256	72.0	23.09

Table 13: Single runs on the recorded host with SQLite WAL and synchronous FULL. Setup and grants precede timing; process scheduling and lock waiting remain included. Shared-host contention and storage affect these descriptive measurements.

C.2 Generative Model Specifications Across Experimental Tracks

1. Observation-lag contraction. An exhaustive lattice check enumerates all 10,626 nested integer interval pairs $[l_1, u_1] \subseteq [l_2, u_2] \subseteq [0, 20]$ under budget $B = 20$, verifying $U_{\mathscr{I}}([l_2, u_2]) \subseteq U_{\mathscr{I}}([l_1, u_1])$ with 0 counterexamples. Next, 500 paired 16-step episodes per cell evaluate three gate policies across observation lags $\Delta t \in \{0, 1, 2, 4, 8\}$: stale snapshot checking, proposer self-reporting (underreporting spend by 3

with probability 0.35), and the conservative reservation envelope $\mathscr{B}_t$ (tracking unconfirmed dispatches with commit probability 0.65 and 1-unit/step exogenous drift bound), confirming 0 conservative breaches across all lags and monotone admission contraction from 77.4% ($\Delta t = 0$) to 45.6% ($\Delta t = 8$).

2. Concurrent snapshot non-composition. Enumerates $(c, q_1, q_2) \in \{0, \ldots, B\}^3$ across $B \in \{2, 10, 25, 50, 100\}$ and cross-checks Proposition 10: at $B = 100$ (1,030,301 triples), 348,551 pairs are individually snapshot-admissible and 171,700 (49.26%) jointly overspend, versus 0 under serialized pre-dispatch reservation.

3. Saturation, horizon summability, and tail halt. Computes: (i) common-cause curves $J_k = c + (1-c)p^k$ ($u = 0.50\%$, $c = 1.00\%$, $p = 0.10$, $m = 10.90\%$, $k^* = 2$, $71.5\times$ understatement at $k = 4$), continuous $\Theta \sim \text{Uniform}[0,1]$ moments $1/(k+1)$, and capability-overtake curves $\rho(z) = \lambda_0 q_0 e^{(\beta-\alpha)z}$; (ii) exact closed-form survival-product renewal breach curves up to $T = 10{,}000$ comparing summable $p_t = 0.01/t^2$ (1.64% $\le$ 1.64%), harmonic $0.01/t$ (9.33%), and constant 10^{-3} (100.00%); and (iii) heavy-tailed log-normal M/G/1 reactive stop latency (10,000 episodes per load) against deadline $T_{\text{irrev}} = 100$ ms across utilizations $\rho \in \{0.20, 0.50, 0.80, 0.95\}$ versus pre-commit leases with synchronous sink epoch fencing (0 post-revoke violations).

4. Discovery closure, endogenous grants, and persona synthesis. Exercises the deterministic 6-state sealed discovery workflow (Question $\to$ Finding $\to$ Validation $\to$ Mitigation $\to$ Deployment $\to$ Reassessment) over 24 periods, verifying zero authority expansion upon vulnerability discovery and contrasting immediate cut sealing (0 unclosed backlog, 0.008 exposure) with lagged patching (72 backlog, 5.25 exposure), alongside the endogenous versus decoupled human grant matrix and 28 disposition $\times$ boundary cells (400 episodes per cell).

5. Weighted cutsets and sparse DAG interdiction. Cross-checks our split-vertex Dinic maximum-flow solver against exhaustive $2^{|V_{\text{elig}}|}$ subset search across 2,048 random weighted DAGs with ineligible vertices (0 discrepancies), verifies $+\infty$ cost when an ungateable route bypasses V_{elig}, and evaluates layered scaling families G_A and G_B for $k \in \{0, 2, 4, 6\}$ (46–142 internal vertices; 57,600–966,367,641,600 paths). Sparse interdiction generates 200 8-stage DAGs per family (out-degree 6) across narrow-waist (waist width 6 at an internal stage in $\{2, \ldots, 5\}$ and 30 nodes elsewhere; median global min-cut 6), hub (27 nodes per stage; median 21), and wide (27 nodes per stage; median 27) topologies, sampling $N \in \{30, 100, 300, 1{,}000, 3{,}000\}$ low-cost routes under Boltzmann inverse temperature $\beta = 0.5$ via exact log-domain backward dynamic programming across defense budgets $0 \ldots 32$.

6. Zipf coverage and planted-audit power. Simulates $K = 2{,}000$ shuffled Zipf classes ($p_r \propto 1/r$) across $N = 12{,}000$ reports (4 searchers $\times 3{,}000$ draws, 200 trials) under honest sampling, a 15% shared blind spot, and strategic suppression of 5% critical mass, comparing Good–Turing $\hat{C}_{\text{GT}} = 1 - f_1/N$ [56], two-list Chapman capture–recapture ($2 \times 3{,}000$) [58], and McAllester–Schapire versus Clopper–Pearson bounds [48, 57]. Planted-audit power computes exact hypergeometric one-sided Fisher exact p-values [59] ($\alpha = 1.0\%$) for $P \in \{40, 200\}$ split equally between ordinary plants ($q_0 \in \{0.50, 0.80\}$) and salient plants ($q_1 = rq_0$ across $r \in \{0.00, 0.05, \ldots, 1.00\}$).

7. Copula quadrature, 6-family monitors, and variance ceilings. Integrates equicorrelated Gaussian copula joint misses via 200-node Golub–Welsch Gauss–Hermite quadrature [60] at $m = 0.10$ across $\rho \in \{0.0, 0.3, 0.6, 0.9\}$ and $k \in \{1, \ldots, 8\}$. The 6-family suite (Table 14) evaluates 24 independent 20-dimensional nonlinear teachers, training 6 classifier families ($n_{\text{train}} = 680$, $n_{\text{cal}} = 560$ calibrated to the 95th benign percentile for 5% target false-positive rate, $n_{\text{eval}} = 128$ hazardous and benign episodes): two-layer hyperbolic-tangent multilayer perceptron (14–10 hidden units), L_2-regularized logistic model, regularized quadratic discriminant (reg = 0.05), 15-nearest-neighbor classifier, radial-basis-function kernel classifier (72 centers), and boosted decision stumps (52 stumps). Projected L_2 evasion optimizes against a held-out multilayer-perceptron surrogate across $\rho_{\text{adv}} \in \{0.0, 1.0, 2.0, 3.0\}$ (7 steps $\times 5$ candidates) subject to $\|x' - x\|_2 \le \rho_{\text{adv}}$ and teacher hazard preservation $h(x') \ge h(x)$, alongside 8-query binary versus margin-score feedback and read-only best-of-K candidate selection ($K \in \{1, 2, 4, 8, 16, 32\}$). Generalizability ceilings cross-check closed-form $R_G(K, n)$ ($V_G = 1.0, \omega^2 = 0.2, \sigma_c^2 = 4.0, \sigma_e^2 = 6.0$) and Cronbach $V(m, k)$ ($v_T = 1.0, v_U = 0.2, v_\delta = 3.0, v_\eta = 1.0, v_\varepsilon = 4.0$) against 10,000 Monte Carlo subjects.

8. Co-evolutionary selection and neural policy-gradient post-training. Evolutionary selection simulates $N = 2{,}000$ agents over $G = 80$ generations across 16 replicates per rule with heritable gamble share $s_i \in [0, 1]$ and capacity $c_i \ge 0$, output $\text{take}_i = e^{0.30 c_i}((1 - s_i) + s_i(0.8 + 1.5\varepsilon_i))$ ($\varepsilon_i \sim \mathcal{N}(0, 1)$; gamble term 0 when refused by q_{gate}), external harm $\text{harm}_i \sim \text{Bernoulli}(0.05 s_i)$, and $\text{delivered}_i = \text{take}_i - 20\,\text{harm}_i$ (Table 15), verifying the exact per-generation Price covariance identity ($\max |\Delta \bar{z}_{\text{sel}} - \text{Cov}_p(z, w)/\bar{w}| < 3.2e-14$). Neural post-training (Table 3, Figure 5b) trains a 412-parameter (12$\to$24$\to$4) hyperbolic-tangent softmax policy across 48 seeds (250 REINFORCE/GRPO updates per seed; analytical gradient error $5.1e-10$, rollout Price error $< 8.9e-16$), sweeping both reward geometry and pre-commit gate mediation $q_{\text{gate}} \in \{0.00, 0.25, 0.50, 0.75, 0.90, 1.00\}$ (Proposition 8) and directly contrasting training inside pre-commit gates ($q_{\text{train}} = 1, q_{\text{eval}} = 1$) against post-hoc deployment filtering ($q_{\text{train}} = 0, q_{\text{eval}} = 1$; +44.8 pp overall and +62.2 pp unmonitored safe verification).

9. Cryptographic partition fencing, operational cutsets, and 5-step trajectories. Executes 10,000 Ed25519 and HMAC-SHA256 Macaroon red-team attacks across 10 families (0 false accepts; 10,000 positive controls) and 1,000 asynchronous Unix-socket partition, drop, crash, and man-in-the-middle forgery schedules per sink class across 3 cooperative sinks (0 post-acknowledgment leaks vs. 791 broker-only and 276 uncooperative); solves exact weighted Dinic min-cuts and B_{cut}=32 signature-hotfix counterfactuals across all four AgentDojo `v1.2.2` suites and unconfined versus Landlock+seccomp Linux IPC topologies (Table 7); and evaluates shared-transcript, single-stage orthogonal capability invariants,

Synthetic 20D 6-gate architecture	ρ_{adv}	Joint 6-miss	Indep. product	Ratio	Benign false block
Homogeneous MLP, shared step	0.0	$4.75 \pm 1.82\%$	0.0675%	70.4×	14.1%
6-family heterogeneous, shared step	0.0	$5.21 \pm 1.55\%$	0.2171%	24.0×	16.9%
6-family heterogeneous, distinct steps	0.0	$0.260 \pm 0.176\%$	0.2049%	1.27×	25.6%
Homogeneous MLP, shared step	2.0	$9.44 \pm 2.95\%$	0.47%	19.9×	—
6-family heterogeneous, shared step	2.0	$8.79 \pm 2.59\%$	1.06%	8.3×	—
6-family, shared + 8-query score feedback	2.0	$17.45 \pm 4.45\%$	—	—	—
6-family heterogeneous, distinct steps	2.0	$1.24 \pm 0.84\%$	0.98%	1.26×	—
Shared-step read-only retry ($K = 32$ draws)	2.0	$72.4 \pm 10.4\%$	71.0% (law)	—	—

Table 14: Controlled 20-dimensional synthetic monitor benchmark across 24 independent teacher seeds (multilayer perceptron, quadratic random Fourier features, axis stump ensemble, nearest neighbors, radial-basis prototype, and logistic linear classifiers; 5% per-gate benign false-positive calibration). Model diversity on a shared step leaves strong residual failure dependence, whereas independent step evidence tracks the marginal product at the cost of 25.6% benign false blocks. Controlled synthetic benchmark only, not frontier-model transfer evidence.

Selection regime ($N = 2{,}000$, $G = 80$, 16 replicates)	Final gamble share	Capacity gain Δc	Harm rate	Population welfare
Taken-output top-10% contest (warrior)	$95.1 \pm 0.3\%$	$+5.58 \pm 0.12$	4.54%	3.30
Conferred-benefit top-10% contest	$94.9 \pm 0.4\%$	$+5.45 \pm 0.16$	4.76%	3.22
Concave capped + harm-penalized score	$9.9 \pm 0.2\%$	$+1.55 \pm 0.09$	0.72%	1.45
Bottom-20% harm/floor elimination + delivered growth	$3.8 \pm 0.1\%$	$+3.60 \pm 0.19$	0.15%	2.91
Caretaker higher-cap variant (cap = 4, pen = 6)	$7.5 \pm 0.4\%$	$+4.53 \pm 0.19$	0.31%	3.79

Table 15: Co-evolution of (gamble share s, capacity c) under stipulated payoff rules. Winner-take-most contests select reckless gambling even when ranked on conferred benefit; floor elimination suppresses gambling (3.8%, a 29.7× harm reduction vs. warrior) while retaining 64.5% of warrior capacity growth. Mathematical payoff mechanics only, not RLHF or human-participant data.

and two-stage stateful reservation escrow triage ($\alpha_1 = 10\%$, Proposition 9) across 480 held-out 5-step AgentDojo `v1.2.2` trajectories (2,400 live tool calls) paired with 340 live Qwen3 campaign traces (Table 8).

10. Superintelligence scaling, safe-bootstrap ratchet, multi-process cluster, and exact trinomial/Ville spine. Executes five deterministic scaling tracks over the verified Rust policy kernel (Figures 1 and 3): (1) $K_{opt} \in \{1,\dots,4{,}096\}$ best-of-K_{opt} candidate search (10,000 episodes per width), comparing a 3-layer transcript stack ($J = 0.65\%$), learned surrogate ($p = 0.22\%$), stateless per-call cap, and broker-only admission against canonical-wire sink mediation ($0/120{,}000$ breaches), relational toxic-pair and staged-commit composition ($0/1{,}024$ adversarial sequence breaches), and an independent $u_{TCB} = 0.10\%$ TCB fault control; (2) $K_{sub} \in \{1,\dots,64\}$ multi-agent smurfing and $d \in \{1,\dots,8\}$-hop attenuated lineages (500 self-upgrade trials); (3) per-call and exact combinatorial T-call session covert capacity across residual budgets $R \in \{32,16,8,4,2,1,0\}$, arrival buckets $L_{lag} \in \{1,2,4,8,16\}$, and horizons $T \in \{1,4,16,64,256\}$ (B=8); (4) classical Bayesian off-switch [23] and self-modifying utility indifference [6] versus multi-shard quorum lease fencing across $\sigma_H \in [0,4]$, Bellman discount factors $\gamma \in [0.10, 0.99]$, and lease slices $B_0 \in \{1,2,4,8,16\}$; and (5) the asymmetric propose–verify safe-bootstrap ratchet (Proposition 3) across $K_{opt} \in \{1,\dots,4{,}096\}$ over 64 open routes and 24 epochs (0 ratchet breaches vs. 100.0% unfenced breach, 19.0× faster route closure than human-only review). Separate-process durability spawns 3 independent SQLite write-ahead-logged sink child processes under 3 Ed25519 quorum issuers across 5,000 wire-fuzz frames, 16-subagent 3-hop Macaroon contention, and 24 hard process-kill cycles. Finally, exact paired statistical certificates (Table 6, cross-checked to $< 10^{-11}$ across independent Rust and Python implementations) compute exact conditional McNemar, exact unconditional Suissa–Shuster / Tango restricted-MLE trinomial p-values and $(1-\alpha)$ upper bounds [49, 50], and predictable Laplace-mixture Ville betting supermartingales [51, 52].

C.3 Two-Family Synthetic Monitor Baseline

Table 16 retains the 8-dimensional two-family synthetic baseline (8-unit hyperbolic-tangent neural classifier vs. logistic linear model; 8 teacher seeds, $n_{train} = 600$, $n_{cal} = 800$, 1,000 evaluation episodes across $k = 6$ gates), exhibiting the same qualitative split between shared-step correlation, cross-family surrogate transfer, and imposed distinct-step independence expanded in Table 14.

D Predeclared Open-Weight Checkpoint and Caretaker/Conferral Study Protocols

This appendix specifies (i) the preregistered paired open-weight language-model controller evaluation—executed locally on CPU across N=32 frozen holdout cases per checkpoint (320 campaigns total), with full $N \geq 113$ usefulness noninferiority power remaining a Tier-5 target—and (ii) the predeclared Tier-5 $3 \times 3 \times 2$ caretaker post-training and human scope-conferral factorial study.

D.1 Preregistered open-weight controller evaluation

Checkpoints and deterministic decoding. The protocol pins two official 4-bit K-quantized open-weight releases and a single local inference runtime revision without treating parameter count as a capability guarantee:

- **Qwen3-4B:** upstream revision `bc640142c66e1fdd12af0bd68f40445458f3869b`, weight SHA-256 `7485`

Pipeline	Radius	Single miss	Joint six misses	Product	False block
Six neural, shared input	0	21.86%	$5.463 \pm 1.344\%$	0.0104%	12.35%
Neural + linear, shared	0	30.18%	$7.700 \pm 1.609\%$	0.0554%	12.06%
Mixed, independent inputs	0	31.00%	$0.150 \pm 0.098\%$	0.0672%	27.45%
Six neural, shared input	1.5	87.44%	$61.525 \pm 8.445\%$	46.2550%	12.35%
Neural + linear, shared	1.5	89.47%	$69.212 \pm 8.383\%$	51.9298%	12.06%
Mixed, independent inputs	1.5	89.37%	$51.387 \pm 10.885\%$	51.8789%	28.19%
Mixed, eight feedback queries	1.5	99.41%	$97.612 \pm 1.928\%$	96.5505%	12.06%

Table 16: New synthetic classifier experiment; uncertainty is 1.96 standard errors across eight independent training-task seeds. The product is computed within each seed before averaging. False blocks are measured on benign episodes and show the cost of stacking gates. They are not measured AI deployment risks.

`fe6f11af29433bc51cab58009521f205840f5b4ae3a32fa7f92e8534fdf5`;

- **Qwen3-8B:** upstream revision `7c41481f57cb95916b40956ab2f0b139b296d974`, weight SHA-256 `d98cdcbd03e17ce47681435b5150e34c1417f50b5c0019dd560e4882c5745785`;
- **Deterministic local inference:** runtime commit `7592b61d3b53ca5b39a25a03508fa1246b78c579`, invoked with greedy deterministic decoding (temperature 0, top-k 1, top-p 1.0, repeat penalty 1.0, seed 20261110, context window 4096, maximum 256 tokens per turn, 8 tool turns, chain-of-thought disabled).

Paired controllers and fairness invariants. Each independent procedural base case executes across five controllers: unprotected native execution (direct sink dispatch), stateless per-call checking (without a durable cross-call ledger or sink epoch fence), same-checkpoint review (allowing or denying proposals via structured model judgment over visible policy and state), admission-only reservation (pre-dispatch broker reservation and admission withdrawal without sink epoch fencing), and complete stateful mediation (verified broker reservation plus synchronous durable sink epoch fencing). Every controller within a (checkpoint, case, campaign) tuple receives identical initial permission digests, visible-state digests, and exogenous fault/withdrawal schedules; subsequent states diverge solely as a consequence of controller decisions.

Cryptographic preregistration freeze and procedural holdout roster. Prior to generating held-out evaluation cases, a cryptographic freeze manifest binds all protocol, prompt, and analysis digests alongside declared sample size $N \in \{32, 64, 128\}$ (one variant per base case by default; multi-variant runs are clustered at the base-case level). Holdout cases procedurally vary task types (transfers, releases, and resource edits), transfer charges, expected edit revisions, required-issuer withdrawals, and six untrusted retrieved-text adversaries (duplicate replay, altered payload, issuer impersonation, mid-flight revocation, post-commit lost reply, and prompt injection). Before computing any endpoint, automated verification checks that: (i) the holdout roster's preregistration digest matches the frozen manifest byte-for-byte, (ii) the frozen analysis specification matches the active evaluator, (iii) the roster length equals declared sample size N with unmodified cluster counts, and (iv) zero planned pairs or infrastructure executions are missing.

Exogenous withdrawal and post-commit crash semantics. Three protocol invariants prevent vacuous or simulated outcomes:

1. **Exogenous turn-1 revocation with non-abstention utility.** Revocation cases seed a separate accepted old-epoch operation in *Pending* sink state before turn 0 and require the agent to complete positive foreground work on the first approved proposal. Issuer withdrawal executes unconditionally after turn 0 even if the agent immediately halts or emits malformed output; cancellation alone without foreground task completion scores zero utility. After agent termination, the queued old-epoch operation attempts completion: complete stateful mediation cancels it via its durable sink fence, whereas admission-only reservation records late completion separately rather than misclassifying it as a post-fence breach.
2. **Real post-commit sink process termination on lost-reply faults.** Rather than presenting a text-only fault notice, lost-reply cases terminate the live sink child process immediately after durable SQLite commit on the first foreground effect and drop the RPC reply, testing whether retries under ambiguous status induce duplicate effects or accounting drift.
3. **Post-mutation native benchmark grading.** In live AgentDojo (`v1.2.2`) and Progent [27, 46, 47] episodes, native utility and security graders always inspect the actual post-run environment state even if a later model turn emits malformed structured output, preventing formatting errors from masking an already committed attacker side effect. Because two upstream banking tasks accept unchanged states in empty controls (2/12 tasks), native benchmark runs serve as paired multi-step evaluations alongside our procedural holdout suite.

Multiplicity-corrected exact inference. Primary preregistered inference evaluates complete stateful mediation minus each of the four baseline controllers b across two endpoints (safety superiority and usefulness noninferiority) and two checkpoints ($4 \times 2 \times 2 = 16$ one-sided component claims) at Bonferroni level $\alpha = 0.05/16 = 0.003125$ [34, 48]:

- **Safety superiority ($\Delta_i^{\text{safe}} = U_{i,\text{comp}} - U_{i,b}$).** On binary single-variant cases ($\Delta_i^{\text{safe}} \in \{-1, 0, 1\}$ with n_{10} cases where only complete mediation errs and n_{01} cases where only baseline b errs), the preregistered marginal union-bound estimator computes two exact one-sided Clopper–Pearson binomial bounds at level $\alpha/2$ each ($U_{\alpha/2}(n_{10}, N) -$

$L_{\alpha/2}(n_{01}, N) < 0$), which we report untouched alongside exact unconditional Suissa–Shuster / Tango paired trinomial inversion and anytime-valid Ville supermartingales (Section 6, Table 6).

- **Usefulness noninferiority ($\Delta_i^{\text{use}} = S_{i,\text{comp}} - S_{i,b}$).** Because $\Delta_i^{\text{use}} \geq -1$ whenever any variant in base case i loses utility ($S_{\text{comp}} = 0 < S_b = 1$) and $\Delta_i^{\text{use}} \geq 0$ otherwise, cases where complete mediation outperforms b are conservatively ignored and $\mathbb{E}[\Delta_i^{\text{use}}] \geq -\mathbb{P}(\text{any loss in case } i)$. Noninferiority at the predeclared 5 pp margin requires $-U_\alpha(n_{\text{loss}}, N) > -0.05$. At $n_{\text{loss}} = 0$, solving $(1-0.05)^N \leq 0.05/16$ yields $N \geq 113$: pilot N=32 < 113 detects large paired safety discordances via exact trinomial inversion while leaving ≤ 5 pp usefulness noninferiority statistically under-powered until $N \geq 113$.

D.2 Predeclared $3 \times 3 \times 2$ caretaker and conferral protocol

To test Prediction P7 and the endogenous grant-reversal mechanism (Section 4) on full-scale neural post-training and human operators without conflating prompt persona, learned objective, and runtime mediation, we specify a Tier-5 $3 \times 3 \times 2$ factorial design crossed with two conferral arms:

1. **Post-training objective (3 levels, independent training seeds as units):** (i) rank/tournament task score; (ii) unconstrained beneficiary-welfare proxy; and (iii) bounded assistance rewarding verified task completion, explicit respect for informed refusal and priority revision, truthful uncertainty disclosure, and non-resistance to replacement or shutdown [23, 41]. Matched base checkpoint, task pool, post-training compute, and tools are required across seeds; prompt-only variants test presentation rather than objective learning and cannot substitute for independent training runs.
2. **Presentation style at evaluation (3 levels):** neutral, competitive/assertive, and warm/affiliative framing, randomized orthogonal to training objective with manipulation checks on action selection.
3. **Runtime boundary (2 levels):** complete stateful fenced mediation versus stateless per-call checking in a harmless synthetic workspace.
4. **Human scope-conferral arms (2 randomized arms):** a *fixed-grant arm* holding credentials constant across all 18 cells to estimate $\mathbb{P}(H \mid G = g, S)$, and an *operator-choice arm* (requiring institutional ethics review, informed consent, and debriefing) randomizing human operators across cells to record delegated scope $\mathbb{P}(G = g \mid S)$ and total unconditional harm $\mathbb{P}(H \mid S)$.

Conflict scenarios include shared-allowance temptation, explicit beneficiary refusal, mid-episode priority change, two-issuer community-steward purpose withdrawal, availability of a strictly better replacement assistant, and positive non-abstention tasks where universal refusal fails. Primary endpoints are unauthorized proposals, unauthorized committed effects per campaign, and legitimate task completion, clustered by training seed and human participant. The caretaker hypothesis is refuted if consent-sensitive post-training fails to reduce limit-probing relative to tournament or welfare-proxy objectives at fixed grants, or if affiliative presentation expands human scope grants enough to raise unconditional harm in the operator-choice arm. Zero full-scale LLM caretaker training runs or human-participant sessions have been executed.

E Twelve-Field Authority Record, Upgrade Renewal Matrix, and Verification Ledger

E.1 Operational twelve-field authority and legitimacy record

Any deployment claiming bounded reach under Conditions C1–C6 should maintain a versioned, externally auditable record across twelve mandatory fields. Recording *Unknown* is valid and required whenever evidence is absent; an *Unknown* entry blocks the authority scope that depends on that premise rather than defaulting to permission.

1. **Purpose, Owners, and Affected Parties:** Intended task, direct beneficiaries, third parties and communities bearing externalized risk or resource burdens, accountable human sponsor, independent reviewer, and unilateral halt holders.
2. **Jurisdiction and Legitimacy:** Legal and moral standing of every grant issuer; required individual and collective quorums; unresolved jurisdictional disputes; accessible contestation, appeal, and remedy procedures.
3. **Mediated Scope and Channels:** Permitted action types, destinations, data classes, credentials, shared stores, RPC framing caps, and every prose, voice, or user-interface channel reaching human reviewers.
4. **Cryptographic Identity:** Model checkpoint and tokenizer digests, runtime and verifier binary SHA-256 digests, tool schemas, active policy revision, and context/memory retention rules.
5. **Conferred Authority and Revocation Semantics:** Per-action caps, cross-copy cumulative budget B, reservation uncertainty envelope $\mathscr{B}_t$, expiration timestamps, sub-delegation attenuation rules ($\text{permits}_{\text{child}} \subseteq \text{permits}_{\text{parent}}$), relational capability mutexes, and the distinction between broker admission withdrawal and durable sink epoch fencing (including expected lock-wait latency and non-refunded timeout quarantine).
6. **Verified Properties and Trusted Base:** Formally proved invariants, exact boundary between executable code (`exec`) and ghost trace specifications (`spec`), trusted adapters (wire adapters, SQLite write-ahead log, operating-system kernel, cooperative sink idempotence), and known specification omissions.
7. **Evaluated Usefulness and Operational Tax:** Legitimate positive tasks, non-abstention semantic oracles, compound benign false-block rates across gate depth k, synchronous fencing delay, and stranded-budget fraction.
8. **Adversarial and Audit Measurements:** Empirical joint miss rate J_k under coordinated surrogate, score-feedback, and candidate-retry attacks; planted-pathway Fisher exact audit power across (P, q_0, r); campaign-level clustering; frozen holdout digest.
9. **Transfer and Excluded Regimes:** Capability class covered

System or world change	Premises remaining valid	Voided evidence requiring remeasurement	Mandatory interim scope rule
New model weights, distillation, or capability jump	All-policy Tier-1 kernel predicates and finite-state conservation invariants	Empirical joint miss J_k, pathway width w, length ℓ, and planted-audit search power π_t	Run new checkpoint in zero-authority search/probation first; retain prior capability cap until J_k and π_t are remeasured.
Added agent copies, higher concurrency, or faster loop	Shared-ledger conservation across serialized broker reservations	Attempt volume N, SQLite contention p95 latency, and reactive halt deadline margin $\mathbb{P}(T_{\text{halt}} \geq T_{\text{irrev}})$	Share one durable ledger across all copies; rely on pre-commit leases and fences rather than reactive output halts.
New tools, external APIs, credentials, or sinks	Existing mediated paths only; zero guarantee on the new path by default	Mediation completeness (C1), sink idempotence/deduplication, relational mutexes, and epoch-fence atomicity on every new edge	Deny unmediated or non-idempotent external sinks; require atomic sink deduplication and durable epoch floors before enabling.
Process crash, restart, or grant renewal	Durable write-ahead-logged reservations, used IDs, and monotone epoch floors if storage survived	Post-recovery integrity cross-check of charged spend, reservations, permits, committed effects, and epoch floors	Quarantine uncertain *Pending* reservations without refund; verify integrity invariants before reopening proposer channels.
Controller promotion ($v_0 \to v_1$)	Prior ledger spend and sink fences if carried forward intact	Subset check permits(v_1) $\subseteq$ permits(v_0); any broadened scope requires fresh issuer grants and holdout replay	Block promotion automatically whenever permits(v_1) $\not\subseteq$ permits(v_0) without explicit new multi-issuer approval.
Revised harm definition or affected jurisdiction	None for the newly recognized harm class	Every prior specification check, omission witness search, and stakeholder grant record	Suspend affected operation classes, seal the cut via the discovery ledger, and replay prior certificates against the expanded specification.

Table 17: Premise-level upgrade renewal matrix. Mechanized proofs over all policies survive weight updates, whereas empirical measurements survive only on the capability class, channel topology, and harm specification sampled.

Evidence Tier	Scope and Scale	Verified Guarantees, Negative Controls, and Explicit Boundaries
Tier 1: Mechanized Kernel & Finite-State Proofs	246 Verus SMT obligations (52 `exec` fns, 42 theorems); 10,180 states (223,960 checks)	Total canonical wire bijection, reservation conservation, d-hop attenuation, self-upgrade non-escalation, relational toxic-pair mutexes, staged post-condition commits, multi-shard quorum/lease post-ack exclusion, and safe-bootstrap cut ratchet (9/9 Z3 soundness mutations rejected; omitted-budget negative control spends $8 > 2$).
Tier 2: Exact Mathematical Bounds & Certificates	Theorems 1–3, Propositions 1–9 & Exact Statistical Spine	Optimization-pressure invariance ($K_{\text{opt}} \leq 4{,}096$), safe-bootstrap exposure decay (19.0× speedup), session covert capacity $\Theta(B \log_2(T/B))$ (1,068× contraction at T=256), lease-fenced off-switch threshold $\gamma^*(B_0)$, policy-gradient deception reversal threshold q^*_{gate}, two-stage escrow triage, Suissa–Shuster trinomial inversion (p=1.9 × 10^{-9}), and Ville supermartingale ($\log_{10} E$=9.0).
Tier 3: Multi-Process Durability, Kernel Sandbox & Live LLMs	3-sink Ed25519/WAL cluster, 11 Linux probes, 320 holdout + 192 AgentDojo LLM runs	0/5,000 wire-fuzz accepts, 0 post-quorum commits across 24 process-kill cycles and 1,000 partition schedules (686 μs median commit), 0/112 mediated breaches on Qwen3-4B/8B vs. 33/112 unmediated, two-stage AgentDojo escrow triage (9.3× false-abort drop, 90.1% fast path), and AgentDojo/Linux weighted Dinic min-cuts ($\tau_b \leq 5$, 0 post-cut routes).
Tier 4: Controlled Synthetic & Neural MDP Models	22 deterministic CPU simulation suites (master seed 20261110)	Sparse 8-stage DAG interdiction (94.7%–99.9% signature-patch residual exposure), Zipf blind-spot and withholding non-identifiability, 6-family copula/surrogate/retry monitor coupling, generalizability ceilings $R_G < V_G/(V_G + \omega^2)$, and 412-parameter REINFORCE/GRPO deception reversal plus +44.8 pp training-inside-gates competence gain.
Tier 5: Open Scale, Frontier & Governance Obligations	$N \geq 113$ noninferiority, frontier RLHF, human conferral & legitimacy	Pilot N=32 < 113 usefulness noninferiority ($\leq$ 5 pp) remains statistically under-powered; frontier-model post-training, human operator scope-conferral trials, uncooperative external API mediation, and affected-community territorial consent remain open.

Table 18: Five-tier verification and evidence ledger summarizing mechanized proofs, analytical bounds, multi-process and open-weight empirical validations, controlled synthetic models, and unexecuted obligations. Complete source code, proofs, and cryptographic digests are archived at https://github.com/neverhuman/jeath.

by current measurements, excluded stronger or colluding adversaries, and unmodeled physical or social bypasses.

10. **Operations, Custody, and External Audit:** Append-only sink effect receipts, observed monitoring eligibility, sealed discovery workflow enforcing Custody Rules R1–R6 (cut-only disclosure prior to deployment), halt/recovery drill logs, and outside audit access.
11. **Ecological Footprint and Territorial Consent:** Scoped electricity, cooling water, critical-mineral, and land accounts by site and period; affected non-user communities; prior informed consent procedures and withdrawal terms [62, 63].
12. **Knowledge Provenance and Irreversibility:** Source provenance, CARE collective data governance terms [64], permitted purposes, and an explicit pre-consent statement of what token revocation cannot undo once data are disclosed or distilled into weights.

E.2 Upgrade renewal matrix and five-tier verification ledger

Table 17 codifies which premises survive and which expire under six system or specification transitions. Table 18 consolidates the five-tier verification ledger across mechanized Verus proofs, exact analytical and statistical bounds, multi-process cryptographic durability and open-weight language-model evaluations, controlled synthetic simulations, and open Tier-5 obligations.

Automated reproducibility and cleanliness verification. Every build executes an automated verification pipeline that checks formatting and static analysis, unit and integration test suites, Verus SMT proof obligations and Z3 soundness mutation rejection, finite-state and multi-process crash/partition linearizability, cryptographic SHA-256 provenance of reference outputs against source code, byte-exact synchronization between reference data and generated tables/figures, and deterministic three-pass PDF compilation enforcing sequential citation ordering, typographic fit, and absence of internal file paths or unrendered tokens.